

Chapter 11 "HEAT"

Heat:

Heat is a form of energy that always flows from high temperature body to low temperature body. OR

"In an isolated system, heat is defined as the sum of kinetic energies of molecules or atoms."

Unit:

The S.I unit of heat is Joule. In C.G.S system unit of heat is Caloric and in British Engineering System the unit of heat is Btu (British thermal unit).

1 Btu = 252 Cal

1 Btu = 1055 Joule

1 Cal = 4.1865 Joule

Different modes of transfer of heat:

1. Convection
2. Conduction
3. Radiation

Convection:

When heat is transferred from hot body to cold body by actual movements of molecules of atoms then it is called convection process. Gases and liquids are heated by this mode of transfer of heat.

Conduction:

In conduction heat is transferred from hot body to cold body by vibration of atoms of molecules when they are in contact with each other. In metals heat is transferred by conduction process.

Radiation:

In radiation heat is transferred from hot body to cold body in form of electromagnetic waves. Sunlight and Laser light are examples of radiation.

Temperature:

"The degree of hotness or coldness of an object or body is called temperature. "

OR

"In an isolated system temperature is the average translational kinetic energy of molecules or atoms."

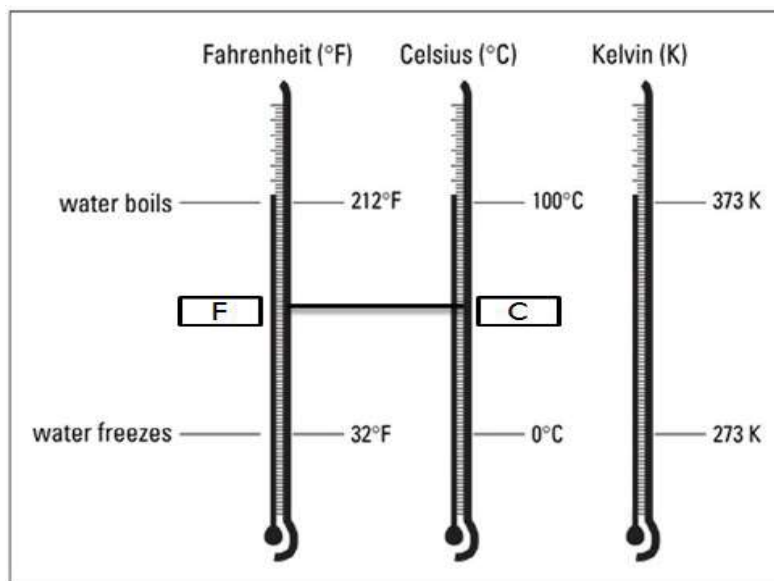
The S.I unit of temperature is Kelvin. Other units are Fahrenheit and Centigrade.

Temperature scales:

There are three different scales of temperature

1. Celsius Scale
2. Fahrenheit Scale
3. Kelvin Scale

Relation between Kelvin scale and Celsius scale



From figure it is clear that

$$K = C + 273$$

Relation between Fahrenheit scale and Celsius scale

Consider a common point on Fahrenheit and Celsius scale then, we have

$$\frac{C - 0}{100 - 0} = \frac{F - 32}{212 - 32}$$

$$\frac{C}{100} = \frac{F - 32}{180}$$

$$C = \frac{100}{180} (F - 32)$$

$$C = \frac{5}{9} (F - 32)$$

Law of Heat Exchange:

In an isolated system the amount of heat lost by the hot body is always equal to the amount of heat gained by the cold body.

$$\text{Heat lost by hot body} = \text{Heat gained by cold body}$$

Thermometric Property:

Property of a substance, which changes uniformly with the change in temperature, is called Thermometric property.

Thermal Expansion:

The change in physical dimension of a body (e-g length, Area or volume) with the change in temperature is called Thermal expansion.

Type of Thermal Expansion:

There are following type of thermal expansion:

1. Linear Expansion.
2. Volume Expansion.
3. Areal Expansion

Linear Expansion:

The phenomenon in which length of an object changes due to change in its temperature is called linear expansion.

Mathematical Expression:

Experimentally it is found that change in length is directly proportional to the original length L and change in temperature ΔT .

$$\Delta L \propto L$$

$$\Delta L \propto \Delta T$$

$$\boxed{\Delta L = \alpha L \Delta T} \text{ ----- (1)}$$

Where α is constant of proportionality and termed as coefficient of linear expansion. Its value changes from material to material and it may be defined as

"The change in length per unit length per unit change in temperature"

$$\boxed{\alpha = \frac{\Delta L}{L \Delta T}}$$

Final Length (L'):

If L' is the final length then change in length can be expressed as

$$\text{Change in length} = \text{final length} - \text{initial length}$$

$$\Delta L = L' - L$$

Substitute this value in equation (1), we have

$$L' - L = \alpha L \Delta T$$

$$L' = L + \alpha L \Delta T$$

$$\boxed{L' = L (1 + \alpha \Delta T)}$$

Volume Expansion:

The phenomenon in which volume of an object changes due to change in its temperature is called volume expansion.

Mathematical Expression:

Experimentally it is found that change in volume is directly proportional to the original volume and change in temperature ΔT .

$$\Delta V \propto V$$

$$\Delta V \propto \Delta T$$

$$\boxed{\Delta V = \beta V \Delta T} \text{ ----- (1)}$$

Where β is constant of proportionality and termed as coefficient of volume expansion and it may be defined as

"Change in volume per unit volume per unit change in temperature."

$$\boxed{\beta = \frac{\Delta V}{V \Delta T}}$$

Final Volume (V'):

If V' is the final volume then change in volume can be expressed as

$$\text{Change in volume} = \text{final volume} - \text{initial volume}$$

$$\Delta V = V' - V$$

Substitute this value in equation (1), we have

$$V' - V = \beta V \Delta T$$

$$V' = V + \beta V \Delta T$$

$$V' = V (1 + \beta \Delta T)$$

Relation between Coefficients of linear and volume expansion:

Consider a box of dimensions l , w and h . Its volume at some temperature T_1 is V . If the temperature changes to T_2 and its volume changes to V' then according to linear expansion

$$L' = L (1 + \alpha \Delta T)$$

Since each dimension of cubic box is linear in nature therefore new dimensions will be

$$l' = l(1 + \alpha \Delta T)$$

$$w' = w(1 + \alpha \Delta T)$$

$$h' = h(1 + \alpha \Delta T)$$

We know that initial volume at temperature T_1

$$V = lwh \text{ _____ } (1)$$

After temperature change to T_2 , final volume will become

$$V' = l'w'h'$$

$$V' = l(1 + \alpha\Delta T) w(1 + \alpha\Delta T) h(1 + \alpha\Delta T)$$

$$V' = lwh(1 + \alpha\Delta T)^3 \text{ _____ } (2)$$

Using the mathematical relation

$$(a + b)^3 = a^3 + b^3 + 3a^2b + 3ab^2$$

And

$$V = lwh$$

Putting values in equation (2), we have

$$V' = V(1 + \alpha^3\Delta T^3 + 3\alpha\Delta T + 3\alpha^2\Delta T^2)$$

As α is very small in magnitude neglecting its higher powers we get

$$V' = V(1 + 3\alpha\Delta T) \text{ ----- } (3)$$

$$V' = V + 3\alpha V\Delta T$$

$$V' - V = 3\alpha V\Delta T$$

$$\Delta V = 3\alpha V\Delta T$$

$$\frac{\Delta V}{V\Delta T} = 3\alpha$$

By definition, we know that

$$\beta = \frac{\Delta V}{V\Delta T}$$

Therefore

$$\boxed{\beta = 3\alpha}$$

Hence, coefficient of volume expansion is three times the coefficient of linear expansion.

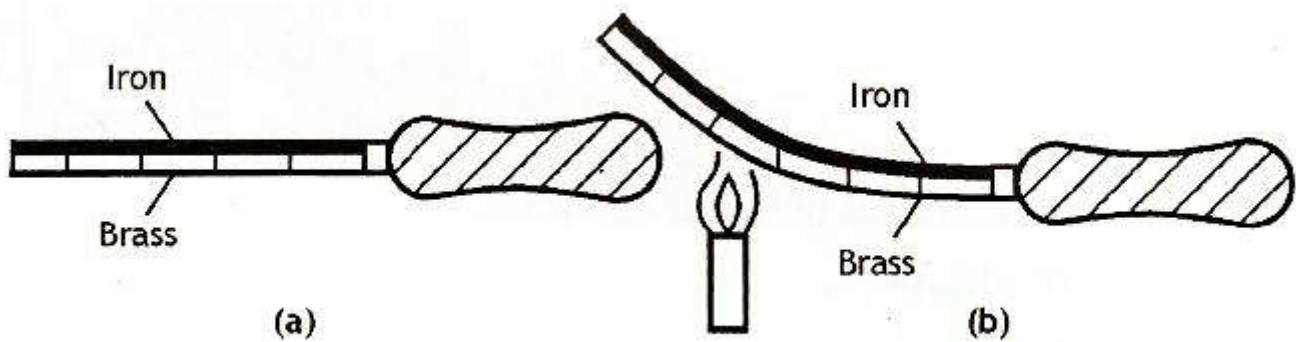
Bimetallic Strip:

A strip which is made of two different metals having different coefficient of linear expansion is called bimetallic strip.

Working Principle:

When the strip is heated the two metals having different coefficient of linear expansion expands at different rates causing the strip to bend.

coefficient of linear expansion $_{Brass} > \text{coefficient of linear expansion}$ $_{Iron}$

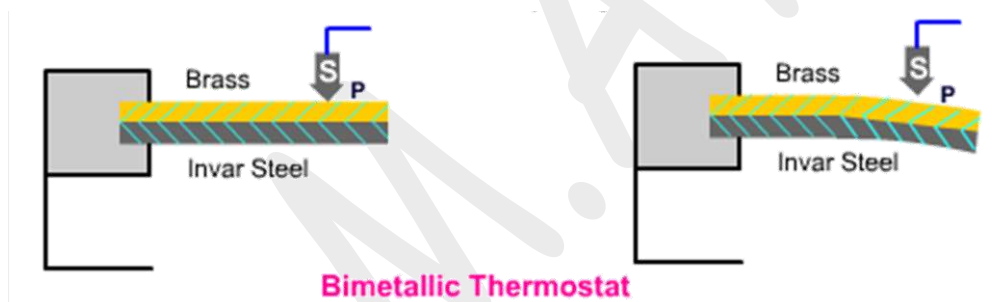


Bimetallic Thermostat:

Bimetallic thermostat is a device used to maintain the temperature. It works on the principle of bimetallic strip.

Working principle:

As the temperature of the strip increases, the two metals expand by different amounts and the strip bends. The change in shape of a bimetallic strip can make or break up electrical connection.

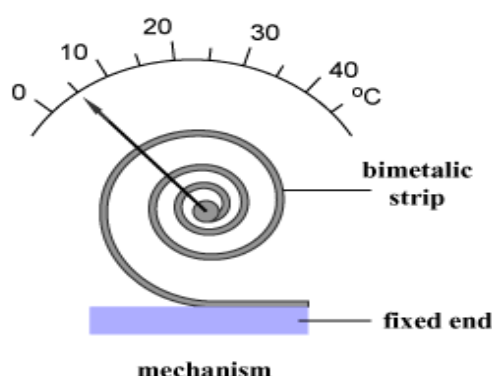


Bimetallic Thermometer:

Bimetallic thermometer is used to measure high range of temperature. It works on the principle of bimetallic strip.

Working Principle:

Thin bimetallic strip is converted into spiral whose one end is fixed and other end is movable which is attached to a pointer which moves over a scale as shown in figure. As the temperature rises the spiral tends to bend and pointer moves over a calibrated scale. The movement of pointer over a calibrated scale measures the temperature.



Kinetic Molecular Theory of Gases:

In 19th century Scientist give assumption on which they define an ideal gas. Some major points of kinetic molecular theory of gases are given below:

1. A finite volume of a gas contains very large number of molecules.
2. The distance between molecules is very large as comparable to the dimensions of molecules.
3. There is no molecular force of attraction between two molecules of a gas.
4. Molecules of a gas are in random motion.
5. The collision between two molecules or with the surface container of a gas is perfectly elastic.

Gas Laws:

Physical laws that relate pressure, volume, mass and temperature of a gas are called gas laws.

Boyles law:

Statement:

According to Boyles Law

"For a given amount of gas, volume of a gas is inversely proportional to its pressure keeping the temperature of gas constant."

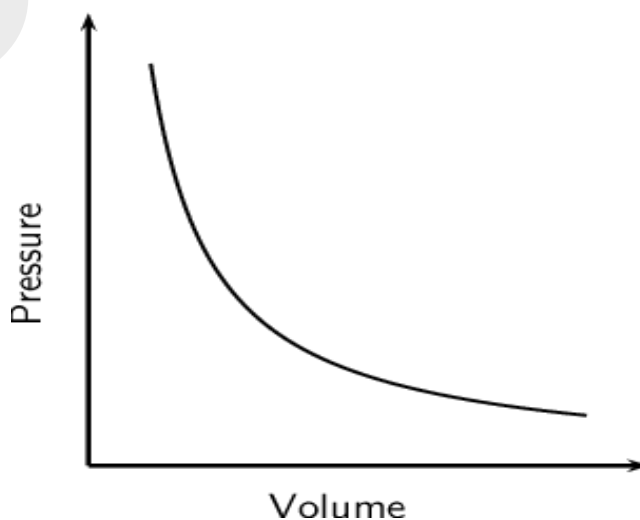
Mathematically:

$$V \propto \frac{1}{P}$$

$$V = \frac{\text{constant}}{P}$$

$$PV = \text{constant}$$

Graphical Representation:



Charles Law:**Statement:**

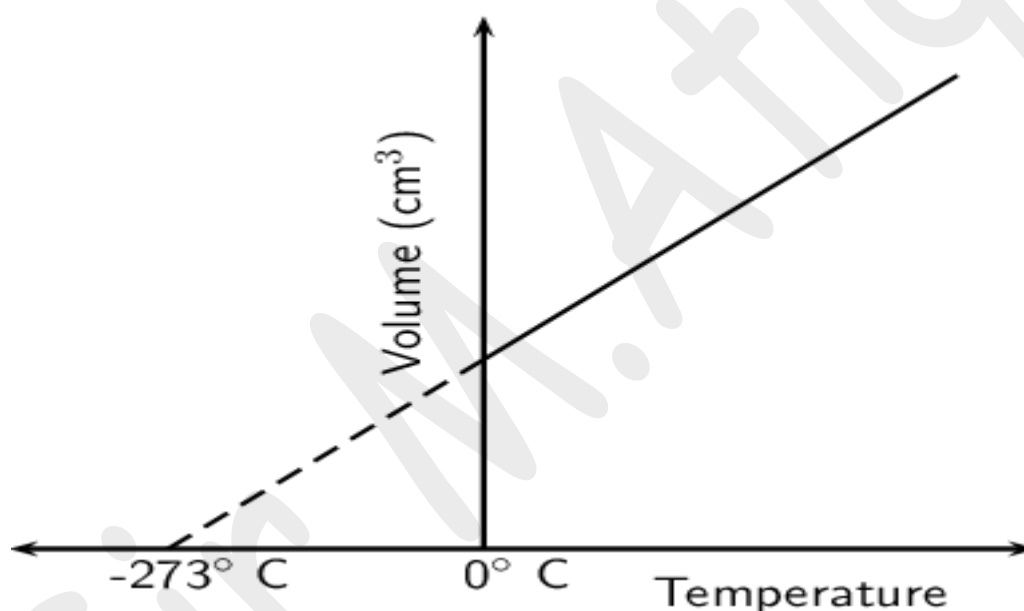
"For a given amount of gas, volume of a gas is directly proportional to the absolute temperature keeping the pressure of gas constant."

Mathematically

$$V \propto T$$

$$V = (\text{Constant})T$$

$$\frac{V}{T} = \text{Constant}$$

Graphical Representation:**Absolute zero temperature:**

Absolute zero can be defined as:

"It is the temperature, at which Volume of an ideal gas is theoretically Zero."

Hence Charles law is not applicable at Zero Kelvin because:

- At Zero Kelvin nothing can exist in gaseous state and Charles law is only applicable on gases.
- According to law of conservation of energy volume of any material cannot be zero at any temperature.

General Gas Law:

According to Boyle's Law:

"Volume of a given mass of a gas is inversely proportional to its pressure keeping the temperature of gas constant."

$$P \propto \frac{1}{V}$$

Or

$$V \propto \frac{1}{P} \text{ --- (1)}$$

According to Charles Law:

"Volume of given mass of a gas is directly proportional to the absolute temperature keeping the pressure of gas constant."

$$V \propto T \text{ --- (2)}$$

According to Avogadro Law

"Volume of given mass of a gas is directly proportional to the number of moles keeping the pressure and temperature of gas constant."

$$V \propto n \text{ --- (3)}$$

Combining (1), (2) and (3), we have

$$V \propto \frac{nT}{P}$$

$$V = \frac{nRT}{P}$$

$$\boxed{PV = nRT}$$

Where

n = number of moles and

R = Universal Gas Constant

$R = 8.314 \text{ J/mol}^{-1} \text{ K}^{-1}$.

Pressure of an Ideal Gas (Pressure Equation):

Consider a cubical container of dimensions equal to "L" containing N number of molecules of an ideal gas. Each is having mass "m". Let one of the N molecules is moving in x- direction with velocity v_x and after striking the wall of the container it bounce back in opposite direction.

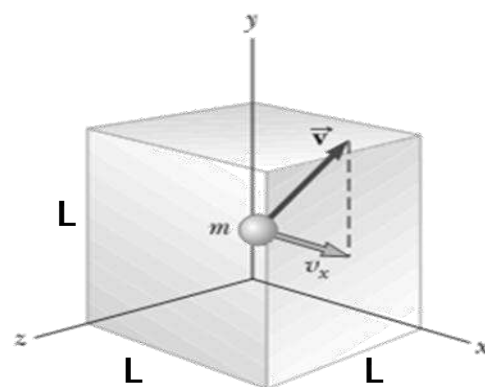
Change in the momentum of the molecule can be written as

Initial momentum: $p_i = -mv_x$

Final momentum: $p_f = mv_x$

Change in momentum = Final momentum - Initial momentum

$$\Delta p = p_f - p_i$$



$$\Delta p = mv_x - (-mv_x)$$

$$\Delta p = mv_x + mv_x$$

$$\Delta p = 2mv_x$$

Time taken during the complete trip can be find out as

$$S = vt$$

Here

$$S = 2L$$

$$v = v_x$$

$$t = \Delta t$$

$$2L = v_x \Delta t$$

$$\Delta t = \frac{2L}{v_x}$$

We know that the rate of change of linear momentum is equal to force, therefore

$$F = \frac{\Delta p}{\Delta t}$$

$$F_x = \frac{2mv_x}{2L/v_x}$$

$$F_x = m \frac{v_x^2}{L}$$

Force due to N molecules can be written as

$$F_x = \frac{m}{L} (v_{1x}^2 + v_{2x}^2 + v_{3x}^2 + \dots + v_{Nx}^2)$$

Since pressure is defined as force per unit area

$$P = \frac{F}{A}$$

$$P_x = \frac{m}{L} (v_{1x}^2 + v_{2x}^2 + v_{3x}^2 + \dots + v_{Nx}^2) \times \frac{1}{L^2}$$

$$P_x = N \frac{m}{L^3} (v_{1x}^2 + v_{2x}^2 + v_{3x}^2 + \dots + v_{Nx}^2)$$

Multiply by N/N

$$P_x = N \frac{m}{L^3} \frac{(v_{1x}^2 + v_{2x}^2 + v_{3x}^2 + \dots + v_{Nx}^2)}{N}$$

Density of the gas " ρ " can be written as:

$$\rho = \frac{m}{V}$$

For N number of molecules

$$\rho = \frac{Nm}{V}$$

Since

$$V = L^3$$

$$\rho = \frac{Nm}{L^3}$$

Similarly square mean of the velocities can be written as

$$\overline{v_x^2} = \frac{(v_{1x}^2 + v_{2x}^2 + v_{3x}^2 + \dots + v_{Nx}^2)}{N}$$

Now pressure can be written as

$$P_x = \rho \overline{v_x^2}$$

Using Pascal principle

$$P_x = P_y = P_z = P$$

$$P = \rho \overline{v_x^2} \text{----- (1)}$$

For molecule having all three components of velocity

$$\overline{v^2} = \overline{v_x^2} + \overline{v_y^2} + \overline{v_z^2}$$

$$\text{For } \overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2}$$

$$\overline{v^2} = \overline{v_x^2} + \overline{v_x^2} + \overline{v_x^2}$$

$$\overline{v^2} = 3\overline{v_x^2}$$

$$\frac{\overline{v^2}}{3} = \overline{v_x^2}$$

Putting in equation (1) we get

$$\boxed{P = \frac{1}{3} \rho \overline{v^2}}$$

Above equation is called Pressure equation of an ideal gas.

Kinetic Energy of Molecules of an Ideal Gas:

Consider the pressure equation for ideal gas

$$P = \frac{1}{3} \rho \overline{v^2}$$

Since

$$\rho = \frac{Nm}{V}$$

$$P = \frac{1}{3} \frac{Nm}{V} \overline{v^2}$$

$$PV = \frac{1}{3} N m \overline{v^2}$$

Using General Gas law $PV = nRT$

$$nRT = \frac{1}{3} N m \overline{v^2} \text{ --- (1)}$$

Number of moles can be expressed in terms of number of molecules (N) and Avogadro number (N_A) as

$$n = \frac{N}{N_A}$$

Equation (1) becomes

$$\begin{aligned} \frac{N}{N_A} RT &= \frac{1}{3} N m \overline{v^2} \\ \frac{R}{N_A} T &= \frac{1}{3} m \overline{v^2} \text{ --- (2)} \end{aligned}$$

Here

$$\frac{R}{N_A} = K_B$$

Where K_B is called Boltzmann Constant

Equation (2) becomes

$$\begin{aligned} K_B T &= \frac{1}{3} m \overline{v^2} \\ 3K_B T &= m \overline{v^2} \end{aligned}$$

Dividing both sides with 2 we get

$$\frac{3}{2} K_B T = \frac{1}{2} m \overline{v^2}$$

Since $K.E = \frac{1}{2} m v^2$

$$\boxed{K.E = \frac{3}{2} K_B T}$$

Above equation represent the kinetic energy of molecules of an ideal gas. From above equation it is clear that kinetic energy of molecule is dependent only on temperature of the gas.

$$\mathbf{K.E \propto T}$$

Derivation of Gas Laws from Pressure equation:

Boyles Law:

Consider the pressure equation for ideal gas

$$P = \frac{1}{3} \rho \overline{v^2}$$

Since

$$\rho = \frac{Nm}{V}, \text{ therefore}$$

$$P = \frac{1}{3} \frac{Nm}{V} \overline{v^2}$$

$$PV = \frac{1}{3} Nm \overline{v^2} \text{ --- (1)}$$

Now from relation for kinetic energy of molecule of an ideal gas

$$K.E = \frac{3}{2} K_B T$$

$$\frac{3}{2} K_B T = \frac{1}{2} m \overline{v^2}$$

$$\overline{v^2} = \frac{3 K_B T}{m}$$

Putting above in equation (1) we get

$$PV = \frac{1}{3} Nm \frac{3 K_B T}{m}$$

$$PV = (N K_B T)$$

$$PV = \text{constant}$$

$$P = \frac{\text{constant}}{V}$$

$$V \propto \frac{1}{P}$$

Hence volume of an ideal gas is found to be inversely proportional to its pressure keeping other factors constant which is according to Boyles Law.

Charles Law:

Consider the pressure equation for ideal gas

$$P = \frac{1}{3} \rho \overline{v^2}$$

Since

$$\rho = \frac{Nm}{V}$$

$$P = \frac{1}{3} \frac{Nm}{V} \overline{v^2}$$

$$V = \frac{1}{3P} Nm \overline{v^2} \text{ --- (1)}$$

Now

$$K.E = \frac{3}{2} K_B T$$

$$\frac{3}{2} K_B T = \frac{1}{2} m \overline{v^2}$$

Separating velocity

$$\overline{v^2} = \frac{3K_B T}{m}$$

Putting above in equation (1)

$$V = \frac{1}{3P} Nm \frac{3K_B T}{m}$$

$$V = \left(\frac{NK_B}{P} \right) T$$

Hence

$$V = (\text{constant}) T$$

Or

$$V \propto T$$

Hence volume of an ideal gas is found to be directly proportional to the absolute temperature which is according to Charles Law.

Root Mean Square Velocity of Molecule of an Ideal Gas (v_{rms}):

Consider the relation for kinetic energy of an ideal gas

$$K.E = \frac{3}{2} K_B T$$

$$\frac{3}{2} K_B T = \frac{1}{2} m \overline{v^2}$$

$$\overline{v^2} = \frac{3K_B T}{m}$$

Taking square root on both sides, we get

$$v_{rms} = \sqrt{\frac{3K_B T}{m}}$$

Heat Capacity:

It is the amount of heat required to raise the temperature of a substance through 1 degree Celsius or 1 degree Kelvin. It is denoted by C .

Mathematically:

$$\Delta Q \propto \Delta T$$

$$\Delta Q = (\text{Constant}) \Delta T$$

Here Constant = Heat Capacity = C

$$\Delta Q = C \Delta T$$

$$C = \frac{\Delta Q}{\Delta T}$$

Specific Heat Capacity:

It is the amount of heat required to raise the temperature of 1 Kg of a substance by 1 degree Celsius or 1 degree Kelvin. It is denoted by c .

Mathematically:

$$\Delta Q \propto \Delta T$$

$$\Delta Q \propto m$$

Combining both, we get

$$\Delta Q \propto m\Delta T$$

$$\Delta Q = (\text{Constant})m\Delta T$$

Here Constant = Specific Heat Capacity = c

$$\Delta Q = c m\Delta T$$

$$c = \frac{\Delta Q}{m\Delta T}$$

The unit of specific heat capacity is Joules per Kilogram per Degree Kelvin $J.Kg^{-1}K^{-1}$

Molar Specific Heat Capacity:

It is the amount of heat required to raise the temperature of 1 mole of a substance by 1 degree Celsius or 1 degree Kelvin.

Mathematically:

Consider relation for specific Heat of a substance

$$c = \frac{\Delta Q}{m\Delta T} \text{ --- (1)}$$

Number of moles (n) can be defined as the ratio of mass of the substance (m) and its molecular mass (M);

$$n = \frac{m}{M}$$

$$m = nM$$

Putting in equation (1) we get

$$c = \frac{\Delta Q}{nM\Delta T}$$

$$(M)(c) = \frac{\Delta Q}{n\Delta T}$$

$$\boxed{M_c = \frac{\Delta Q}{n\Delta T}}$$

Here M_c is called molar specific heat of the substance and its unit is $J.mol^{-1}K^{-1}$.

Molar Specific Heat Capacity at Constant Volume:

It is the amount of heat required to raise the temperature of a gas by 1 degree Celsius or 1 degree Kelvin in such a way that gas is not allowed to expand (i.e. at constant volume). It is represented by C_v .

$$\Delta Q_v = nc_v\Delta T$$

Molar Specific Heat Capacity at Constant Pressure:

It is the amount of heat required to raise the temperature of a gas by 1 degree Celsius or 1 degree Kelvin at constant pressure. It is represented by C_p .

$$\Delta Q_p = nc_p\Delta T$$

Thermodynamics:

Thermodynamics is the branch of physics which deals with the transformation of heat energy into other forms of energy.

Work Done in Thermodynamics:

Consider a thermodynamic system such as a gas in a cylinder with insulated walls and fitted with a movable frictionless piston. When the gas is heated the gas in container is allowed to expand and results in increase in height of the piston (h) (due to increase in volume) to Work done by the system can be written as

$$\begin{aligned} \text{Mechanical Work done} &= \Delta W \\ &= F \cdot d \quad \text{---(1)} \end{aligned}$$

Since

$$P = \frac{F}{A}$$

Or

$$F = PA, \text{ put in (1)}$$

Equation (1) becomes

$$\Delta W = PAh$$

Since $Ah = \Delta V$ (change in volume), therefore

$$\text{Mechanical Work done} = \Delta W = P\Delta V$$

Above equation, represent work done in thermodynamics. It shows that if volume is increased, the work will be positive and if volume is decreased, the work will be negative.

First Law of Thermodynamics:

Statement:

"In thermodynamic process, heat energy added to a system is equal to the increase in the internal energy stored in the system plus the work done by the system."

Mathematically:

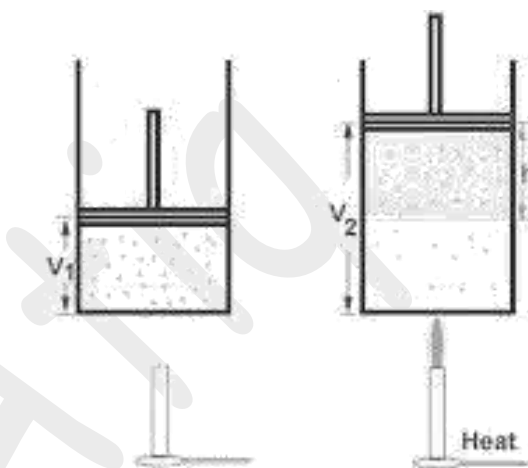
Let

ΔQ = amount of heat energy absorbed by the system

ΔW = work done by the system

ΔU = change in internal energy of the system

Then according to 1st law of thermodynamics:



$$\Delta Q = \Delta U + \Delta W$$

Sign convention:

- ΔQ is positive when heat is added to a system.
- ΔQ is negative when heat is rejected from the system.
- ΔW is positive when work is done by the system.
- ΔW is negative when work is done on the system.
- ΔU is positive when internal energy is increased.
- ΔU is negative when internal energy is decreased.

Applications of First Law of Thermodynamics:

There are four applications of thermodynamics

1. Isobaric process (constant pressure)
2. Isochoric process (constant volume)
3. Isothermal process (constant temperature)
4. Adiabatic process (heat, $\Delta Q = 0$)

Isobaric Process:

A thermodynamic process that takes place at constant pressure is called isobaric process.

Explanation:

Consider a gas cylinder with insulated walls and frictionless movable piston. If heat, ΔQ is added to the system then according to the first law of thermodynamics

$$\Delta Q = \Delta W + \Delta U \quad \text{--- (1)}$$

When the gas is heated the gas in container is allowed to expand and results in increase in height of the piston (h) (due to increase in volume). The Work done by the system can be written as

$$\Delta W = F \cdot d \quad \text{--- (2)}$$

Since
$$P = \frac{F}{A}$$

Or

$$F = PA$$

Equation (2) becomes

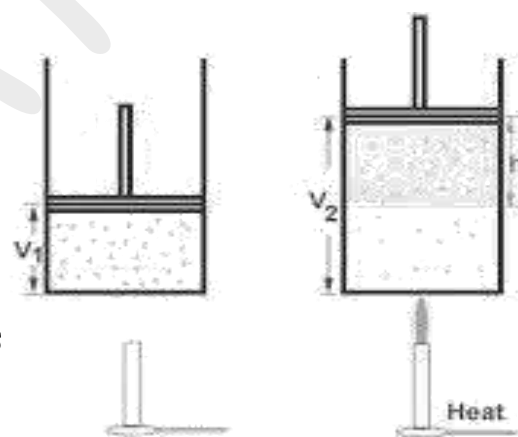
$$\Delta W = PAh$$

Since $Ah = \Delta V$ (change in volume), therefore

$$\Delta W = P\Delta V$$

Putting value of ΔW in equation (1), we get

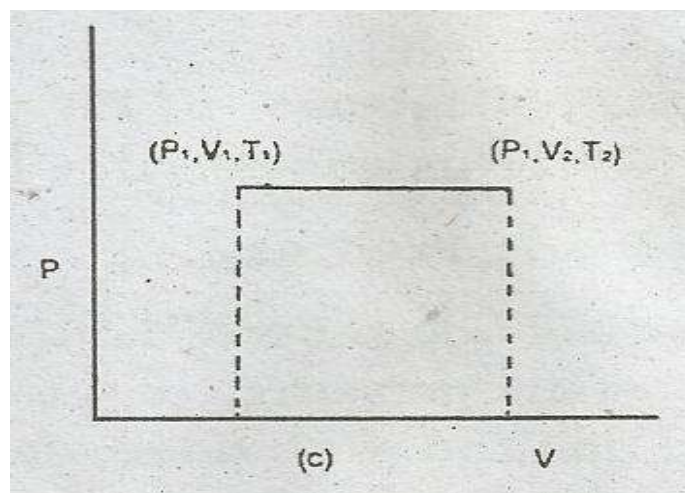
$$\Delta Q = \Delta U + P\Delta V$$



Above equation represent Isobaric process.

Conclusion:

1. Heat supplied to the system is used partially to change the internal energy and partially to do work.
2. Mechanical Work, $\Delta W = P\Delta V$
3. Graph between pressure and volume in isobaric process is given by



Isochoric Process:

A thermodynamic process that takes place at constant volume is called isochoric process.

Explanation:

Consider a gas cylinder with insulated walls and fixed piston. If heat, ΔQ is added to the system then according to the first law of thermodynamics

$$\Delta Q = \Delta W + \Delta U \quad \text{--- (1)}$$

Hence when the gas is heated it is not allowed to expand and no work is done. As piston is fixed and $\Delta V = 0$, therefore

$$\Delta W = P\Delta V = P(0) = 0$$

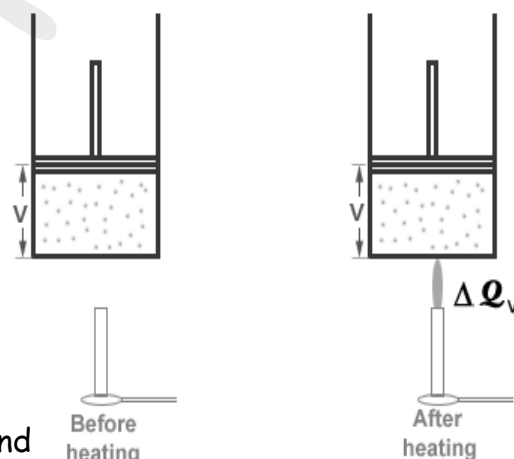
Now equation (1) will become

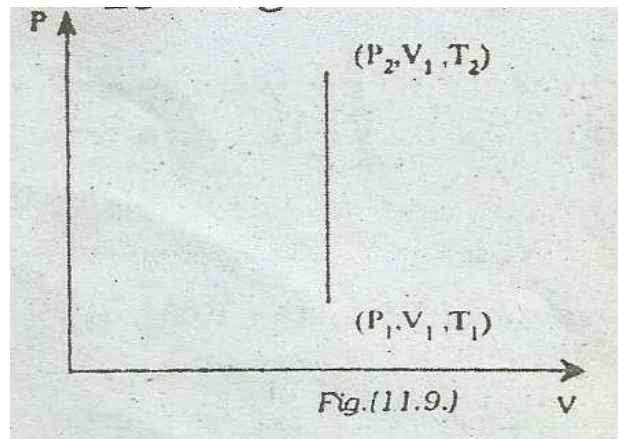
$$\boxed{\Delta Q = \Delta U}$$

Above equation represent Isochoric process.

Conclusion:

1. In isochoric process heat supplied to the system is used completely to change the internal energy.
2. Work done is zero i.e. $\Delta W = 0$
3. Graph between pressure and volume in isochoric process is given by





Isothermal Process:

Such a thermodynamic process that takes at constant temperature is called isothermal process.

Explanation: (Isothermal expansion)

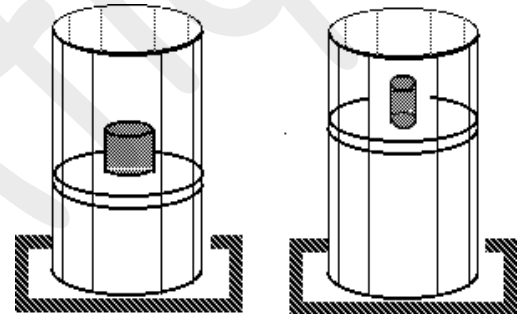
Consider a gas cylinder with insulated walls and frictionless movable piston. Let the gas is allowed to expand by decreasing the pressure. At the same time heat is added to the system by heat reservoir to maintain the constant temperature. According to 1st law of thermodynamics

$$\Delta Q = \Delta W + \Delta U \text{ --- (1)}$$

Since there is no change in temperature, therefore

$\Delta U = 0$, substitute in (1), we have

$$\Delta Q = \Delta W$$



Conclusion:

1. Work done is maximum in this process, $\Delta Q = \Delta W$
2. For an isothermal process all the heat energy absorbed is utilized in doing work with no change in internal energy.
3. Pressure and volume are inversely proportional at constant temperature which holds Boyle's law.
4. The graphical representation of isothermal process is given by



Adiabatic Process:

A thermodynamic process in which no heat flows into or out of the system is called adiabatic process.

Explanation: (Adiabatic expansion)

Consider a gas cylinder with insulated and non conducting walls. The walls and base of the container are perfectly insulated from its surrounding and no heat flow in or out of the system.

If the piston is released by decreasing the pressure then the gas will expand at the cost of internal energy of the system because no heat will flow in or out of the system.

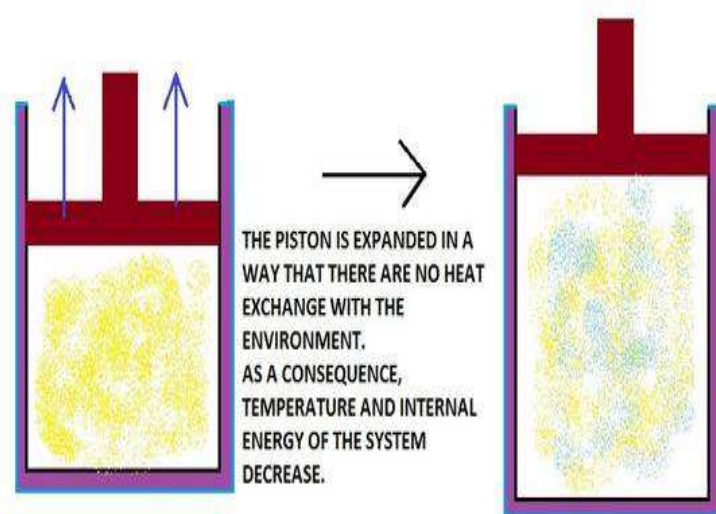
Now according to first law of thermodynamics

$$\Delta Q = \Delta U + \Delta W \text{ ----- (1)}$$

Since no heat flow in or out, therefore $\Delta Q = 0$, put in (1), we have

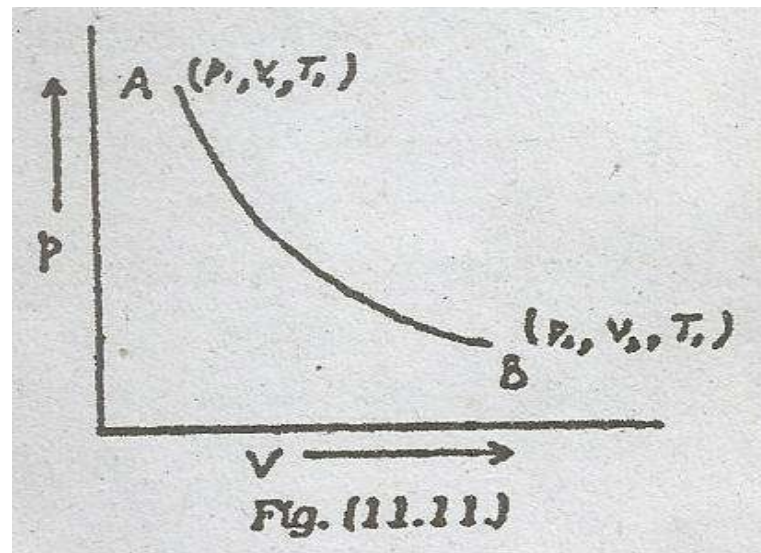
$$0 = \Delta U + \Delta W$$

$$\boxed{-\Delta U = \Delta W}$$



Conclusion:

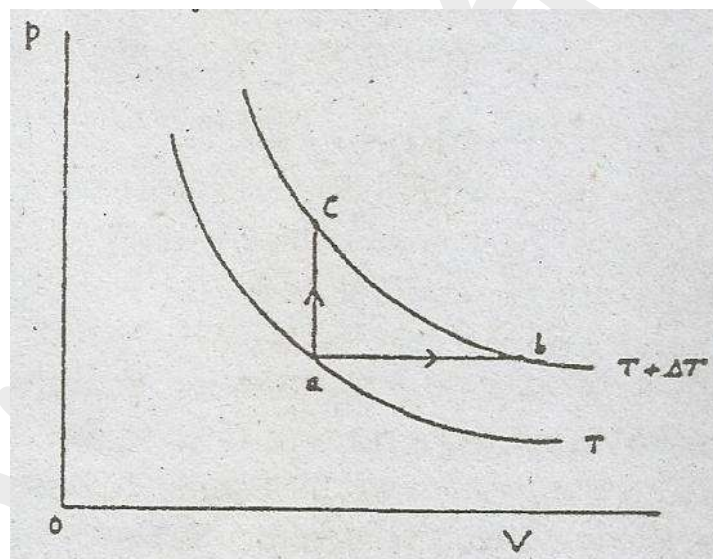
1. Mechanical work is done at the cost of internal energy, $\Delta W = -\Delta U$
2. $P V^\gamma = \text{constant}$, where $\gamma = \frac{C_P}{C_V}$
3. Graphical representation of adiabatic process is given by



Question: Prove that $C_p - C_v = R$

Relation between Specific Heat at Constant Pressure (C_p) and Specific Heat at Constant Volume (C_v):

Consider two isotherms of an ideal gas at temperature T and $T + \Delta T$.



From a → b:

Suppose that heat is provided to the gas at constant pressure along path a → b. The heat transformed to the gas in this process is given by

$$\Delta Q_p = nc_p \Delta T$$

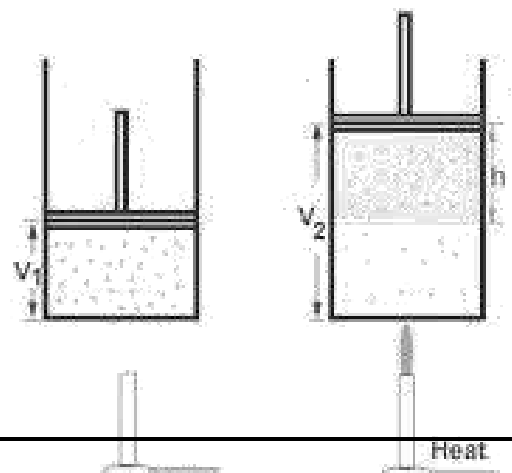
[since $\Delta Q = nc \Delta T$]

The mechanical work done in this process is given by

$$\Delta W = P \Delta V$$

According to 1st law of thermodynamics

$$\Delta Q_p = \Delta U + \Delta W \text{ --- (1)}$$



Substitute values of ΔQ_p and ΔW in above equation (1), we have

$$nC_p\Delta T = \Delta U + P\Delta V \text{----- (2)}$$

Using general gas law

$$PV = nRT$$

$$P\Delta V = nR\Delta T$$

Putting this value in Equation (2), we get

$$nC_p\Delta T = \Delta U + nR\Delta T \text{----- (3)}$$

From a $\rightarrow$ c:

Now suppose that heat is provided to the gas at constant volume along path a $\rightarrow$ c as shown in graph. The heat transferred in this process is given by

$$\Delta Q_v = nc_v\Delta T$$

We know that at constant volume, $\Delta W = 0$, and

According to 1st law of thermodynamics

$$\Delta Q_v = \Delta U + \Delta W \text{----- (4)}$$

Substitute $\Delta W = 0$ and $\Delta Q_v = nc_v\Delta T$ in above equation (4), we have

$$nc_v\Delta T = \Delta U$$

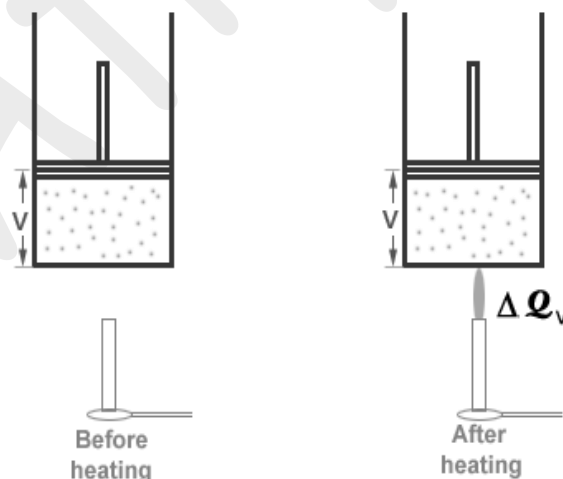
Substitute this value ΔU in equation (3), we get

$$nC_p\Delta T = nc_v\Delta T + nR\Delta T$$

$$nC_p\Delta T = n\Delta T(c_v + R)$$

$$c_p = c_v + R$$

$$\boxed{c_p - c_v = R}$$



Hence the difference of specific heat at constant pressure and specific heat at constant volume is equal to the general gas constant. Above equation also shows that the specific heat capacity of any ideal gas at constant pressure is always greater than specific heat capacity of the same gas at constant volume. I-e

$$\boxed{c_p > c_v}$$

Heat Engine:

Any device which converts heat energy into mechanical energy is called heat engine.

Essentials of Heat Engine:

Essential of heat engines are:

1. Furnace or hot body.
2. Working substance.

3. Condenser or Cold body.

Carnot Engine:

Introduction:

Carnot engine is an ideal heat engine which converts heat energy into mechanical energy. It is an imaginary heat engine free from friction & heat losses due to radiations or conduction.

Efficiency of Carnot engine:

Suppose that heat Q_1 is extracted from source (hot reservoir) of temperature T_1 by Carnot engine and heat Q_2 is rejected to sink (cold reservoir) of temperature T_2 .

Then useful mechanical work in one cycle, $W = Q_1 - Q_2$

Input heat in each cycle = Q_1

Heat rejected in each cycle = Q_2

The ratio of output mechanical work to the input energy is called efficiency of Carnot engine.

$$\text{Efficiency} = \frac{\text{Output work}}{\text{Input heat}}$$

Therefore

$$E = \frac{W}{Q_1}$$

From figure, $W = Q_1 - Q_2$

Therefore

$$E = \frac{Q_1 - Q_2}{Q_1}$$

$$E = 1 - \frac{Q_2}{Q_1}$$

In percentage form

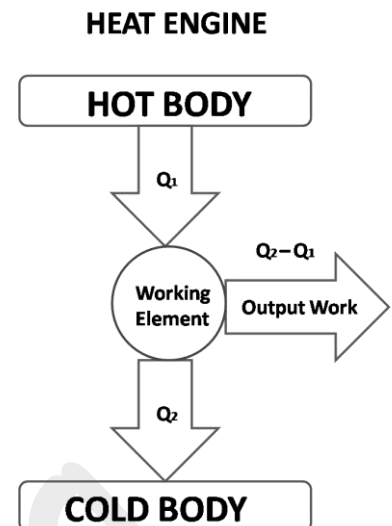
$$E\% = \left(1 - \frac{Q_2}{Q_1}\right) \times 100$$

Since $Q \propto T$

Therefore

$$E\% = \left(1 - \frac{T_2}{T_1}\right) \times 100$$

Refrigerator:



The device that makes heat to flow from cold body to hot body is called Refrigerator.

Essential of Refrigerator:

- Condenser or cold body.
- Working substance.
- Furnace or hot body.

Second Law of Thermodynamics:

Two statements can state the second law of thermodynamics.

Kelvin Statement:

"It is impossible for a heat engine to take heat from a source and converts it completely into work."

Clausius Statement:

"It is impossible to cause heat flow from a cold body to a hot body without the expenditure of work."

Equivalency of Kelvin and Clausius Statements:

Suppose that Kelvin's statement is false then we could have a heat engine which takes heat from a source and converts it completely into work. If we connect this perfect heat engine to an ordinary refrigerator we can take heat from the hot body and convert it completely to work. This work can be used to operate the refrigerator which conveys heat from the cold body to hot body. The net result is a transfer of heat from a cold body to a hot body without the expenditure of work which is contrary to Clausius statement. Therefore the two statements are equivalent.

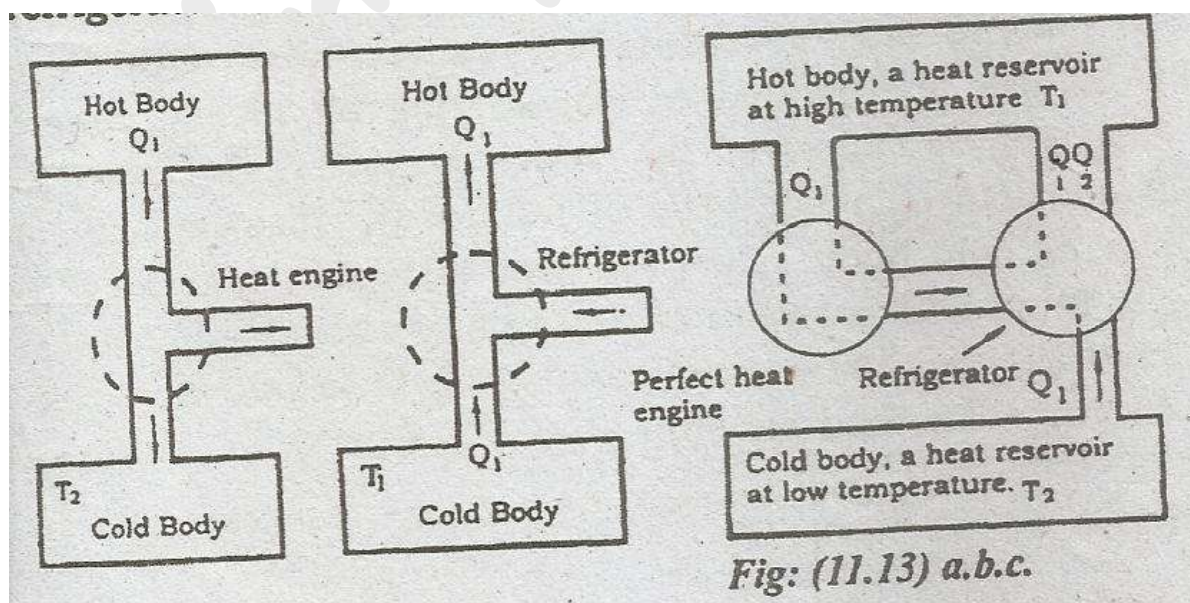
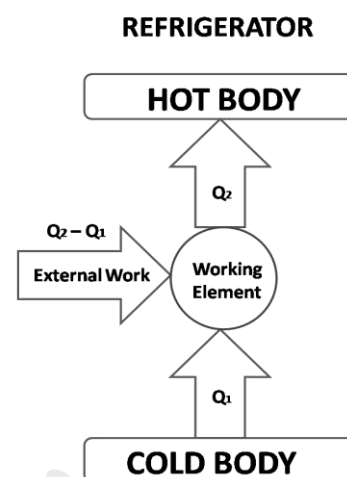


Fig: (11.13) a,b,c.

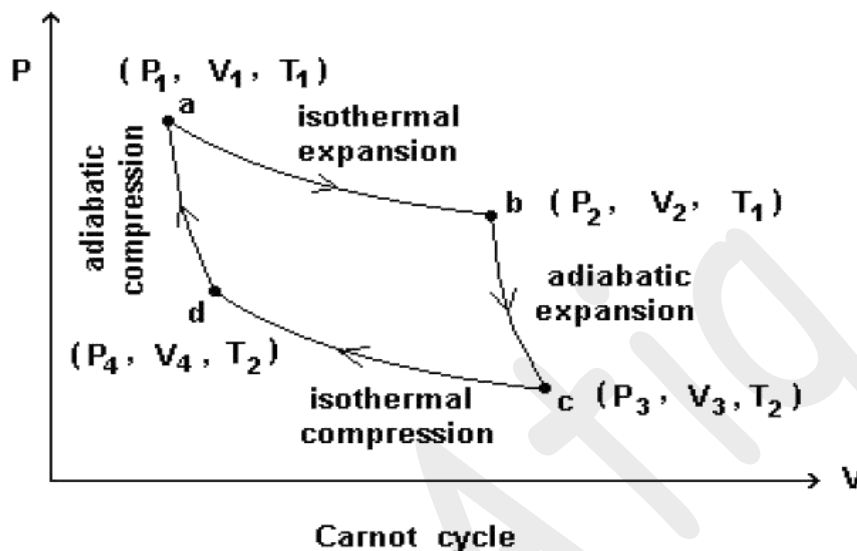
Carnot Cycle:

Sadi Carnot in 1840 described an ideal engine using only isothermal and adiabatic processes. He showed that a heat engine operating in an ideal reversible cycle between

two heat reservoirs at different temperature would be the most efficient engine. Carnot cycle uses an ideal gas as working substance.

Working:

Let we have an ideal gas with initial pressure P_1 , volume V_1 and temperature T_1 . Working of Carnot engine consist of steps as follows



Step-1

(Isothermal

expansion, a → b)

The gas is allowed to expand isothermally at constant temperature T_1 , absorbing heat Q_1 from the hot reservoir. Therefore initial state (P_1, V_1, T_1) of a gas changes to final state (P_2, V_2, T_1) .

Step-2 (Adiabatic expansion, b → c)

At this stage (P_2, V_2, T_1) the gas is then allowed to expand adiabatically until its temperature drops to T_2 . Therefore the final state will be (P_3, V_3, T_2) .

Step - 3 (Isothermal compression, c → d)

The gas at this stage is compressed isothermally at constant temperature T_2 rejecting heat Q_2 to the cold reservoir. The temperature is constant and the final state in this stage becomes (P_4, V_4, T_2) .

Step - 4 (Adiabatic compression, d → a)

Finally the gas is compressed adiabatically to restore its initial state (P_1, V_1, T_1) at temperature T_1 . Since there is no change in temperature in complete cycle and working substance returns to its initial state so the change in internal energy in complete cycle is zero (i.e. $\Delta U = 0$). Thermal and mechanical equilibrium is maintained all the time so that each process is perfectly reversible.

Entropy:

The measure of molecular disorder of a system during a process is called Entropy.

In any process entropy increases or remains constant.

Mathematically:

If ΔQ be the heat transferred to a system at constant kelvin temperature T , then

$$\Delta S = \frac{\Delta Q}{T}$$

The S.I unit of entropy is J/K.

Second law of thermodynamics in terms of entropy:

"When an isolated system undergoes a change, the entropy of the system either remains constant or it increases."

Chapter#12 "Electrostatics"

Electrostatics:

Electrostatics is the branch of physics which deals with the study of charges in rest condition.

Coulomb's Law:

Statement:

"The electrostatic force of repulsion or attraction between two charges is directly proportional to the product of magnitude of charges and inversely proportional to the square of separation of charges."

Mathematical Representation:

Consider two charges q_1 and q_2 placed at a distance r from each other. The magnitude of electric force between two charges F can be written as

$$F \propto q_1 q_2$$

$$F \propto \frac{1}{r^2}$$

Combining above expressions, we get

$$F \propto \frac{q_1 q_2}{r^2}$$

$$F = k \frac{q_1 q_2}{r^2}$$

Where, k = constant of proportionality and it depends upon the medium between charges and its value for free space is defined as

$$k = \frac{1}{4\pi\epsilon_0}$$

Here, ϵ_0 is the permittivity of free space. Numerical value of k for free space is

$$k = 8.98755 \times 10^9 \text{ Nm}^2\text{C}^{-2} \cong 9 \times 10^9 \text{ Nm}^2\text{C}^{-2}$$

Electric force can now be written as

$$F = \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r^2}$$

For medium other than free space value of k can be defined as

$$k = \frac{1}{4\pi\epsilon}$$

Here, ϵ is the permittivity of medium and it is defined as

$$\epsilon = \epsilon_0 \epsilon_r$$

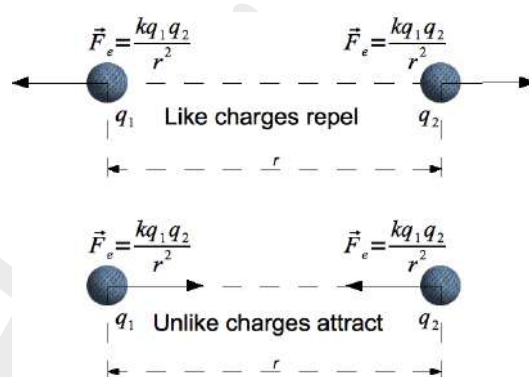
Here, ϵ_r is called relative permittivity. Electric force between two charges can now be written as

$$F = \frac{1}{4\pi\epsilon} \times \frac{q_1 q_2}{r^2}$$

$$F = \frac{1}{4\pi\epsilon_0 \epsilon_r} \times \frac{q_1 q_2}{r^2}$$

In vector form force exerted by charge q_1 on q_2 can be written as

$$\vec{F} = \frac{1}{4\pi\epsilon_0 \epsilon_r} \times \frac{q_1 q_2}{r^2} \hat{r}$$



Where: $\hat{r}$ = unit vector in the direction of $\vec{F}$.

Electric Field:

"The region around the charge in which another charge experiences electrostatic force is called electric field."

Test Charge:

A charge with very small magnitude (approaching to zero) so that the effect of its electric field is negligible is called test charge.

Electric Field Intensity:

The electric field intensity is defined as the electric force experienced by test charge divided by the magnitude of test charge. It is a vector quantity and its direction is same as the direction of electric force.

Mathematical Representation:

The electric field $\vec{E}$ intensity due a charge q at a point in space in presence of a test charge q_0 can be written as

$$\vec{E} = \frac{\vec{F}}{q_0}$$

Since

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \times \frac{qq_0}{r^2} \hat{r}$$

Therefore

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \times \frac{qq_0}{r^2} \frac{1}{q_0} \hat{r}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2} \hat{r}$$

$$\vec{E} = \frac{kq}{r^2} \hat{r}$$

Here $\hat{r}$ is the unit vector in the direction of electric field.

The magnitude of electric field intensity can be written as

$$E = \frac{kq}{r^2}$$

Unit:

The SI unit of electric field intensity is Newton per Coulomb N/C or volt per meter V/m .

Electric Flux:

Number of Electric lines of force passing through surface area is called electric flux.

Electric flux is a scalar quantity. The S.I unit is Nm/c .

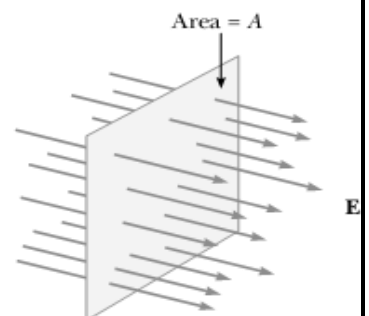
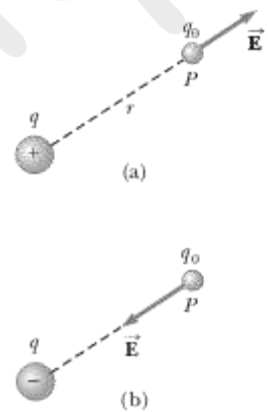
Mathematically:

"The dot product of Electric field intensity and Vector area is called Electric Flux."

$$\phi = \vec{E} \cdot \vec{A} \quad \text{Or}$$

$$\phi = EA \cos \theta$$

θ = angle between electric field intensity $\vec{E}$ and area vector $\vec{A}$.



Vector Area is defined as

"A vector Perpendicular to surface having magnitude equal to the Area of the surface."

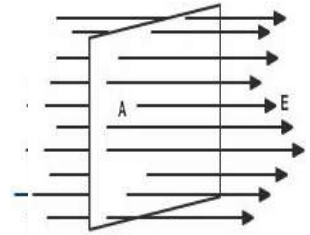
Maximum Flux:

From figure electric intensity and area vector are parallel i.e. $\theta = 0^\circ$

$$\phi = EA \cos \theta$$

$$\phi = EA \cos(0^\circ) \quad [\text{since } \cos(0^\circ) = 1]$$

$$\phi = EA$$



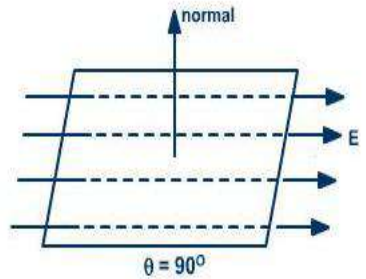
Minimum Flux:

From figure electric intensity and area vector are perpendicular i.e. $\theta = 90^\circ$

$$\phi = EA \cos \theta$$

$$\phi = EA \cos(90^\circ) \quad [\text{since } \cos(90^\circ) = 0]$$

$$\phi = 0$$



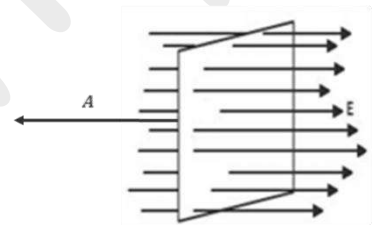
Negative Flux:

From figure electric intensity and area vector are anti parallel i.e. $\theta = 180^\circ$

$$\phi = EA \cos \theta$$

$$\phi = EA \cos(180^\circ) \quad [\text{since } \cos(180^\circ) = -1]$$

$$\phi = -EA$$



Electric flux due to a point charge in a closed sphere:

Consider an isolated positive point charge q placed at the center of sphere. The electric lines of forces from q will spread uniformly in space around it cutting the surface of an imaginary sphere. To determine the flux due to a point charge we divide the whole sphere into small sections such that area of each section is ΔA .

Now the total flux over the surface of sphere is the summation of individual flux of all sections.

$$\phi = \sum E \Delta A \cos \theta$$

As the angle between vector area and electric field is zero for each section, $\theta = 0^\circ$

$$\phi = \sum EA \cos(0)$$

$$\phi = \sum E \Delta A$$

Since E is uniform for all sections, therefore

$$\phi = E \sum \Delta A$$

Where $\sum \Delta A = \text{Area of sphere} = 4\pi r^2$

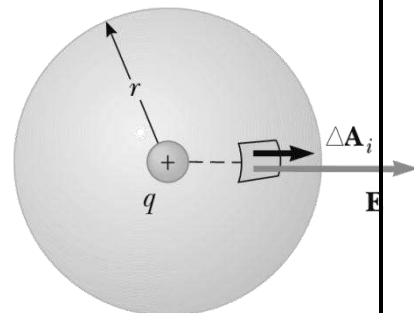
Therefore

$$\phi = E (4\pi r^2)$$

We know that the electric field intensity due to a point charge q

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

Substitute this value in above equation, we get



$$\phi = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2} (4\pi r^2)$$

$$\phi = \frac{q}{\epsilon_0}$$

The electric flux depends on the charge enclosed by the sphere.

Gauss's Law:

Statement:

"The total electric flux in an arbitrary closed surface called Gaussian surface is equal to the total charge enclosed by that surface divided the permittivity of free space.

Mathematically:

Mathematically Gauss's Law can also be defined as

$$\phi = \frac{q}{\epsilon_0}$$

If there are n positive spherical charge bodies of different enclosed in a Gaussian surface then flux due to each charge body can be written as

$$\phi_1 = \frac{q_1}{\epsilon_0}$$

$$\phi_2 = \frac{q_2}{\epsilon_0}$$

$$\phi_n = \frac{q_n}{\epsilon_0}$$

If total flux diverging out of the surface is ϕ then it may be written as

$$\phi = \sum \phi_n$$

$$\phi = \sum \frac{q_n}{\epsilon_0}$$

$$\phi = \frac{1}{\epsilon_0} \sum q_n$$

Hence, total flux diverging out is equals to the $\frac{1}{\epsilon_0}$ times the total charge enclosed q .

$$\phi = \frac{q}{\epsilon_0}$$

Application of Gauss's Law:

a) The Electric Field Due to a Thin Spherical charged Shell:

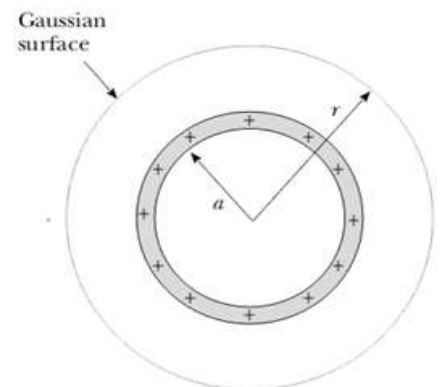
Consider a uniformly charged spherical shell having radius a as shown in figure below. Due to the symmetrical charge distribution the electric field has the same value on all the points of the surface. To calculate the electric field outside charged spherical shell considers a Gaussian's surface or radius r .

Now, Electric flux due to the charges can be written as

$$\phi = EA$$

Since according to Gauss's law

$$\phi = \frac{q}{\epsilon_0}$$



We have

$$\frac{q}{\epsilon_0} = EA \quad \text{---(1)}$$

Field intensity inside the Shell:

For Field intensity inside the Shell consider Gaussian surface consider a Gaussian surface inside the shell. Since charged inside the shell is zero there fore

Equation (1) will become

$$\frac{0}{\epsilon_0} = EA$$

$$\boxed{E = 0}$$

Field intensity on the surface of Shell:

For field intensity on the surface Equation (1) will become

$$\frac{q}{\epsilon_0} = E(4\pi a^2)$$

$$\boxed{E = \frac{q}{4\pi\epsilon_0 a^2}}$$

Field intensity outside the Shell:

For Field intensity outside the Shell consider Gaussian surface consider a Gaussian surface outside the shell having radius r such that $r > a$.

Equation (1) will become

$$\frac{q}{\epsilon_0} = E(4\pi r^2)$$

$$\boxed{E = \frac{q}{4\pi\epsilon_0 r^2}}$$

Hence Electric field intensity decreases with increase in distance from the center of shell.

b) Electric Field Due to a infinite charged Sheet:

Consider an infinite positive charged sheet as shown in figure. Consider a cylindrical Gaussian surface passing through the charged sheet as shown in figure.

Let ϕ_1 = flux at curved surface of cylinder

ϕ_2 = flux at cross-sections of cylinder

Since, there is no electric line of force through the surface area of Gaussian surface there is no electric flux through the surface area

i-e

$$\phi_1 = (0)A_1 \cos \theta$$

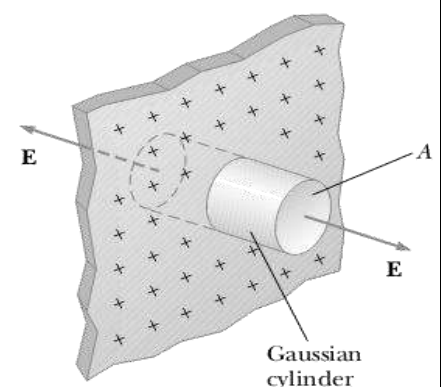
$$\boxed{\phi_1 = 0}$$

Electric flux through the two cross section of the Gaussian Surface

$$\phi_2 = EA \cos \theta + EA \cos \theta$$

Since the angle between electric line of forces and vector Area is 0°

$$\phi_2 = 2EA \cos(0^\circ)$$



$$\phi_2 = 2EA$$

Total flux ϕ can be written as

$$\phi = \phi_1 + \phi_2$$

$$\phi = 0 + 2EA$$

$$\phi = 2EA$$

According to Gauss law

$$\phi = \frac{q}{\epsilon_0}$$

$$2EA = \frac{q}{\epsilon_0}$$

$$E = \frac{q}{2A\epsilon_0} \text{ --- (1)}$$

Charge density σ can be defined as

$$\sigma \equiv \frac{q}{A}$$

Equation (1) will become

$$E = \frac{\sigma}{2\epsilon_0}$$

c) Electric Intensity Due To Two Oppositely Charged Sheets:

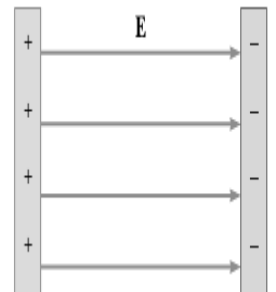
Consider two metal plates of same area placed parallel to each other such that the distance between them is much smaller than their area. If the plates are carrying the equal amount of charges the flux leaving the positive plate is same as the flux entering the negative plate.

Electric field intensity for both plates will be same as for infinitely charged sheet i.e. $\frac{\sigma}{2\epsilon_0}$

Net Electric field intensity at any point between the plates will be

$$E = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0}$$

$$E = \frac{\sigma}{\epsilon_0}$$



Electric Potential Energy:

The work done against the electric field is stored as energy known as electric potential energy.

Electric Potential Difference:

The work done in moving a test charge against the electric field divided by the magnitude of test charge is called electric potential difference.

Mathematical Explanation:

Consider test charge q_0 placed at a point in an electric field. If the Work done in carrying the that test charge to that point is W then electric potential V can be written as

$$V = \frac{W}{q_0}$$

Unit:

The S.I unit of Electric Potential is Volts (V).

Relation between Electric Field and potential Difference:

Consider a positive test q_0 charge moved from point a to point b against the electric field. Such that distance from a to b is ΔS . Suppose that the field has the uniform value over distance ΔS .

The work done ΔW against the electric field in moving that test charge

$$\Delta W = F \Delta S \cos \theta \text{ _____ (1)}$$

Since electric force experienced by test charge

$$F = q_0 E$$

Therefore equation (1) becomes

$$\Delta W = q_0 E \Delta S \cos \theta$$

$$\Delta W = q_0 E \Delta S \cos 180^\circ$$

$$\Delta W = q_0 E \Delta S (-1)$$

$$\Delta W = -q_0 E \Delta S$$

Since

$$\Delta V = \frac{\Delta W}{q_0}$$

$$\Delta V = \frac{-q_0 E \Delta S}{q_0}$$

$$\Delta V = -E \Delta S$$

$$E = -\frac{\Delta V}{\Delta S}$$

The quantity $\Delta V / \Delta S$ is called the potential gradient.

Electron-Volt:

Electron volt is a unit of energy and it can be defined as "The work done by electron charge against the electric field such that 1 volt of potential difference is obtained."

$$1eV = 1.6 \times 10^{-19} J$$

Capacitor:

System of two conductors separated by an insulating medium such as air or any other forms a capacitor.

Capacitor is an electrical device that can store electrical charge. The distance between the conductors should be very small as compared to their size.

Mathematical Explanation:

If V is the potential difference between the plates of capacitor and q is the charge stored on each plate then it is found that

$$q \propto V$$

$$q = CV$$

Here, C is called the capacitance of capacitor.

Capacitance:

The capacity of capacitor to store electric charges is called capacitance.

Unit:

The unit of capacitance is Farad(F)

Farad is very large unit so for practical purpose convenient units are

Micro farad $\mu F = 10^{-6} F$

Pico farad $pF = 10^{-12} F$

Parallel Plate capacitor:

When two oppositely charged parallel plates are separated by a small distance such that the distance between the plates is very small as compare to the facing area of plates then this kind of arrangement is called capacitor.

Derivation for Capacitance:

Consider to oppositely charge metal plates having charge q placed at the distance d from each other.

If the medium between the plates is air the electric field intensity between the plates can be written as

$$E = \frac{\sigma}{\epsilon_0}$$

Now since

$$V = E \Delta S = Ed$$

$$V = \frac{\sigma d}{\epsilon_0}$$

Now expression for capacitance C_o can be written as

$$C_o = \frac{q}{V} \text{ ————— (1)}$$

Since

$$\sigma = \frac{q}{A}$$

$$q = \sigma A$$

Putting values of q and V in equation (1), we get

$$C_o = \frac{A\epsilon_0}{d}$$

From above expression it is clear that the capacitance of a capacitor is directly proportional to the area of the plates and inversely proportional to the distance between the plates.

If the medium between the plates is other than air the capacitance of the capacitor will be

$$C = \frac{A\epsilon}{d} = \frac{A\epsilon_0\epsilon_r}{d}$$

$$C = C_o\epsilon_r$$

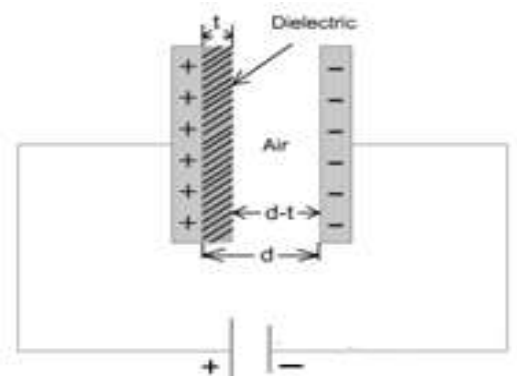
Hence the presence of a dielectric medium increases the capacitance.

Compound Capacitor:

In parallel plate capacitor if the space b/w the plates is partially filled with the dielectric medium then it is called compound capacitor.

Derivation:

Consider a compound capacitor partially filled with dielectric medium and partially filled with air.



Let

d = Distance between the plates of capacitor

t = Thickness of dielectric medium

$d - t$ = Thickness of air

ϵ_r = Relative permittivity of the medium

V = Potential difference across capacitor

Electric intensity when air is present

$$E_1 = \frac{\sigma}{\epsilon_0}$$

Electric intensity when dielectric is present

$$E_2 = \frac{\sigma}{\epsilon_0 \epsilon_r}$$

The potential difference between plates is

$$V = V_1 + V_2$$

Since $V = E\Delta S$ or $V = Ed$

$$V = E_1(d - t) + E_2 t$$

$$V = \frac{\sigma(d - t)}{\epsilon_0} + \frac{\sigma}{\epsilon_0 \epsilon_r} t$$

$$V = \frac{\sigma}{\epsilon_0} \left\{ (d - t) + \frac{t}{\epsilon_r} \right\}$$

$$V = \frac{\sigma}{\epsilon_0} \left\{ d + t \left(\frac{1}{\epsilon_r} - 1 \right) \right\}$$

$$V = \frac{\sigma}{\epsilon_0} \left\{ d + t \left(\frac{1}{\epsilon_r} - 1 \right) \right\}$$

Since charge density $\sigma = \frac{q}{A}$

$$V = \frac{q}{A\epsilon_0} \left\{ d + t \left(\frac{1}{\epsilon_r} - 1 \right) \right\}$$

Since $q = CV$

$$V = \frac{CV}{A\epsilon_0} \left\{ d + t \left(\frac{1}{\epsilon_r} - 1 \right) \right\}$$

$$1 = \frac{C}{A\epsilon_0} \left\{ d + t \left(\frac{1}{\epsilon_r} - 1 \right) \right\}$$

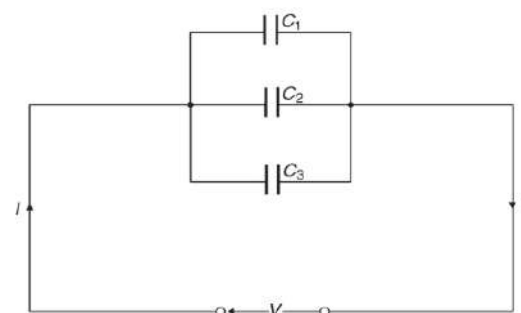
$$C = \frac{A\epsilon_0}{\left\{ d + t \left(\frac{1}{\epsilon_r} - 1 \right) \right\}}$$

Combination of Capacitors:

Capacitors connected in parallel:

When capacitors are connected in a way such that the plates of capacitors are connected to common terminals of battery then this type of arrangement is called parallel combination.

Derivation for equivalent capacitance:



Consider three capacitors C_1, C_2, C_3 connected in parallel. A charge q given to a point divide itself resides on the plates of individual capacitor as q_1, q_2, q_3 respectively.

Since

$$q = CV$$

Hence, the total charge stored in parallel combination is equal to the sum of charge stored across each capacitor in combination.

$$q = q_1 + q_2 + q_3 \text{ -----(1)}$$

Since $q = CV$

$$\text{Therefore, } q_1 = C_1V_1, q_2 = C_2V_2, q_3 = C_3V_3$$

substituting all values in equation (1), we get

$$C_eV = C_1V_1 + C_2V_2 + C_3V_3$$

Since $V = V_1 = V_2 = V_3$

$$C_eV = C_1V + C_2V + C_3V$$

$$C_eV = (C_1 + C_2 + C_3)V$$

$$C_e = C_1 + C_2 + C_3$$

Conclusion:

1. Equivalent capacitance is equal to the algebraic sum of the individual capacitances.
2. Equivalent capacitance is greater than the larger individual capacitance.
3. For parallel combination of n capacitors above expression will become.

$$C_e = C_1 + C_2 + C_3 + \dots + C_n$$

Capacitors connected in Series:

When capacitors are connected end to end then this type of arrangement is called series combinations of capacitors.

Derivation for equivalent capacitance:

Consider three capacitors C_1, C_2, C_3 are connected in series. If potential difference V is applied across combination then total voltage will divide in voltages across individual capacitors. Hence, Sum of individual voltage across each capacitor is equal to the Voltage of source.

i-e

$$V = V_1 + V_2 + V_3 \text{ -----(1)}$$

Since

$$\frac{q}{C} = V$$

Therefore

$$\frac{q_1}{C_1} = V_1, \frac{q_2}{C_2} = V_2, \frac{q_3}{C_3} = V_3$$

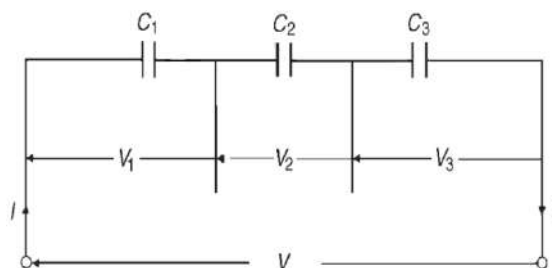
Substituting all the values in equation (1), we get

$$\frac{q}{C_e} = \frac{q_1}{C_1} + \frac{q_2}{C_2} + \frac{q_3}{C_3}$$

In series combination charge is same on each capacitor, i.e.

$$q = q_1 = q_2 = q_3$$

Above expression will become



$$\frac{q}{C_e} = \frac{q}{C_1} + \frac{q}{C_1} + \frac{q}{C_3}$$

$$\frac{q}{C_e} = q \left(\frac{1}{C_1} + \frac{1}{C_1} + \frac{1}{C_3} \right)$$

$$\frac{1}{C_e} = \frac{1}{C_1} + \frac{1}{C_1} + \frac{1}{C_3}$$

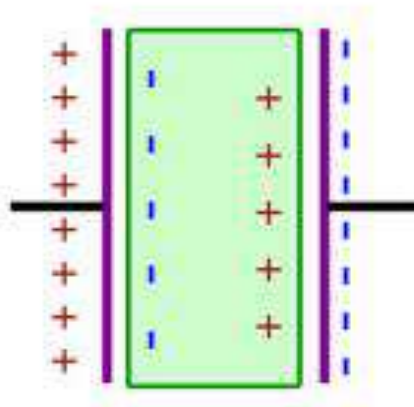
Conclusion:

1. The reciprocal of equivalent capacitance is equal to the algebraic sum of the reciprocals of individual capacitances in the network.
2. Equivalent capacitance is less than the smallest individual capacitance.
3. For series combination of n capacitors above expression will become.

$$\frac{1}{C_e} = \frac{1}{C_1} + \frac{1}{C_1} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$$

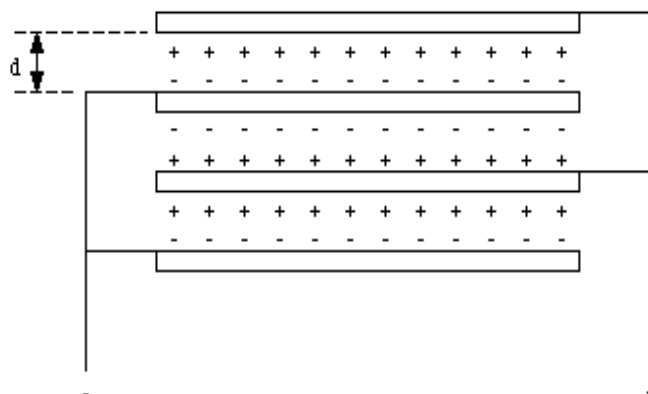
Dielectric effect in Capacitor:

In capacitor, the dielectric medium between the plates increases the capacitance. The potential difference b/w the plates distorts the molecules of the dielectric medium and hence polarized these molecules, one end becomes positive and another end becomes negative. The potential difference b/w the plates decreases and hence the capacitance of the capacitor increases.



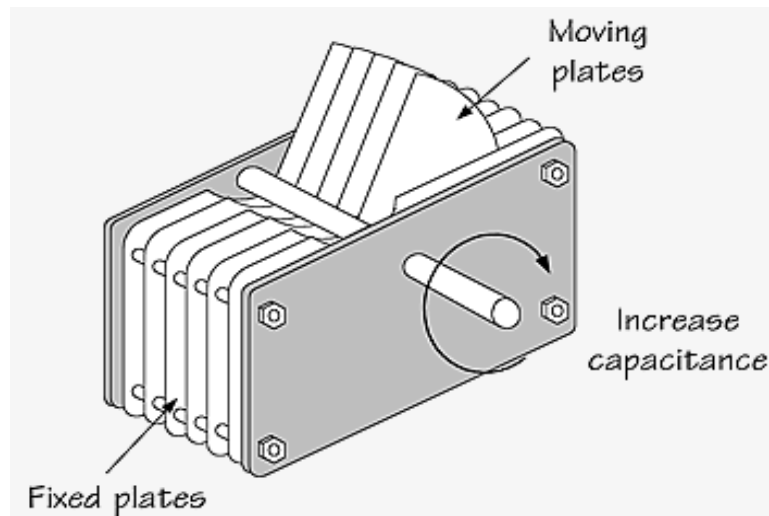
Multi plate Capacitor:

A multi plate capacitor is designed to have large capacitance. It consists of a large number of plates each of large area. If there are N plates then we have $N-1$ capacitors in parallel.



Variable Capacitor

Variable capacitor is a kind of capacitor used for tuning radio receiver. It consists of two sets of semi-circular plates separated by air. These plates are usually made by aluminum or brass. In order to change the capacitance, one set of plates is fixed and other is rotated by a knob to change the effective area of the plates. Thus by changing the area of the plates we can change the capacitance and hence tuning frequency of the radio receiver.



Electrolyte Capacitor:

All electrolytic capacitors are polarized capacitors whose anode (+) is made of a particular metal on which an insulating oxide layer forms by anodization, acting as the dielectric of the electrolytic capacitor. A non-solid or solid electrolyte which covers the surface of the oxide layer in principle serves as the second electrode (cathode) (-) of the capacitor.



Chapter#13 "Current Electricity"

Electric Current:

The amount of charge passing through the conducting material in unit time is called electric current.

Mathematically:

Let q is the amount of charge passing through the cross section of a conductor in time t . The electric current I can be expressed mathematically as

$$I = \frac{q}{t}$$

The S.I unit of electric current is Amperes and it is equal to Coulomb/second.

Electronic Current:

It is the current in the direction of motion of electron flowing in conductor under the influence of a potential source. Its direction is from negative to positive.

Conventional Current:

It is the current in the direction in which positive charges would have been flown in conductor under the influence of a potential source. Its direction is from positive to negative.

Ohm's Law:

Significance:

It is an empirical law discovered by Georg Simon Ohm in 1827. It is fundamental law in electrical and electronics.

Statement:

"The current passing through a conductor is directly proportional to the potential difference applied across the conductor keeping the other physical factors constant such as resistance and temperature."

Mathematically:

If I is the current passing through a conductor and V is the voltage dropped across it then mathematically Ohm's law can be expressed as

$$V \propto I$$

$$V = (\text{constant})I$$

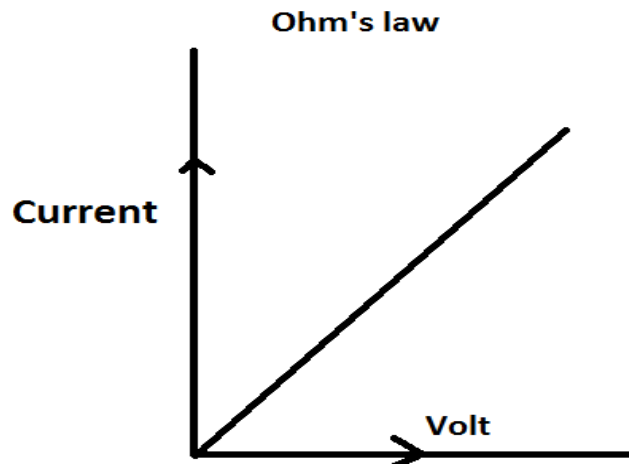
Here, Constant = R = Resistance of conductor

$$V = (R)I$$

$$V = IR$$

Graphical representation:

The graph between the voltage and current is shown in figure.



Resistance:

The opposition to the flow of charge is called resistance. The S.I unit of resistance is ohm (Ω)

Conductance:

The reciprocal of the resistance is called conductance. It is the facility to the flow of charge.

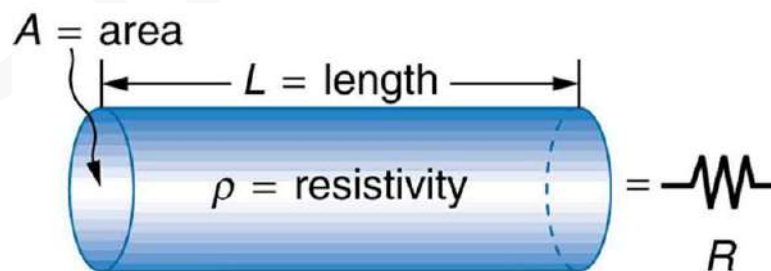
Resistivity:

Resistance per unit length per unit cross sectional area of material is called Resistivity or specific resistance.

$$\rho = \frac{RA}{L}$$

Derivation:

Resistance of the conductor depends upon the dimensions of the conductor such as length and cross-sectional area. To determine the expression for the resistivity of the conductor suppose that we have conductor of length 'L' and area of cross-section 'A'.



It is found that the resistance of conductor increases with increasing length and decreases when area of cross-section is increased i.e.

$$R \propto L$$

$$R \propto \frac{1}{A}$$

Combining above expression, we have

$$R \propto \frac{L}{A}$$

$$R = \frac{\rho L}{A}$$

Here ρ is called resistivity or Specific resistance and can be expressed as.

$$\rho = \frac{RA}{L}$$

The S.I unit of Specific resistance or resistivity is Ohm-meter ($\Omega.m$).

Q. Explain the effect of temperature on resistance.

Dependence of Resistance on temperature:

The resistance in flow of electric current is due to the collision of electrons with the ions or atoms of the metal (solid). The electrical resistance of metals increases with the increase in temperature. As the temperature of a body increases its atoms or molecules start vibrating with the higher amplitude and the probability of collision of electrons with the ions or atoms increases which ultimately affect the drift velocity of free electrons for a given applied voltage. This causes the increase in resistance of material and hence with rising the temperature, the decrease in drift velocity is attributed to the decrease in the electric current or increase in the resistance for a given applied voltage.

Q. Derive the expression for the effect of temperature on the resistance of a conductor.

Derivation:

Let R_o is the resistance of a material (metal) at temperature T_1 and if the temperature of metal is changed to T_2 the final resistance of the material will changed to R_t . Therefore,

Change in resistance, $\Delta R = R_t - R_o$

Change in temperature, $\Delta T = T_2 - T_1$

Experimentally it is found that the change in resistance (ΔR) is directly proportional to the original resistance R_o and temperature difference ΔT

$$\Delta R \propto R_o$$

$$\Delta R \propto \Delta T$$

$$\Delta R \propto R_o \Delta T$$

$$\Delta R = \alpha R_o \Delta T$$

$$\alpha = \frac{\Delta R}{R_o \Delta T}$$

Here α is called the temperature co-efficient of resistance and it is defined as

"Change in resistance per unit original resistance per unit change in temperature is called temperature co-efficient of resistance."

Positive Temperature Coefficient:

When resistance increases with the increase in temperature as in case of metals, then temperature coefficient is positive.

Negative Temperature Coefficient:

When resistance decreases with the increase in temperature as in case of semiconductor, then temperature coefficient is negative.

Final Resistance:

Since

$$\Delta R = \alpha R_o \Delta T$$

As we know that

$$\Delta R = R_t - R_o$$

We get

$$R_t - R_o = \alpha R_o \Delta T$$

$$R_T = R_o + \alpha R_o \Delta T$$

$$R_T = R_o(1 + \alpha \Delta T)$$

In terms of resistivity

As we know that resistance is directly proportional to resistivity therefore,

$$\rho_T = \rho_o(1 + \alpha \Delta T)$$

Combination of Resistance:

Series Combination of Resistors:

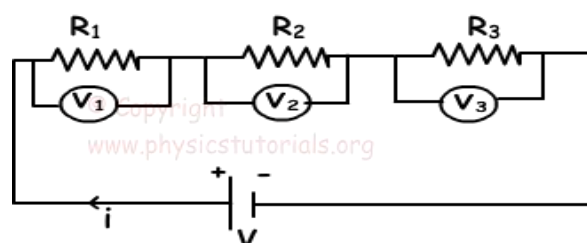
When resistances are connected end to end, then this kind of arrangement is called series combination of resistors.

Current and Voltage in series combination:

In series combination of resistors there is only one path for the flow of electric current. Electric current passing through each resistor is same. The sum of potential differences across the resistors is equal to the total potential difference applied across the combination.

Derivation for Equivalent Resistance

Consider three resistances R_1 , R_2 and R_3 connected to one another in series circuit as shown below.



Let the circuit is connected to a power supply of voltage 'V' and an electric current 'I' is passing through the circuit. Let

Potential difference across R_1 is V_1

Potential difference across R_2 is V_2

Potential difference across R_3 is V_3

The sum of these Potential differences is equal to 'V'.

$$V = V_1 + V_2 + V_3 \text{ ----- (1)}$$

According to Ohm's law

$$V = IR$$

$$V_1 = I_1 R_1 \quad V_2 = I_2 R_2 \quad V_3 = I_3 R_3$$

Substituting these values in equation (1), we have

$$IR_E = IR_1 + IR_2 + IR_3$$

$$IR_E = I(R_1 + R_2 + R_3)$$

$$R_E = R_1 + R_2 + R_3$$

Conclusion:

1. In series combination the equivalent resistance is the sum of the individual resistances.
2. Equivalent resistance is greater than largest resistance connected in series.

Parallel Combination of Resistors:

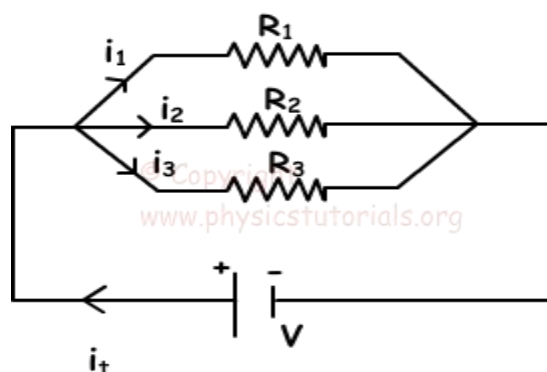
When resistors are connected to the same terminals of the battery, then this kind of arrangement is called parallel combination of resistances.

Current and voltage in parallel combination:

In parallel combination of resistors there is more than one path for the flow of electric current. Electric current passing through each resistor is different and it depends upon the value of resistance. Potential difference across each resistor is the same.

Derivation for Equivalent Resistance

Consider three resistances R_1 , R_2 and R_3 connected to the same terminals of battery as shown in figure below



Let the circuit is connected to a power supply of voltage 'V' and an electric current 'I' is passing through the circuit.

Electric current passing through R_1 is I_1

Electric current passing through R_2 is I_2

Electric current passing through R_3 is I_3

The sum of all three currents is equal to 'I'.

$$I = I_1 + I_2 + I_3 \text{----- (1)}$$

According to Ohm's law

$$I = \frac{V}{R}$$

$$I_1 = \frac{V_1}{R_1}, I_2 = \frac{V_2}{R_2}, I_3 = \frac{V_3}{R_3}$$

substituting all these values in equation (1), we have

$$\frac{V}{R_E} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$$

Since in parallel combination potential difference across each resistor is same i.e.

$$V = V_1 = V_2 = V_3$$

$$\frac{V}{R_E} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{1}{R_E} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Equivalent resistance of circuit is always smaller than any of the resistance connected parallel in the circuit.

Conclusion:

1. The reciprocal of the equivalent resistance is the sum of the reciprocals of the individual resistances.
2. The equivalent resistance is less than the smallest resistance in combination.

Power Dissipation in Resistors:

The power dissipated as heat when electric current is passing through the resistor is called power dissipation in resistor.

Explanation:

When an electric current passes through a conductor, some useful electrical energy is dissipated in the form of heat energy. This loss of electrical energy is due to the collision of charges with the atoms of conductor. Loss of electrical energy in unit time is referred to as "power dissipation in resistor".

Unit:

Its unit is Watt (W) which is equivalent to Joule per second (J/s).

Mathematically:

The work done in transferring charge 'q' to the resistor is given by

$$W = qV$$

Power is defined as Work done per unit time

$$P = \frac{W}{t}$$

$$P = \frac{qV}{t}$$

Since

$$I = \frac{q}{t}$$

$$\boxed{P = VI} \text{--- (1)}$$

Using Ohms law $V = IR$

$$P = (IR)I$$

$$\mathbf{P = I^2R} \text{--- (2)}$$

Again using Ohm's Law

$$I = \frac{V}{R}$$

$$P = \left(\frac{V}{R}\right)^2 R$$

$$P = \frac{V^2}{R^2} R$$

$$\boxed{P = \frac{V^2}{R}} \text{--- (3)}$$

Combining equation (1), (2) and (3)

$$\boxed{P = VI = I^2R = \frac{V^2}{R}}$$

Heat energy in resistor:

If a current "I" flows steadily through a resistor "R" for time "t" then, total heat energy supplied through the resistor "R" is given by

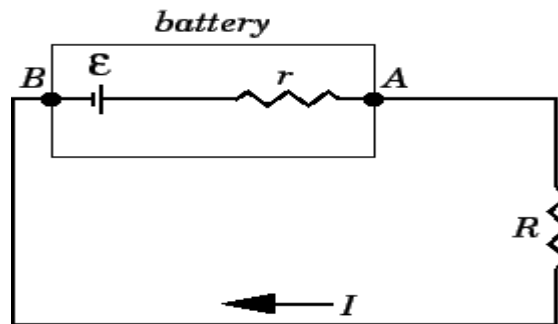
Heat energy = Power dissipated x time

$$H = (I^2R) (t)$$

$$\mathbf{H = I^2Rt}$$

Electromotive Force (E.M.F):

The potential difference exists between the two terminals of a battery when it is not connected in external circuit is called electromotive force.



Mathematically:

Suppose a battery of electromotive force ' E ' is connected to external resistor ' R ' and let ' r ' is the internal resistance of battery as shown in figure.

Using Ohm's law

Voltage of source = Current $\times$ Resistance in circuit

$$E = I(r + R)$$

$$E = Ir + IR$$

Here

$V = IR$ = Potential difference across external resistor

$$E = Ir + V$$

Here E is the E.M.F. of the battery and r is the internal resistance of battery and I is the Maximum current that a battery can provide.

Electromagnetism:

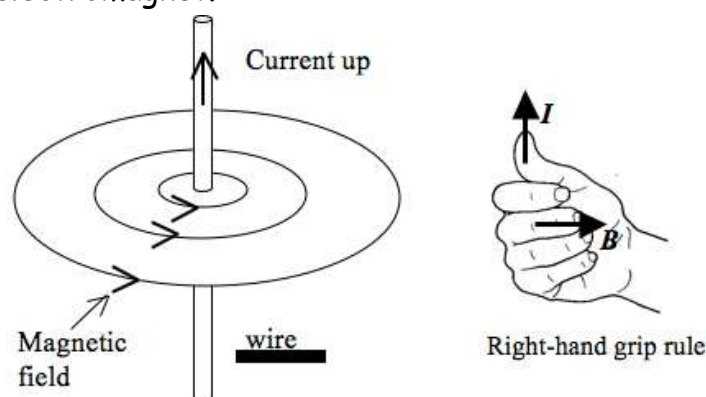
Electromagnetism is the branch of physics which relates the electricity and magnetism.

Introduction:

In 1819 Christian Orested discovered that magnetic field is generated around wire when current is passing through it, this wire treated as electromagnet. In 1829 Ampere's observed that two current carrying wires exerts force on each other and suggest that the magnetic forces are produced due to the currents passing through the wires.

Electromagnet:

When charges flow in a wire, magnetic field is produced around the wire which is called electromagnet.

**Magnetic Field:**

The region around magnet in which it can experiences force on another magnet or moving charge particle is called magnetic field. The S.I unit of magnetic field is Tesla. It is a vector quantity.

Direction of Magnetic Field:

The direction of magnetic field is given by right hand grip rule in which if the thumb represents current the fingers will represent the direction of magnetic field.

Magnetic Force:

When charge particle is moving in magnetic field making some angle with the field then it experiences force known as magnetic force.

Force on a charge moving in uniform magnetic field:

Consider a charge particle of magnitude ' q ' is moving in a magnetic field ' $\vec{B}$ ' with velocity ' $\vec{v}$ '. Let the charge particle makes an angle ' θ ' with the field. Charge particle experiences magnetic forces " $\vec{F}$ " when it is moving in a uniform magnetic field at some angle with the field. This magnetic force is directly proportional to the

1. Magnitude of charge ' q '
2. Magnitude of velocity of charge particle ' v '

3. Strength of the magnetic field ' B '

4. Sine of the angle between the velocity of charge particle and field ' $\sin\theta$ '

Mathematically:

$$F \propto q$$

$$F \propto v$$

$$F \propto B$$

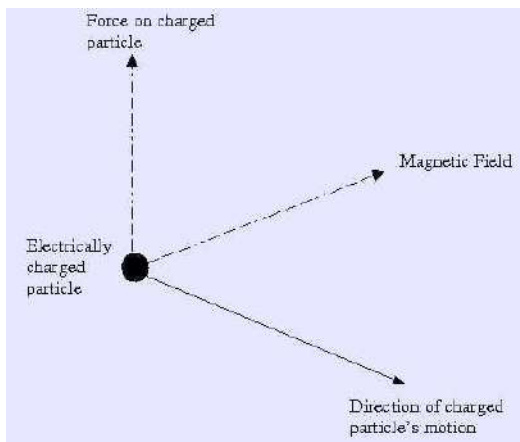
$$F \propto \sin\theta$$

Comparing above expressions, we get

$$F = qvB\sin\theta$$

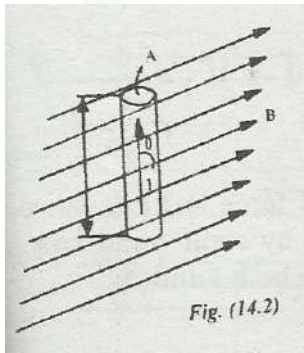
In vector form

$$\vec{F} = q(\vec{v} \times \vec{B})$$



Force on a current carrying conductor in uniform magnetic field

Consider a current carrying conductor placed in a uniform magnetic field ' $\vec{B}$ '. Let some electric current ' I ' is passing through the conductor. The current carrying conductor experiences magnetic force ' $\vec{F}$ ' if the conductor is placed at some angle ' θ ' with the field.



Since we know that the magnetic force on a charge particle when it is moving in uniform magnetic field is given by

$$\vec{F} = q(\vec{v} \times \vec{B}) \text{ --- (1)}$$

Let

L = length of conductor

n = Total number of charges per unit volume

A = Area of cross – section of conductor

e = charge on single electron

The total charge

$$q = Ne \text{ --- (2)}$$

$$n = \frac{N}{V}$$

$$n = \frac{N}{AL}$$

$$N = nAL$$

Substitute this value in equation (2), we get

$$q = nALe$$

Substitute this value in equation (1), we get

$$\vec{F} = nALe (\vec{v} \times \vec{B}) \text{ --- (3)}$$

Consider a unit vector $\vec{u}$ in the direction of velocity and length since both vectors contain same directions, therefore

$$\vec{u} = \frac{\vec{v}}{v} \text{ or } \vec{u} = \frac{\vec{L}}{L}$$

$$\vec{v} = \vec{u}v$$

Substitute this value in equation (3), we get

$$\vec{F} = nALe (\vec{u}v \times \vec{B})$$

$$\vec{F} = nALe (\vec{u}v \times \vec{B})$$

$$\vec{F} = nALev (\vec{u} \times \vec{B})$$

Now substitute $\vec{u} = \frac{\vec{L}}{L}$ in above equation, we get

$$\vec{F} = nALev \left(\frac{\vec{L}}{L} \times \vec{B} \right)$$

$$\vec{F} = nAev (\vec{L} \times \vec{B})$$

$$\vec{F} = \frac{nALe}{t} (\vec{L} \times \vec{B})$$

Since we know that $nALe = It$, therefore above equation will becomes

$$\vec{F} = \frac{q}{t} (\vec{L} \times \vec{B})$$

$$\vec{F} = I(\vec{L} \times \vec{B})$$

In magnitude form

$$F = ILB\sin\theta$$

$$F = BIL\sin\theta$$

Force on a current carrying rectangular coil in uniform magnetic field:

Consider a current carrying rectangular coil 'ABCD' suspended in a uniform magnetic field ' $\vec{B}$ ' such that the plane of the coil is parallel to the magnetic field. Let current 'I' is passing through the coil or loop in clockwise direction as shown in figure

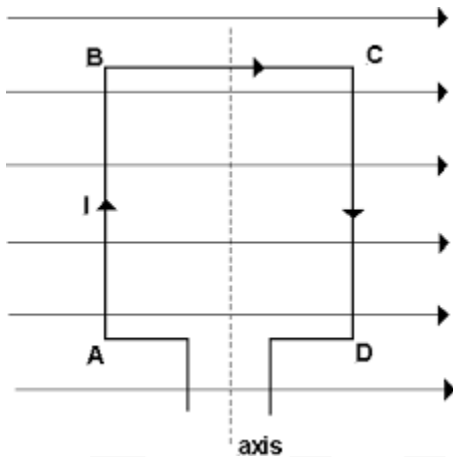
Let

L = Length of rectangular coil

b = width of the rectangular coil

A = b × L = area of the rectangular coil

N = number of turns in the coil



Since we know that the magnetic force on a current carrying conductor in uniform magnetic field

$$F = BIL\sin\theta$$

The sides of the rectangular loop can be treated as the current carrying conductors. The vertical sides of the rectangular loop experiences magnetic force as these sides are perpendicular to the field.

$$F = BIL\sin 90^\circ$$

$$F = BIL(1)$$

$$F = BIL$$

The horizontal sides of the loop experiences no magnetic force as these sides are either parallel or anti parallel to the field because $\theta = 0^\circ$ or 180° and $\sin 0^\circ = 0$ and $\sin 180^\circ = 0$

$$F = 0$$

The magnetic forces on two vertical sides of the conductor are identical in magnitude but opposite in direction due to the currents flowing in opposite direction in vertical sides. The two forces constitute a torque of couple about axis of rotation such that the width of rectangular coil 'b' becomes the arm of the couple. Mathematically it is given by

Torque of Couple = (Magnitude of force) $\times$ (Arm of couple)

$$\tau = BIL \times b$$

$$\tau = BI(L \times b)$$

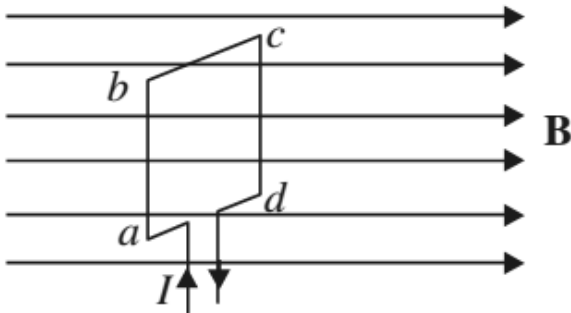
$$\tau = BIA$$

For 'N' number of turns, we have

$$\tau = BIAN$$

$$\tau = BINA$$

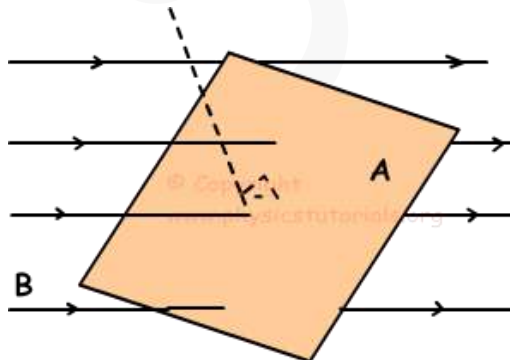
If the plane of the rectangular coil makes an angle α with the field then



$$\tau = BIN \cos \alpha$$

Magnetic Flux

The number of magnetic lines of forces passing through the surface normally is called magnetic flux. It is denoted by ϕ_B .



Mathematically:

It is the dot product of magnetic field " $\vec{B}$ " and area vector " $\vec{\Delta A}$ ".

$$\varphi_B = \vec{B} \cdot \vec{\Delta A}$$

$$\varphi_B = B\Delta A \cos\theta$$

Unit:

The S.I unit of magnetic flux is Weber. It can be defined as "if magnetic field of 1 Tesla passing through the surface area of 1 m^2 normally, then magnetic flux will be 1 Weber."

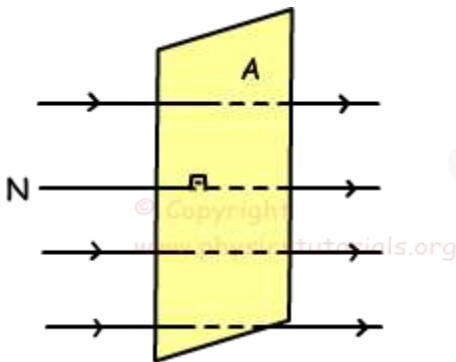
$$\varphi_B = B\Delta A \cos\theta$$

Substituting S.I units in above equation

$$\text{Weber} = \text{Tesla m}^2$$

Maximum Flux

The magnetic flux will be maximum if the angle between the magnetic field is parallel to area vector i.e. magnetic lines of forces crossing the surface normally. $\theta = 0^\circ$



$$\varphi_B = B\Delta A \cos\theta$$

$$\varphi_B = B\Delta A \cos 0^\circ$$

$$\varphi_B = B\Delta A(1)$$

$$\varphi_B = B\Delta A$$

Minimum Flux

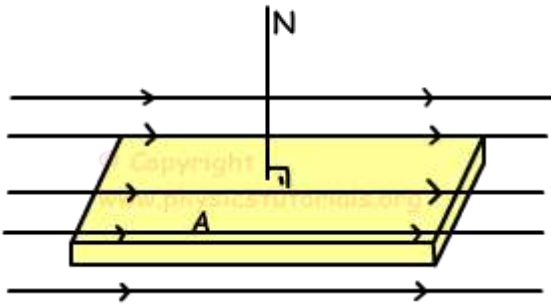
The magnetic flux will be minimum if the angle between the magnetic lines of forces and area vector is 90° i.e. no magnetic line of force passing through the surface as shown in figure

$$\varphi_B = B\Delta A \cos\theta$$

$$\varphi_B = B\Delta A \cos 90^\circ$$

$$\varphi_B = B\Delta A(0)$$

$$\varphi_B = 0$$



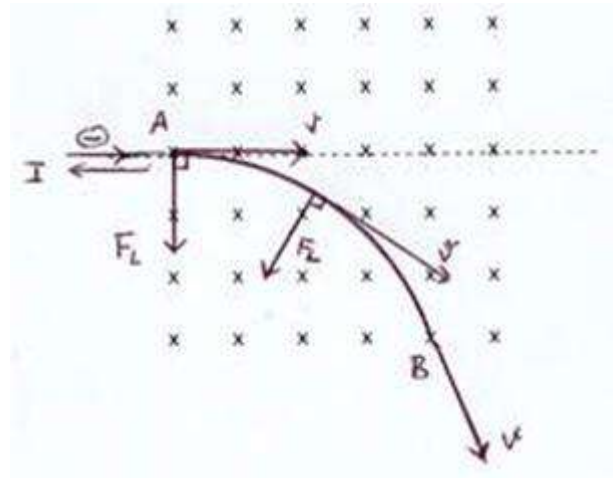
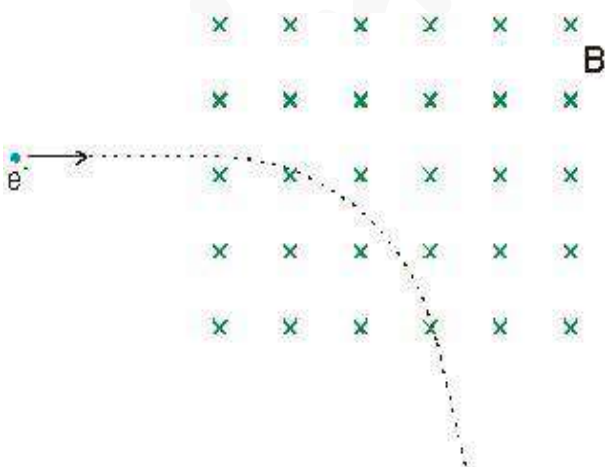
Negative Flux

The magnetic flux will be minimum if the magnetic lines of forces are opposite to area vector i.e. $\theta = 180^\circ$

$$\begin{aligned}\phi_B &= B\Delta A \cos\theta \\ \phi_B &= B\Delta A \cos 180^\circ \\ \phi_B &= B\Delta A (-1) \\ \phi_B &= -B\Delta A\end{aligned}$$

Motion of a charged particle in uniform magnetic field:

Consider a charge particle " q " with mass " m " moving in uniform magnetic field " B " with velocity " v ". The charge particle experiences magnetic force as it enters the field and in the result moves in a curve path. The magnitude of velocity remain unchanged but its direction changes from point to point and finally the charge particle moves in a circular path. The magnetic force which is acting towards the centre of circle at any position on the circle provides the necessary centripetal force to the charge particle.



Mathematically:

Let " r " is the radius of circle and " θ " is the angle at which charge particle enters the field, then

Magnetic force = Centripetal force

$$qvB\sin\theta = \frac{m(v_{\text{perpendicular}})^2}{r}$$

$v\sin\theta$ = perpendicular component of velocity

$$qvB\sin\theta = \frac{m(v\sin\theta)^2}{r}$$

$$evB\sin\theta = \frac{mv^2\sin\theta^2}{r}$$

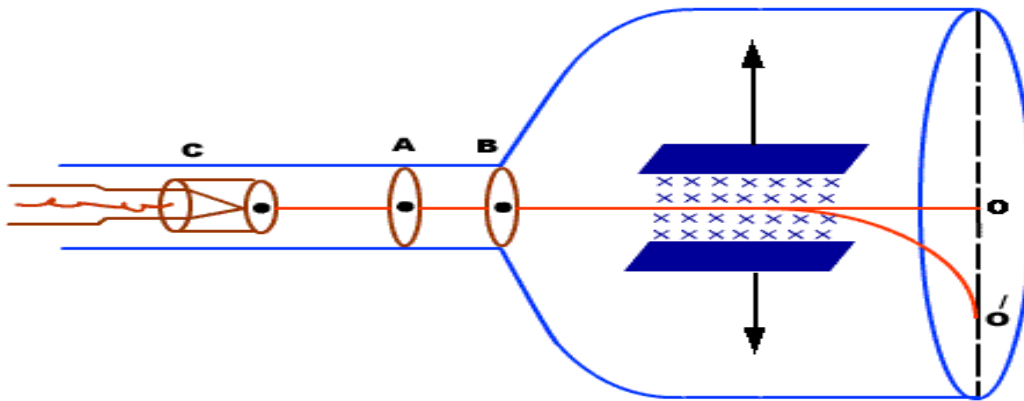
$$eB = \frac{mv\sin\theta}{r}$$

$$r = \frac{mv\sin\theta}{eB}$$

The e/m ratio of electron by J.J. Thomson method:

Introduction:

The e/m ratio of electron was found by Sir J.J. Thomson. In his method electrons are produced by heating a tungsten filament by passing a current through it. The electrons moving sideways are also directed towards the screen by applying negative potential on a hollow cylinder open on both sides surrounding a filament. The electrons are then accelerated by applying high potential positive voltage to disks A and B. The electron beam produced travels in a straight path and produced the spot on the screen at point o.



Velocity of electron:

If "V" is the total potential difference applied and "v" is the velocity of electron then loss in potential energy is equal to the gain in kinetic energy of electron. The energy equation is given by

Loss in Potential Energy = Gain in Kinetic energy

$$qV = \frac{1}{2}mv^2$$

$$eV = \frac{1}{2}mv^2$$

$$v^2 = \frac{2eV}{m}$$

$$v = \sqrt{\frac{2eV}{m}}$$

e/m ratio of electron:

If there is a magnetic field " $\vec{B}$ " present b/w the plates directed inside to the page as shown in figure the electron deviated from its straight path and moves in a curve path. The force due to magnetic field on moving electrons makes them move in a circular path a the light spot shifts from o to o' on the screen. The magnetic force on electron provides the necessary centripetal force and

$$evB = \frac{mv^2}{r}$$

$$eB = \frac{mv}{r}$$

$$e/m = \frac{v}{rB}$$

Radius of circular path:

In figure

a = shift of electron from o to o'

b = distant travelled in magnetic fields

r = radius of circle

Applying Pythagoras theorem in right angle triangle $\Delta O'CD$

Hypotenuse² = Base² + Perpendicular²

$$r^2 = (r - a)^2 + b^2$$

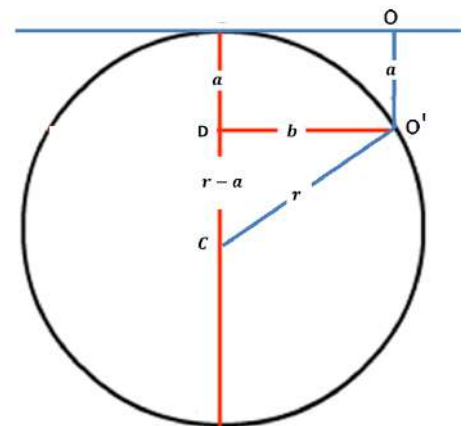
$$r^2 = r^2 - 2ar + a^2 + b^2$$

Since a is very small, therefore neglecting a^2

$$r^2 = r^2 - 2ar + b^2$$

$$b^2 = 2ar$$

$$r = \frac{b^2}{2a}$$



Neutral Condition:

By applying potential difference between the two horizontal plates electric field is produced to apply electric force on electron in a direction opposite to the direction of magnetic force. The potential difference is adjusted so that the two forces neutralize each other and electron beam pass through the region of electric and magnetic fields without deviation. In this condition
Electric force = Magnetic force

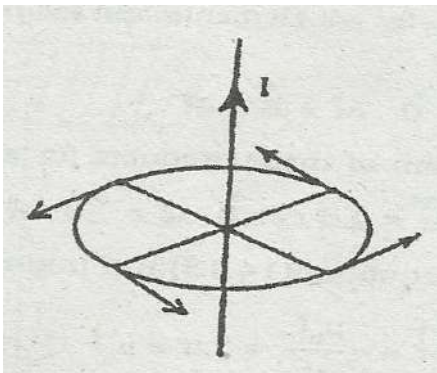
$$qE = qvB$$

$$E = vB$$

$$v = \frac{E}{B}$$

Biot and Savart law:**Statement:**

"The magnitude of magnetic field around a straight current carrying conductor is directly proportional to the twice of current and inversely proportional to the distance from the conductor."

**Mathematically:**

Consider a long straight wire carrying a current "I" in the direction as shown in figure. The magnetic field at all the points on the curve taken in form a circle round the wire is tangential and of the same magnitude.

$$B \propto 2I$$

$$B \propto \frac{1}{r}$$

Comparing both expressions, we get

$$B \propto \frac{2I}{r}$$

$$B = (\text{constant}) \frac{2I}{r}$$

$$\text{Here constant} = \frac{\mu_0}{4\pi}$$

where μ_0 = permeability of free space

Substitute the value of constant in above equation, we get

$$B = \left(\frac{\mu_0}{4\pi}\right) \frac{2I}{r}$$

$$B = \frac{\mu_0 I}{2\pi R}$$

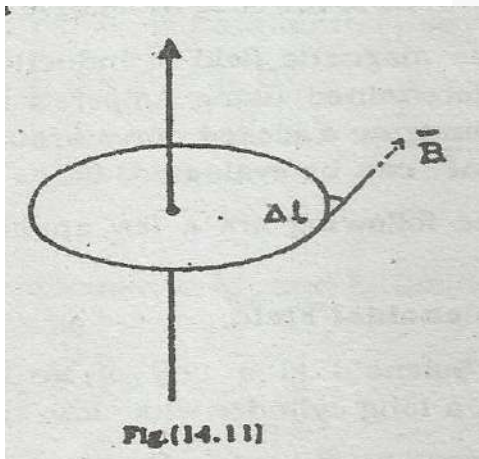
Ampere's law:

Statement:

"The sum of the products of the tangential components of magnetic field of induction and the length of an element of a closed taken in the magnetic field is μ_0 times the current which passes through the area bounded by this curve."

Mathematically (Proof):

Consider a long straight wire carrying a current "I" in the direction as shown in figure. The magnetic field at all the points on the curve taken in form a circle round the wire is tangential and of the same magnitude.



Let the circle be divided into small elements each of length Δl , such that at all elements the angle between the field and length element is 0° .

Since we know that

$$\vec{B} \cdot \vec{\Delta l} = B \Delta l \cos \theta$$

$$\vec{B} \cdot \vec{\Delta l} = B \Delta l \cos 0^\circ$$

$$\vec{B} \cdot \vec{\Delta l} = B \Delta l$$

The sum of these products for all the elements

$$\sum (\vec{B} \cdot \vec{\Delta l}) = \sum B \Delta l \cos 0^\circ$$

$$\sum (\vec{B} \cdot \vec{\Delta l}) = B \sum \Delta l$$

$\sum \Delta l = \text{sum of the lengths of all elements} = \text{circumference of the circle} = 2\pi r$

$$\sum (\vec{B} \cdot \vec{\Delta l}) = B (2\pi r) \text{ --- (1)}$$

According to Ampere's law

$$\sum (\vec{B} \cdot \vec{\Delta l}) = \mu_0 I \text{ --- (2)}$$

Comparing (1) and (2), we get

$$B (2\pi r) = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

Applications of Ampere's law:

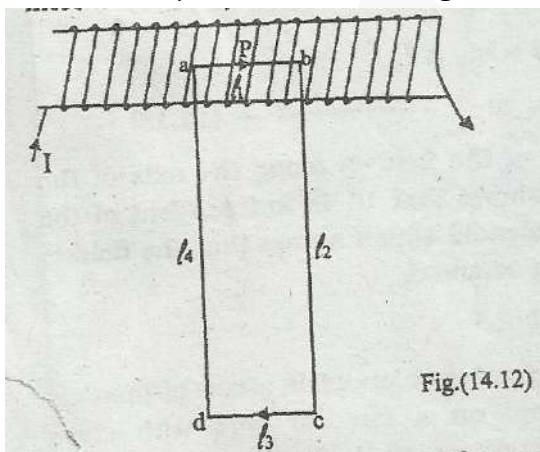
Ampere's law is used to calculate the magnetic field of solenoid and toroid.

Solenoid:

Solenoid is a coil of insulated copper wire wound on a long cylindrical with close turns.

Magnetic field of a Solenoid:

Consider a long solenoid with "N" number of turns. Let current "I" is passing through the solenoid and strong magnetic field is generated inside the solenoid. According to right hand rule the field is parallel to the length of solenoid.



To determine the magnetic field of solenoid consider an imaginary rectangular loop abcd with the side ab on the axis and the side cd far away where the field is zero as shown in figure. The rectangular loop is divided into four elements l_1 , l_2 , l_3 and l_4 .

The total magnetic flux of a closed rectangular loop is given by

$$\sum (\vec{B} \cdot \vec{l}) = (\vec{B} \cdot \vec{l}_1) + (\vec{B} \cdot \vec{l}_2) + (\vec{B} \cdot \vec{l}_3) + (\vec{B} \cdot \vec{l}_4) - - - (1)$$

$(\vec{B} \cdot \vec{l}_1)$:

$$\vec{B} \cdot \vec{l}_1 = Bl_1 \cos \theta$$

Since side $\vec{l}_1$ is parallel to the magnetic field $\vec{B}$, therefore magnetic flux is maximum for side ab

$$\vec{B} \cdot \vec{l}_1 = Bl_1 \cos 0^\circ$$

$$\vec{B} \cdot \vec{l}_1 = Bl_1 (1)$$

$$\vec{B} \cdot \vec{l}_1 = Bl_1$$

$(\vec{B} \cdot \vec{l}_2)$ and $(\vec{B} \cdot \vec{l}_4)$:

Since both sides $\vec{l}_2$ and $\vec{l}_4$ are perpendicular to the magnetic field $\vec{B}$ i.e. $\theta = 90^\circ$, $\cos 90^\circ = 0$

Therefore the magnetic flux is zero for both perpendicular side bc and ad.

$$\vec{B} \cdot \vec{l}_2 = 0$$

$$\vec{B} \cdot \vec{l}_4 = 0$$

$(\vec{B} \cdot \vec{l}_3)$:

Since side cd is far away from solenoid where the field is very small. It can be neglected and put equal to zero.

$$B = 0$$

Therefore

$$\vec{B} \cdot \vec{l}_3 = 0$$

Substituting all values in (1), we get

$$\sum (\vec{B} \cdot \vec{l}) = Bl_1 + 0 + 0 + 0$$

$$\sum (\vec{B} \cdot \vec{l}) = Bl_1$$

In general,

$$\sum (\vec{B} \cdot \vec{l}) = Bl - - - (2)$$

According to Ampere's law the total magnetic flux around closed loop abcd

$$\sum (\vec{B} \cdot \vec{l}) = \mu_0 I$$

For N number of turns

$$\sum (\vec{B} \cdot \vec{l}) = \mu_0 NI - - - (3)$$

Comparing equations (2) and (3), we get

$$Bl = \mu_0 NI$$

$$B = \mu_0 \frac{N}{l} I$$

$$\frac{N}{l} = n = \text{number of turns per unit length}$$

$$B = \mu_0 nI$$

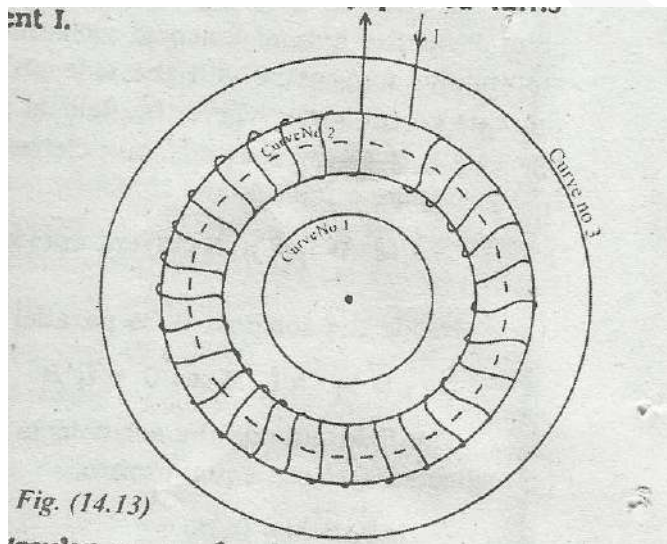
The magnetic field is independent of the position within the solenoid which shows that the field is uniform within a long solenoid. The direction of magnetic field is along the axis of the solenoid.

Toroid:

Toroid is a coil of insulated copper wire wound on a circular core with close turns. It is also called circular solenoid.

Magnetic field of a Toroid:

Consider a toroid with "N" number of turns. Let current "I" is passing through the toroid winding and strong magnetic field is generated inside the core of toroid.



To calculate the magnetic field of toroid consider three imaginary curves as shown in figure. Consider the following cases:

- 1 If the circular path (curve 1) is outside the core on the inner side of the toroid it encloses no current, therefore the magnetic field is zero at curve 1.

$$B = 0$$

2. If the circular path (curve 2) is within the core the area bounded by the curve will be threaded by N turns each carrying a current I. It is evident from symmetry that the field at all points of the curve must have the same magnitude and it should be tangential to the curve at all points.

The sum of these products for all the elements

$$\sum (\vec{B} \cdot \vec{\Delta l}) = \sum B \Delta l \cos 0^\circ$$

$$\sum (\vec{B} \cdot \vec{\Delta l}) = B \sum \Delta l$$

$$\sum \Delta l = \text{Sum of the lengths of all elements} = \text{Circumference of the circle} = 2\pi r$$

$$\sum (\vec{B} \cdot \vec{\Delta l}) = B (2\pi r)$$

3. If the circular path (curve 3) is outside of the core on the outer side of toroid the area bounded by the curve will be threaded by each turns twice but in opposite directions and the algebraic sum of all the currents is zero.

$$B = 0$$

Therefore the magnetic flux at curve 1 and curve 3 is zero and the total magnetic flux of toroid is the magnetic flux at curve 2 i.e.

$$\sum (\vec{B} \cdot \vec{\Delta l}) = B (2\pi r) \text{ --- (1)}$$

According to Ampere's law

$$\sum (\vec{B} \cdot \vec{\Delta l}) = \mu_0 NI \text{ --- (2)}$$

Comparing (1) and (2), we get

$$B (2\pi r) = \mu_0 NI$$

$$B = \frac{\mu_0 NI}{2\pi r}$$

A toroid, therefore, produced a uniform magnetic field of induction which is confined in the space occupied by the core.

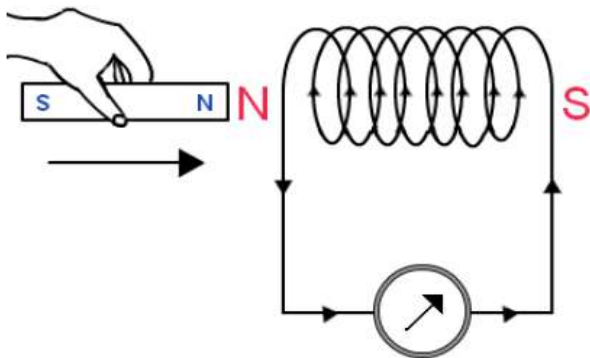
Electromagnetic Induction:

Electromagnetic induction is a process in which conductor placed in a magnetic field causes the production of a voltage across the conductor.

Introduction:

In 1830 Joseph Henry in the United States and a year later Faraday in England independently observed that an emf is setup in a coil placed in a magnetic field whenever the flux through the

coils changes. This effect is called electromagnetic induction and if the coil forms a part of a closed circuit, the induced emf causes a current to flow in the circuit.



Faraday's law of induction:

Statement:

"The induced emf in a coil is directly proportional to the rate at which magnetic flux is changing and the number of turns in the coil."

Mathematically:

$$\xi \propto \frac{d\phi_B}{dt}$$

$$\xi \propto N$$

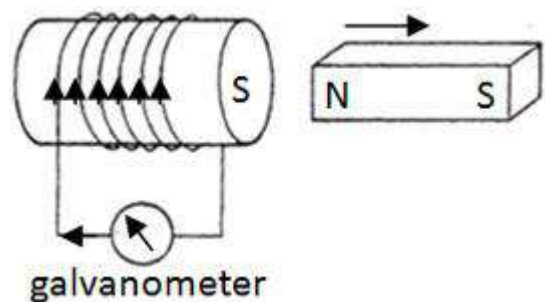
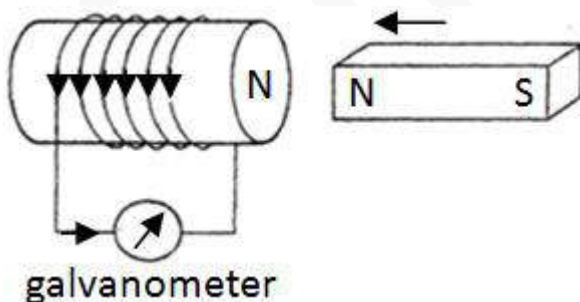
Comparing both expressions, we get

$$\xi = N \frac{d\phi_B}{dt}$$

Lenz's law:

Statement:

"The induced current always flows in such a direction as to oppose the change which is giving rise to it."

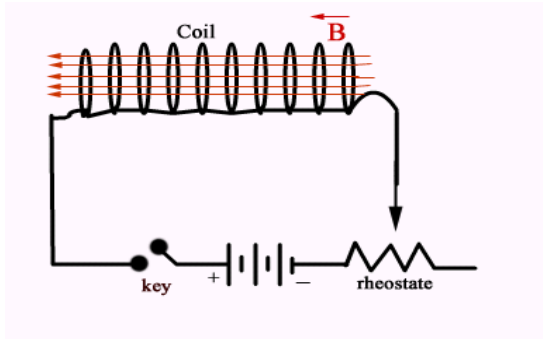


Mathematically:

$$\xi = -N \frac{d\phi_B}{dt}$$

Self Induction:

When a magnetic flux changes through the coil and emf induces in the same coil then this phenomenon is called self induction.

**Mathematically:**

A coil through which a current is flowing has an associated magnetic field. If the current changes then magnetic flux does also changes and an emf is induced in the coil. This induced emf is directly proportional to the rate at which magnetic flux is changing

$$\xi \propto \frac{dI}{dt}$$

$$\xi = (\text{constant}) \frac{dI}{dt}$$

Here, constant = L = Self Inductance of the coil

$$\xi = L \frac{dI}{dt}$$

If the current is increasing, the back emf opposes the increase, if the current is decreasing, it opposes the decrease. Therefore using the concept of Lenz's law

$$\xi = -L \frac{dI}{dt} \text{ --- (1)}$$

According to Faraday's law of induction the induced emf in the coil

$$\xi = -N \frac{d\phi}{dt} \text{ --- (2)}$$

Comparing (1) and (2), we get

$$-L \frac{dI}{dt} = -N \frac{d\phi}{dt}$$

$$L dI = N d\phi$$

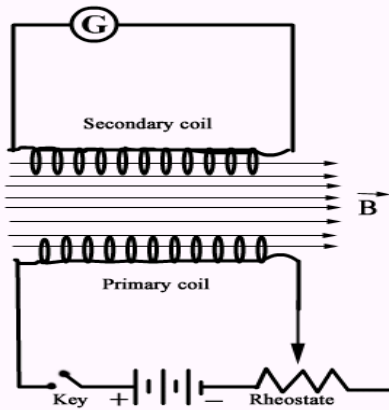
$$d(LI) = d(N\phi)$$

$$LI = N \phi$$

The self inductance of the coil depends on the dimensions of the coil, the number of turns and the permeability of the core material. The S.I unit of self inductance is Henry.

Mutual Induction:

If two coils are close together, then changing magnetic flux in one coil (primary coil) setup a changing magnetic flux in secondary coil and hence produces an induced emf in secondary coil. This phenomenon is called mutual inductance.



Mathematically:

The induced emf in the secondary coil is directly dependent on the rate at which magnetic flux is changing in the primary coil.

$$\xi_s \propto \frac{dI_p}{dt}$$

$$\xi_s = (\text{constant}) \frac{dI_p}{dt}$$

Here, constant = M = Mutual Inductance of the coil

$$\xi_s = M \frac{dI_p}{dt}$$

The S.I unit of mutual inductance is Henry. It depends upon the dimensions of the coil, number of turns and permeability of the core material. According to Lenz's law the induced emf always oppose the change which causes this emf, therefore

$$\xi_s = -M \frac{dI_p}{dt} \text{ --- (1)}$$

Since we know that the Faraday's law of induction for induced emf in a secondary coil

$$\xi_s = -N_s \frac{d\phi_s}{dt} \text{ --- (2)}$$

Comparing (1) and (2), we have

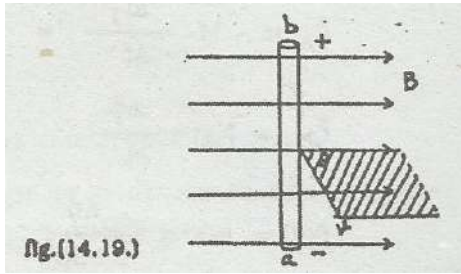
$$\begin{aligned}
 -N_s \frac{d\phi_s}{dt} &= -M \frac{dI_p}{dt} \\
 N_s d\phi_s &= M dI_p \\
 d(N_s \phi_s) &= d(M I_p) \\
 N_s \phi_s &= M I_p
 \end{aligned}$$

Motional EMF:

When a conductor is moved across a magnetic field, a potential difference appears across the ends. This potential difference is known as motional emf.

Derivation:

Consider a wire of length "l" moving across the magnetic field of induction "B" with a velocity "v" as shown in figure. Each free electron of the conductor is moving with the conductor and thus experiences a force which is give by



$$\begin{aligned}
 \vec{F} &= q(\vec{v} \times \vec{B}) \\
 F &= qvB\sin\theta
 \end{aligned}$$

The electrons are flows toward end a leaving the end b with positive charge till the force of electric field balance the force due to the motion of conductor. Thus a potential difference is setup b to a

Potential difference = work done by unit charge

$$\begin{aligned}
 V &= \frac{W}{q} \\
 V &= \frac{\text{Force} \times \text{distance}}{q} \\
 V &= \frac{qvB\sin\theta \times l}{q} \\
 V &= vBL\sin\theta
 \end{aligned}$$

If the conductor is moving at right angles to the field i.e. $\theta = 90^\circ$.

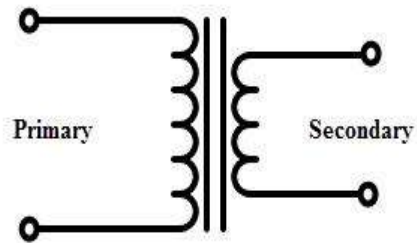
$$\begin{aligned}
 V &= vBL\sin 90^\circ = vBL(1) \\
 V &= vBL
 \end{aligned}$$

Transformer:**Definition:**

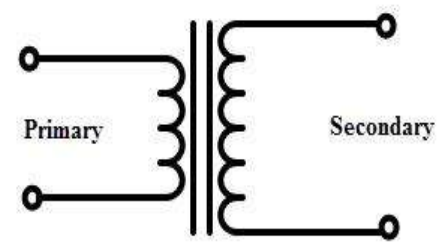
Transformer is a device which is used to step up or step down an alternating emf.

Principle:

It works on the principle of mutual induction and explained on the basis of Faraday's law of electromagnetic induction.



Step-down Transformer



Step-up Transformer

Types of Transformer

Types of Transformer

There are two types of transformers based on the ratio of output to the input voltage.

1. Step-up Transformer
2. Step-down Transformer

Step up Transformer:

In step-up transformer the number of turns in secondary coil is greater than the number of turns in primary coil.

$$N_S > N_P$$

Step down Transformer:

In step-down transformer the number of turns in secondary coil is less than the number of turns in primary coil.

$$N_S < N_P$$

Construction:

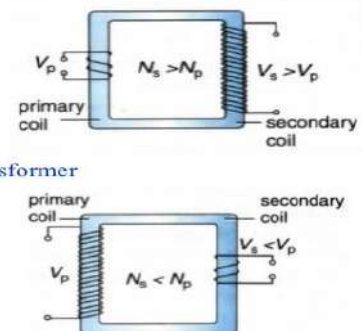
Two coils of insulated copper wire, the primary and secondary, are wound, one on top of the other, on a laminated soft iron core and thus these are linked magnetically. The presence of the soft iron ensures that all the flux associated with primary coil also passes through the secondary coil.

Working:

Suppose that an alternating emf V_p is applied to the primary coil. Since the input voltage is A.C. i.e. varying current is flowing in primary coil therefore magnetic flux is changing in primary coil.

Step-up Transformer

Step-down Transformer



The changing flux associated in the primary coil also passes through the secondary coil and induces emf in the secondary coil.

Ratio of Output to the input voltage:

According to Faraday's law of inductance the voltage V_p in primary coil is given by

$$V_p = N_p \frac{d\phi}{dt} \text{ --- (1)}$$

Let V_s is the voltage induced in the secondary coil then Faraday's law of induction for secondary coil is given by

$$V_s = N_s \frac{d\phi}{dt} \text{ --- (2)}$$

Divide equations (1) and (2), we get

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Efficiency of Transformer:

When a load (resistance) is connected across secondary, a current I_s flows in the secondary. Suppose that the current in the primary is I_p . If the transformer is 100% efficient.

Power output = Power input

$$V_s I_s = V_p I_p$$

Efficiency of a transformer, $\eta = \frac{\text{Power Output}}{\text{Power Input}}$

$$\eta = \frac{V_s I_s}{V_p I_p}$$

In terms of percentage

$$\eta\% = \frac{V_s I_s}{V_p I_p} \times 100$$

Power losses in Transformer:

There are four types of power losses in transformer

- Eddy current losses
- Hysteresis losses
- I^2R losses
- Flux leakage losses

1. Eddy Current losses (Circulating Current losses):

In transformer A.C. voltage is supplied to the primary windings which produce alternating emf in secondary winding. When changing flux in primary coil links with secondary at the same time some part of this flux also get linked with some other parts of the transformer (e.g. soft iron core) which results small circulating current in them. This current is called eddy current. This current produce heating effect in the core of the transformer.

Correction method:

To reduce eddy losses the core is laminated i.e. made up of thin sheets of soft iron each separated from the next by a layer of insulating varnish. These very nearly eliminate eddy current losses.

2. Hysteresis losses (Core losses):

Each time the direction of magnetization of the core is reversed, some energy is wasted in overcoming in internal friction. This is known as hysteresis losses or core losses and it produce heating in the core.

Correction method:

It is minimized by using core of high permeability.

3. I^2R losses (Copper losses):

Some energy is dissipated as heat in the primary and secondary coils of transformer which is called copper losses or I^2R losses.

Correction method:

This is reduced by using suitably thick wire.

4. Flux Leakage losses (Stray losses):

Some loss of energy occurs because a small amount of the flux associated with the primary coils fails to pass through the secondary coil and this small flux passes through winding insulation and transformer insulating oil.

Correction method:

Flux leakage losses can be reduced by making some special designs of core for primary and secondary windings.

Dynamo:

Dynamo is a device which converts either mechanical energy into electrical energy or electrical energy into mechanical energy e.g. generator or motor.

Generator:

An electric generator is a device which converts mechanical energy into electrical energy.

A.C. Generator:

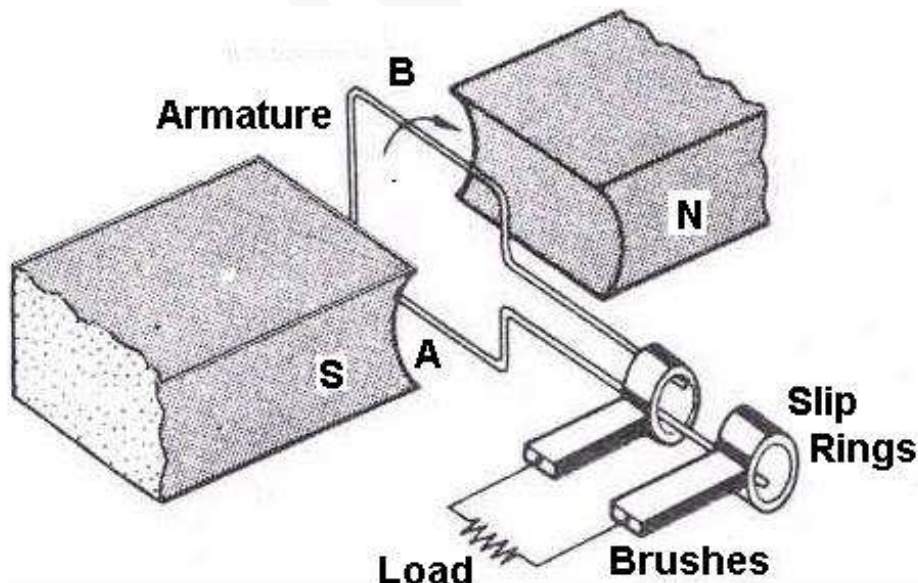
AC generator is a device which converts mechanical energy into alternating current.

Principle:

The principle of the generator is that an emf is induced in the coil due to changing magnetic flux.

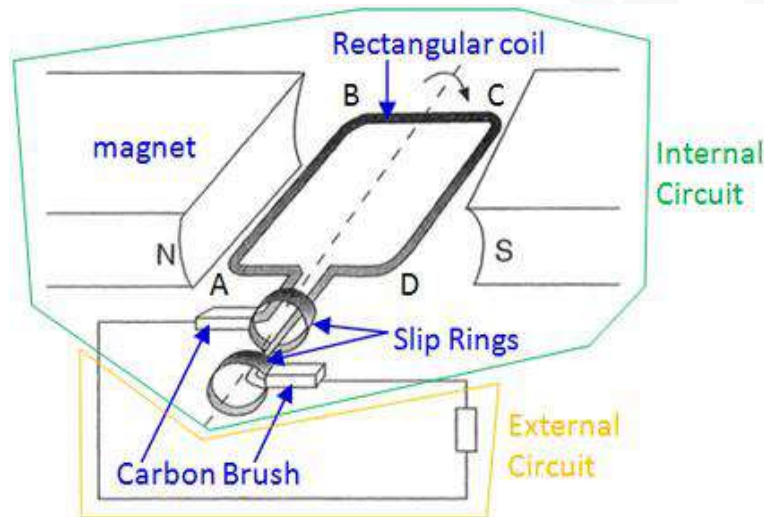
Construction:

1. **Field Magnet:** It is a strong magnet to produce a strong and uniform magnetic field $\vec{B}$.
2. **Armature:** Armature of the coil consists of a large number of turns of insulated copper wire wound on a laminated soft iron core.
3. **Slip rings and** The ends of the coil are joined to two separate copper rings fixed on the axle. These slip rings rotate along with the armature.
4. **Collecting Brushes:** Carbon brushes make contact between the armature and external circuit. These brushes remain pressed against each of the rings which form the terminals of the external circuit.

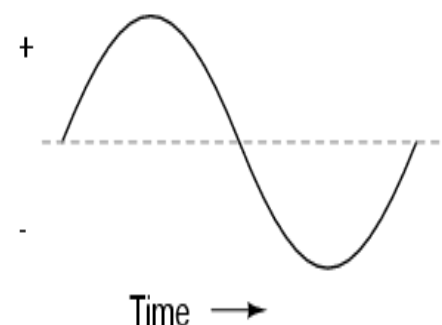
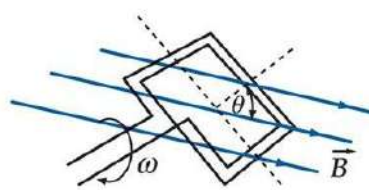
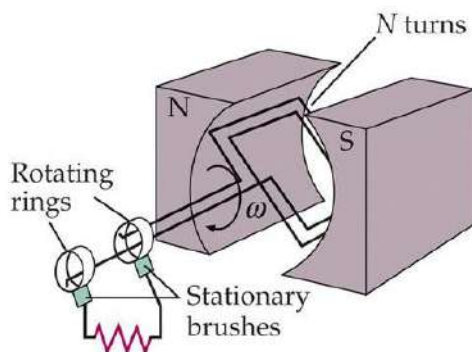


Working:

As the coil rotates clockwise, one side of the rectangular coil moves inward and other side outward. Therefore changing magnetic flux induces emf and current flows in the coil. The slip ring also rotates with the coil and sliding over carbon brushes. Carbon brushes are used to provide stationary contacts between external circuit and rectangular coil. The induced current thus flows in the external circuit through stationary carbon brushes. As the face of the coil changes from one side to another the magnetic flux changes in the reverse direction and the direction of current becomes reverse. We therefore see that the induced current in the external circuit changes direction after every half rotation of the coil.

**Derivation:**

A motional emf is setup in each of the two sides AB and CD in opposite directions when the coil is rotated because these sides are moving in opposite sense with respect to the magnetic field. Since the other two sides AD and BC are moving in the same sense with respect to field. The emfs are induced in the same direction and cancel each other.



Let

v = linear velocity

B = Magnetic field of induction

N = Number of turns

l = length of the vertical side of the coil

ξ = emf produced in the coil

ω = angular velocity

b = width of the coil

A = area of the rectangular coil

The motional emf produced in the two sides of the coil is given by

$$\xi = 2vBNl\sin\theta \text{ --- (1)}$$

Since we know that angular velocity, $\omega = \frac{\theta}{t}$

Therefore, $\theta = \omega t$, substitute this value in equation (1), we get

$$\xi = 2vBNl\sin\omega t \text{ --- (2)}$$

The relation between linear velocity and angular velocity

$$v = r\omega \text{ --- (3)}$$

Since each particle of the both side rotates in a circle of radius half of the width of the rectangular coil, therefore

$$r = \frac{b}{2}$$

Substitute this value in (3), we get

$$v = \left(\frac{b}{2}\right)\omega$$

Put this value in equation (2), we get

$$\xi = 2 \left(\frac{b}{2} \right) \omega B N l \sin \omega t$$

$$\xi = b \omega B N l \sin \omega t$$

$$\xi = (bl) \omega B N \sin \omega t$$

$$\xi = (A) \omega B N \sin \omega t$$

$$\xi = N \omega A B \sin \omega t$$

Since we know that $\omega = 2\pi f$, therefore

$$\xi = N \omega A B \sin 2\pi f t$$

This equation shows that the alternating emf changes with the time.

Problem 14.14:

A 100 turns coil in a generator has an area of 500 cm² rotates in a field with B = 0.06 Weber / m². How fast must the coil rotated in order to generate a maximum voltage of 150 volts.

Ans.

Given Data

$$N = 100$$

$$A = 500 \text{ cm}^2 = 500 \times 10^{-4} \text{ m}^2$$

$$B = 0.06 \text{ Weber / m}^2$$

How fast must the coil rotated, $\omega = ?$ (angular speed)

Maximum voltage, $\xi = 150 \text{ volts}$

Solution:

Since we know that

$$\xi = N \omega A B \sin \omega t$$

$$\xi_{\max} = N \omega A B$$

$$150 = (100)(\omega)(500 \times 10^{-4})(0.06)$$

$$150 = (0.3)(\omega)$$

$$\frac{150}{0.3} = \omega$$

$$\omega = 150 \text{ rad/second}$$

Magneto:

A small generator employing a permanent magnet is called magneto and it is used in ignition system of petrol engines, motor bikes and motor boats etc.

Alternators:

The field magnets of large generators are electromagnets and these generators are called alternators.

Stator and Rotor:

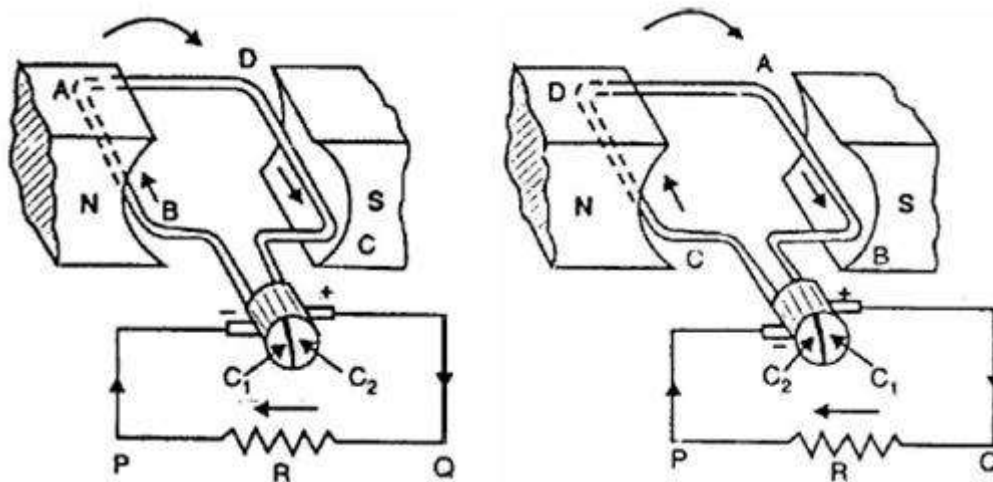
When the armature is stationary and the field magnet rotates around the armature then this is called stator and rotating magnet is called rotor.

D.C. Generator:

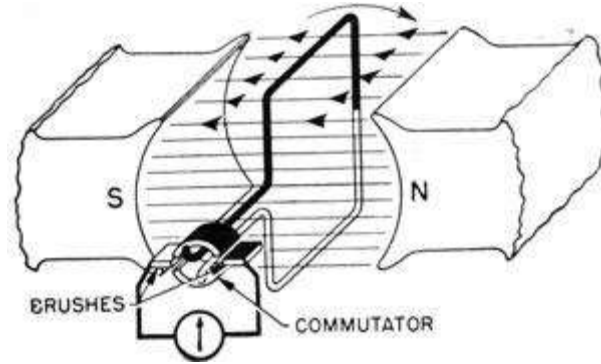
DC generator is a device which converts mechanical energy into direct current.

Commutator:

By replacing the slip rings of an A.C. generator by split ring, or Commutator, the generator can be made to produce a direct current through the external circuit.

**Working of Commutator or D.C. Generator:**

As the coil rotates past its next vertical position, an emf is induced in the coil in the reverse direction. At the same time the two halves of the split rings exchange their contacts with the external i.e. carbon brushes so that the direction of induced emf in the external circuit remains unchanged.



Electric Motor:

An electric motor is a device which converts electrical energy into mechanical energy.

D.C. Motor:

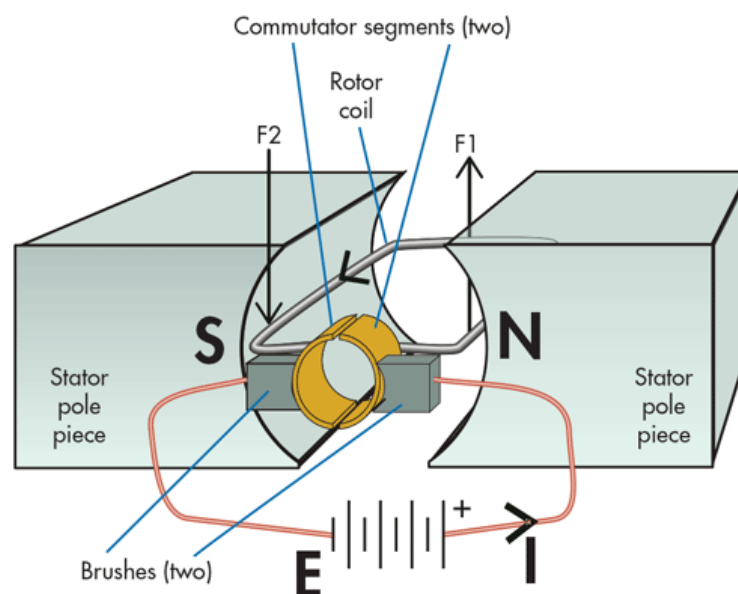
A simple D.C. motor convert electrical energy into mechanical energy when direct current is applied to it.

Working:

When a current is passed through a coil capable of rotation in a magnetic field of induction then it experiences a couple which is given by

$$\tau = BIN\cos\alpha$$

This couple rotate the coil in anticlockwise direction. The couple becomes zero when the face of the coil is perpendicular to the field. As the coil complete half rotation the Commutator reverse the direction of the current. It follows that the current in the rectangular loop always flow in the same direction i.e. clockwise and therefore the coil rotates in anticlockwise direction.



Chapter 15 "Electrical Measuring Instruments"

The Moving Coil Galvanometer:

The moving coil galvanometer is a basic electrical instrument. It is used for the detection or measurement of small currents.

Principle

When current flows in a rectangular coil placed in a magnetic field it experience a magnetic torque and if it is free to rotate under a controlling torque, it rotates through an angle proportional to the current flowing through it. The rotation or deflection thus indicates a current through it.

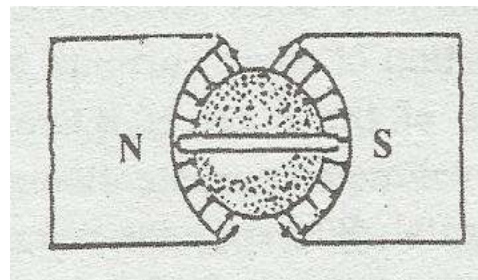
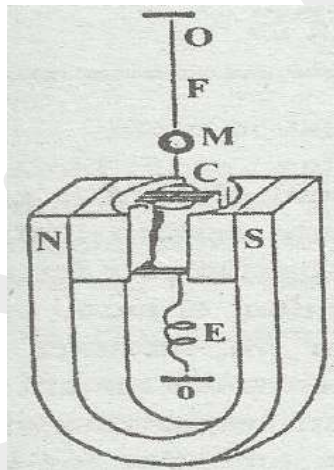
Construction:

The moving coil galvanometer is usually consists of the following parts:

1. A U-shaped permanent magnet with cylindrical concave pole-pieces to produce radial magnetic field.
2. A flat rectangular coil of thin enamel insulated wire.
3. A soft iron cylinder to make the magnetic field stronger and radial.
4. A scalar lamp and scale arrangement to measure current.
5. A pointer to show deflection.

Working:

When a current passes through the galvanometer coil, it experiences a magnetic deflecting torque, which tends to rotate it from its rest position. As the coil rotates it produces a twist in the suspension strip. The twist in the strip produces an elating restoring torque. The coil rotates until the elastic restoring torque due to which the strip does not equalize and cancel the deflecting magnetic torque and then it attains equilibrium and stops rotating further.



Since the deflection torque on a current carrying rectangular torque is given by

$$\boxed{\text{Deflection torque} = BINA \cos \alpha}$$

Where

B = Strength of the magnetic field.

I = Current in the coil.

A = Area of the coil.

N = Number of turns in the coil.

θ = The angle of deflection of the coil.

The restoring elastic torque is proportional to the angle of twist of the suspension strip and according to Hook's Law

$$\boxed{\text{Elastic restoring torque} = C\theta}$$

Where

θ = Angle of twist.

C = Elasticity constant of spring.

Under equilibrium condition:

Deflecting magnetic torque = Elastic restoring torque

$$BINA \cos\alpha = C\theta$$

$$I = \frac{C\theta}{BNA \cos\alpha} \text{ --- (1)}$$

In radial magnetic field the plane of the coil is always parallel to the field irrespective of the position of the coil rotates. So the angle between the plane of the coil and direction of the field is always zero i.e. $\alpha = 0^\circ$ and $\cos(0^\circ) = 1$

The equation (1) therefore becomes:

$$I = \frac{C\theta}{BNA}$$

Since elastic constant " C ", magnetic field " B ", area " A " and number of turns " N " are constant for a galvanometer, therefore,

$$I = (\text{constant})\theta$$

$$\boxed{I \propto \theta}$$

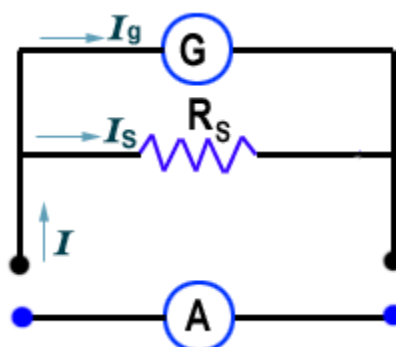
Therefore the current passing through the coil is directly proportional to the angle of twist of the suspension (or deflection).

Ammeter:

An ammeter is an electric instrument, which is used for measuring electric current.

Conversion of Galvanometer into Ammeter:

A galvanometer can be converted into ammeter to measure large amount of current. For this purpose the small resistance called a shunt is connected in parallel with the galvanometer. This resistance allows the large excess current through itself while a fraction of the current passes through the galvanometer coil. The scale of the instrument is so calibrated that it can measure the total input current directly.



Derivation:

Consider a galvanometer "G" whose resistance is R_g and which gives full scale deflection when current " I_g " flows through it. To convert the galvanometer into an ammeter which can measure a maximum current " I ", a shunt resistance " R_s " of small resistance is connected in parallel with the galvanometer such that the current " I_s " must flow through the shunt resistance " R_s ".

The potential difference " V_g " across the galvanometer is given by

$$V_g = I_g R_g$$

The potential difference " V_s " across the shunt is given by

$$V_s = I_s R_s$$

$$I = I_g + I_s$$

$$I_s = I - I_g$$

$$V_s = (I - I_g) R_s$$

$I_s = I - I_g =$ current through the shunt.

Since R_g and R_s are in parallel combination, therefore the potential difference across these resistances will be same

$$V_s = V_g$$

$$(I - I_g) R_s = I_g R_g$$

$$R_s = \frac{I_g}{I - I_g} R_g$$

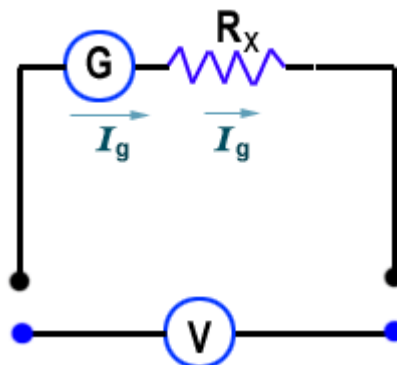
Ammeter is always connected in series with the circuit.

Voltmeter:

Voltmeter is an electric instrument which is used for measuring potential difference.

Conversion of Galvanometer into Voltmeter:

A galvanometer can be converted into voltmeter by connecting a high value resistance in series combination with the galvanometer. This resistance allows the large potential difference drops across it while a fraction of total potential difference drops across galvanometer. The scale of the instrument is so calibrated that it can measure the total voltage directly.



Derivation:

Consider a galvanometer "G" whose resistance is " R_g " and which deflects full scale for the current " I_g ". To convert this galvanometer into a voltmeter measuring a potential difference up to " V " volts a high value resistance " R_x " must be connected in series with it. Let " V " is the total potential difference applied across the two ends of the combination. Since the total potential difference in series combination is the sum of the individual potential differences

$$V = V_g + V_x$$

Since we know that the current is same in series combination therefore the current " I_g " must flow through the series combination.

$$V = I_g R_g + I_g R_x$$

$$V = I_g (R_g + R_x)$$

$$V - I_g R_g = I_g R_x$$

$$R_x = \frac{V - I_g R_g}{I_g}$$

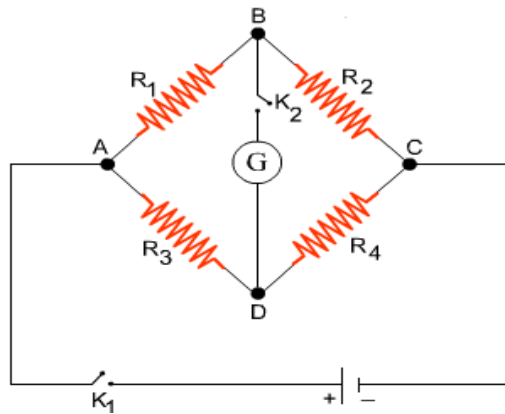
$$R_x = \frac{V}{I_g} - \frac{I_g R_g}{I_g}$$

$$R_x = \frac{V}{I_g} - R_g$$

Voltmeter is always connected in parallel with the circuit.

Wheat Stone Bridge:

If four resistances R_1 , R_2 , R_3 and R_4 are connected end to end in order to form a closed mesh ABCDA and between one pair of opposite junctions. A and C cell is connected through a key K_1 while between the pair of opposite junctions B and D a sensitive galvanometer "G" is connected through another key K_2 . The circuit so formed is called a "Wheatstone-bridge".



Working:

In the above bridge if the key is closed first, some current flows through the cell and the resistance R_1 , R_2 , R_3 and R_4 . If the key K_2 is also closed the current will usually be found to flow through the galvanometer indicated by its deflection. However if the resistance R_1 , R_2 , R_3 and R_4 . (or at least one of them) are adjusted, a condition can always be attached in which the galvanometer show no deflection at all i.e. no current passes through it. Then the potential difference between B & D must be zero i.e. B & D must be at the same potential. This implies that

R_1 & R_2 are connected in series.

Reason: (only one path for the flow of current)

R_3 & R_4 are connected in series.

R_1 & R_3 are connected in parallel.

Reason: (two paths for the flow of current)

R_2 & R_4 are connected in parallel.

Let current I_1 flows through R_1 & R_2 and I_2 through R_3 & R_4 .

When bridge is balanced,

Electric potential of point 'B' = Electric potential of point 'D'

Therefore,

$$V_{AB} = V_{AD}$$

Since according to Ohm's law, $V = IR$

$$I_1 R_1 = I_2 R_3 \text{ --- (1)}$$

Similarly,

$$V_{BC} = V_{CD}$$

$$I_1 R_2 = I_2 R_4 \text{ --- (2)}$$

Dividing equation (1) by equation (2), we get

$$\frac{I_1 R_1}{I_1 R_2} = \frac{I_2 R_3}{I_2 R_4}$$

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Under balanced condition if any three resistances are known then the fourth can be found easily (i.e. whetstone principle).

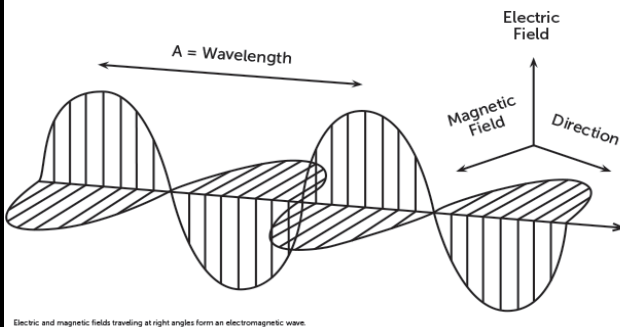
Sir M. Atiq

Chapter 16 "Electromagnetic Waves and Electronics" Compiled by: Sir M. Atiq

Electromagnetic Waves:

The moving electric field (arises from charge) and magnetic fields (arise from charge motion) combined together known as electromagnetic waves. OR

If an electric field and magnetic field is changing through any space or region, the two fields will propagate out of this region in the surrounding space. Such oscillating electric and magnetic field are known as electromagnetic waves.

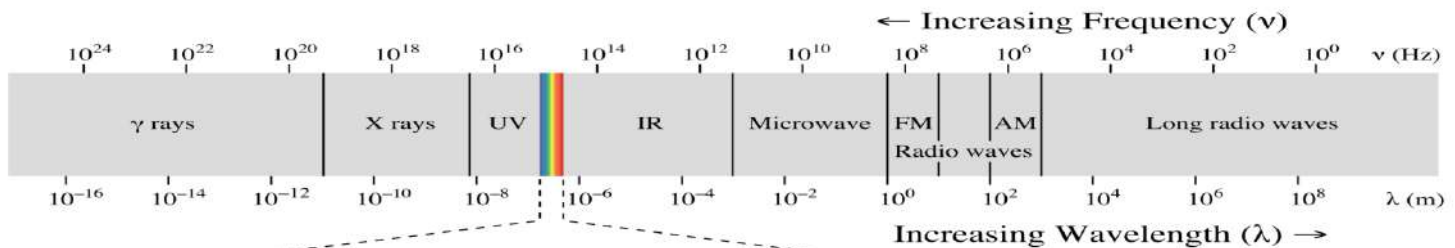


Properties:

In EM wave's electric field, magnetic field and direction of propagation are mutually perpendicular to each other.

Electromagnetic waves do not required medium to propagate. EM waves can pass through the vacuum.

EM waves are transverse waves in which the particle motion is perpendicular to the direction of propagation.



Uses and Applications:

1. EM waves are used in communication and navigation.
2. Ariel, receiver sets detects the radio waves of EM Spectrum for several purposes.
3. Microwave oven, photocell, light sensing diodes uses the EM waves.
4. Electromagnetic waves are also used in crystallography, radiography.
5. A famous experiment photoelectric effect also uses the EM waves.
6. Treatment of cancer, detecting the flaws in metals, radiation detectors (e.g. Geiger counter).

Electromagnetic Spectrum:

Spectrum formed by electromagnetic radiation is known as electromagnetic spectrum.

Waves in Electromagnetic spectrum

1. Radio waves (large range of wavelength)
2. Microwaves (wavelengths from 1mm to 300 mm)
3. Infrared waves (mean value is 10 micrometer and are also called heat waves)

4. Visible waves (wavelength from 400 nm to 700 nm and are visible to normal human eye)
5. Ultraviolet waves (60 nm to 300 nm having very high frequencies and very short wavelengths)
6. Gamma rays (very short wavelengths even less than 10^{-11} m and very high frequencies)
7. The speed of Electromagnetic waves is give by

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

$$c = 2.998 \times 10^8 \text{ms}^{-1} = 3 \times 10^8 \text{ms}^{-1}$$

Information Carried By Electromagnetic Waves:

In the field of telecommunication it is common practice to send information by wireless mean to do it electromagnetic waves are used.

Modulation:

The superimposing of a low energy EM signal on a high frequency signal so that information can be transmitted is called Modulation.

Modulating Signal:

The low frequency signal that contain message to be transmitted is called modulating signal.

Carrier Signal:

The high frequency signal on which modulating signal is super imposed is called carrier signal.

Modulated signal:

The signal obtain after the modulation is called modulated signal.

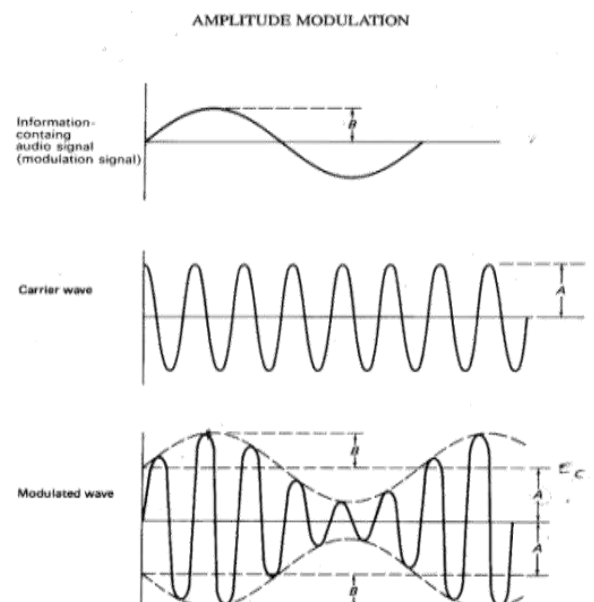
Types of Modulation:

There are following two (basic) types of (analog) modulation:

- 1) Amplitude Modulation.
- 2) Frequency Modulation.

Amplitude Modulation:

A type of modulation, in which amplitude of a carrier wave is modulated by an impose signal, usually at audio frequency. The variation in the amplitude of the carrier signal is proportional to



the variation of the modulating signal while the frequency of the carrier remains constant.

Percentage Modulation:

In amplitude modulation it is common to calculate percentage of modulation (M).

Mathematically percentage modulation (A.M) can be expressed as

$$M = \frac{\text{Amplitude of modulating signal (volts)}}{\text{Amplitude of carrier signal (volts)}} \times 100$$

$$M = m_a \times 100$$

Here, m_a is called Modulation index and it may be defined as

"It is the ratio peak amplitude of modulating signal to the peak amplitude of carrier signal."

For modulation for error free purposes m_a should be less than or equal to 1.

Application:

Amplitude modulation is used in variety of telecommunication application such as

1. Transmitting video signals in television transmission.
2. Armature AM radio.
3. Walky Talkies.

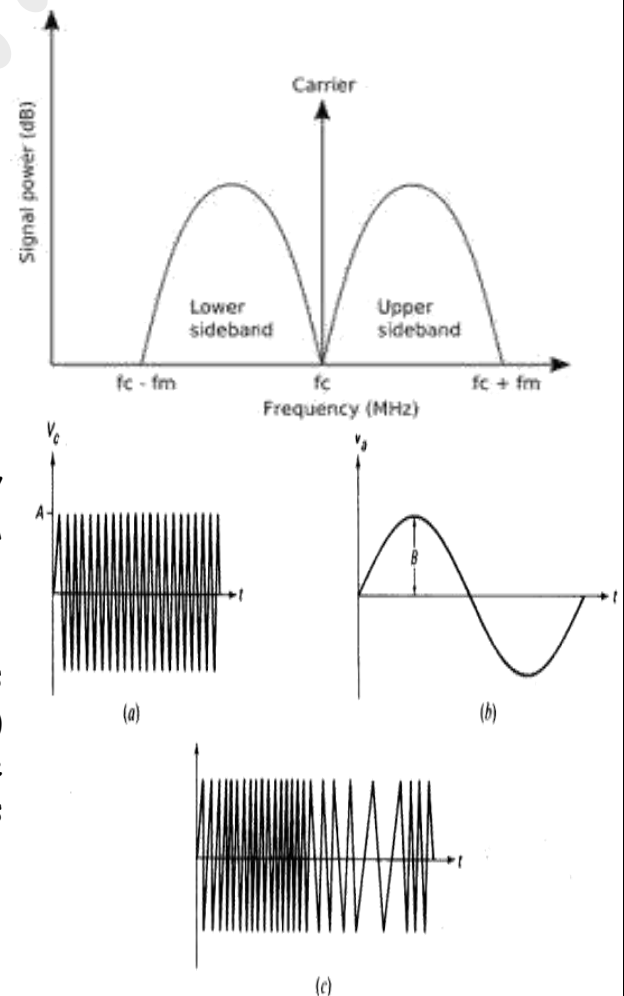
Sidebands:

In radio communications sidebands is a band of frequencies higher than or lower than the carrier frequency, containing power as a result of the modulation process.

Amplitude modulation of a carrier wave normally results in two mirror-image sidebands. The signal components above the carrier frequency constitute the upper sideband (USB), and those below the carrier frequency constitute the lower sideband (LSB). In conventional A.M transmission

Frequency Modulation:

A type of modulation in which in which a carrier wave is made to carry the information in a signal (audio or visual) by fluctuation in the frequency of the carrier wave. The variation in the frequency of the carrier signal is



proportional to the frequency of the modulating signal while the amplitude of the carrier remains constant.

Application:

Frequency modulation is used in variety of telecommunication application such as

- Transmitting Audio signals in television transmission.
- Commercial FM radio.
- Cellular communication.

Reception of Electromagnetic signals:

Electromagnetic waves carrying the modulated carrier signal are received at the receiving end with help of an antenna and then Demodulated to obtain the message signal.

Demodulation:

Signal processing process use to extract out the message from the modulated carrier signal is called demodulation.

Demodulator:

The circuit used to demodulate the Modulated carrier signal is called Demodulator.

Construction:

A simple Demodulator circuit for A. M waves is shown in figure. It consists of a Rectifier diode, capacitor with High capacitance and a high value resistance.

Working:

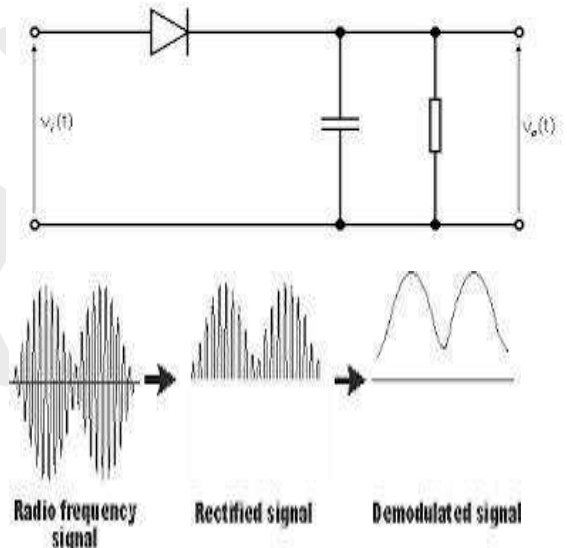
Working of the demodulator circuit is as follows:

1. Amplitude modulated electromagnetic wave is received at antenna.
2. the voltage signal induced into the antenna by EM radiation is fed to the rectifier diode
3. The diode acts as a non linear device and produces a totally positive signal at its output.
4. This varying positive voltage is fed to a capacitor which acts as a low pass filter and joins peak voltage of each cycle with peak of next cycle and hence produced message signal containing information.

Solid State Physics:

Solid state physics is the branch of physics that deals with the structure and properties of solids.

Two types of solids are 1. Crystalline & 2. Amorphous



Crystalline:

Crystalline state of solids is characterized by regular or periodic arrangement of atoms or molecules. Most of the solids are crystalline in nature. E.g. diamond, sodium chloride

Non Crystalline (Amorphous):

Non crystalline state of solids is characterized by disordered arrangement of atoms or molecules. E.g. is glass, carbon

Properties of Crystal:

1. A crystal has sharp melting point.
2. It is incompressible.
3. It has definite shape.
4. It has characteristic latent heat of fusion.
5. It consists of identical blocks. These identical blocks are the smallest volume of the crystal and called unit cell.

Valance Band:

The outer most shell of the electrons in every element is called valance band. Electrons in valance band are not free to the applied electric field. Electrons in the valance band are bounded to the atom.

Conducting Band:

Next to the valance band is the conduction band. Electrons in this band are free to the applied electric field.

Forbidden Band:

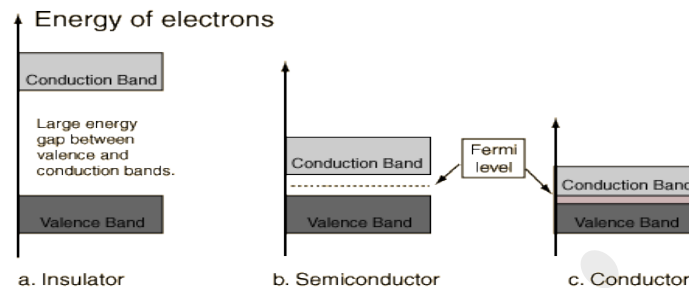
The band which lies between the conduction band and valance band is called forbidden band. Electrons cannot lie in this band it is forbidden.

As far as concerning the band theory the substances are dividing into three categories.

1. Conductors
2. Semi-Conductors
3. Insulators

Conductors:

There is no forbidden band in the conductors and the valance band and the conduction band are overlap. Electrons from the valance band can easily jumps to the valance band when it is subjected to the electric field.



The resistance of conductors increases with the increase in temperature. E.g. are copper, gold

Semi-Conductors:

The forbidden band of very small width exists between the valance band and conduction band of semi-conductors. E.g. are silicon and germanium.

The resistance of semi-conductors decreases with the increase in temperature.

Insulators:

A very wide energy gap exists between the valance band and conduction band of the insulator. Electrons in the valance band cannot jumps to the conduction band when electric field is applied to the insulator. E.g. is wood, rubber, plastic.

The resistance remains same with the increase in temperature.

Intrinsic Semiconductors:

1. The pure semiconductors are called "intrinsic semi-conductors".
2. These intrinsic semi-conductors located in the group 4 of the periodic table. E.g. are silicon and germanium.
3. They have 4 valance electrons.
4. The number of charge carriers is determined by the properties of the material itself.

Extrinsic Semi-Conductors:

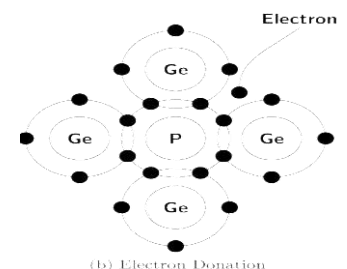
1. Doped semiconductors or impure semiconductors are called "extrinsic semi-conductors".
2. Extrinsic semiconductor formed when a small amount of impurity from group 3 or group 5 of the periodic table is added to the silicon or germanium (intrinsic semiconductors).
3. There are two types of extrinsic semiconductors. N-type & P-type
4. The number of charge carriers is determined by the doping level.

Doping:

The process of addition of impurity to a pure semiconductor is called doping.

N-type Semiconductor:

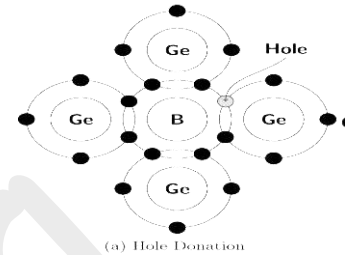
1. When extremely small amount of pentavalent substance (e.g. arsenic (As) or phosphors (P) located in group 5 of the periodic table) are added to the germanium or silicon crystal (group 4 element), "n-type semiconductor" is formed.
2. The process of doping in this case is called donor doping.
3. Donor doping donate electrons. As excess electrons are achieved in each covalent bonding.
4. Majority charge carriers are electrons while minority charge carriers are holes.



5. Negative charge carriers are available in N-type semiconductor.

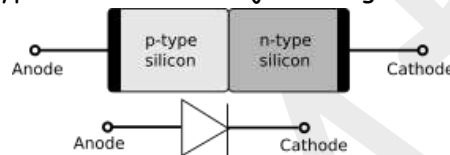
P-type Semiconductor:

1. When extremely small amount of trivalent substance (e.g. boron (B) or aluminum (Al) located in the group 3 of the periodic table) are added to the germanium or silicon crystal (group 4 elements), "p-type semiconductor" is formed.
2. The process of doping in this case is called acceptor doping.
3. Acceptor doping accept electrons. As excess hole are created in each covalent bonding.
4. Majority charge carriers are holes while minority charge carriers are electrons.
5. Positive charge carriers are available in P-type semiconductor.



PN-Junction Diode

When both p-type semiconductor and n-type semiconductors joined together then PN-junction is formed.



It is also called junction diode, PN-junction diode or simply diode. OR

If a crystal of germanium or silicon is doped in such a way that half is doped with a trivalent impurity and the other half is doped with a pentavalent impurity, then a PN-junction is formed.

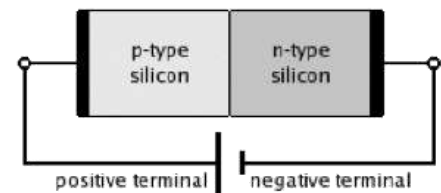
The PN-junction is extremely thin of the order of micrometer.

Biasing

The application of electric potential across a semiconductor diode is called biasing.

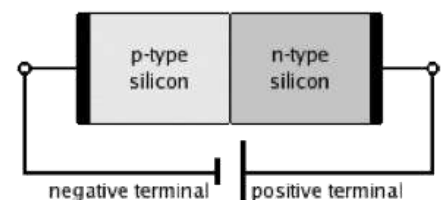
Forward Biasing:

1. In forward biasing positive terminal of a battery is attached with the p type layer of PN-junction and negative terminal of a battery is attached with the n type layer of PN-junction.
2. The applied forward potential establishes an electric field which reduces the height of the potential barrier at the junction.
3. The junction resistance is very low becomes almost zero.
4. Considerable amount of current flows due to the majority charge carriers.
5. Forward Biasing permits the flow of current through PN-junction diode.



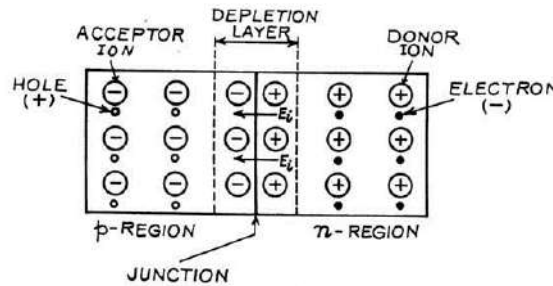
Reverse Biasing:

1. In Reverse biasing negative terminal of a battery is attached with the p type layer of PN-junction and positive terminal of a battery is attached with the n type layer of PN-junction.
2. The applied reverse potential establishes an electric barrier which increases the height of the potential barrier at the junction.
3. The junction resistance becomes very high.
4. A very small amount of current flows due to the minority charge carriers.
5. Reverse Biasing does not permit the flow of current.



Formation of the Potential barrier in PN-Junction:

In PN-Junction diode holes are majority charge carriers in p-side of the junction and electrons are majority charge carriers in the n-side of the junction. When diffusion of these electrons and holes take place across the junction the positive charge (due to the positive ions) developed at the n-side of the junction and negative charge (due to the negative ions) developed at the p-side of the junction. These newly formed positive and negative ions give an electric field which prevents the majority charge carriers flow across the junction is called "depletion region". Potential difference due to this electric field is setup at the junction which is called "potential barrier" or "junction barrier". This junction barrier is small in forward biasing and large in reverse biasing.

**Rectifier:**

The device which is used to convert alternating current into direct current is known as rectifier. In most application of the electronics diode is uses as a rectifier so called diode rectifier.

Types of Rectifier:

1. Half wave rectifier

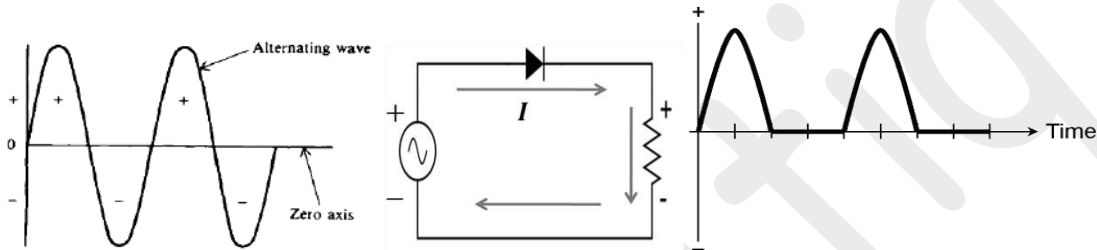
2. Full wave rectifier

Half wave rectification:

Such type of rectification in which unidirectional current flows in the output load resistance only for the positive half cycle of the input AC voltage is called half wave rectification.

Construction:

It consists of semiconductor diode (PN-Junction) D and resistance R which is connected in series. The output is taken across the load resistance R. In this circuit DC current is obtained only for the positive half cycle of the input AC current that's why it is called half wave rectifier.

**Working:**

During the positive half cycle of the input AC voltage the anode (p-side) of the diode is positive with respect to the cathode (n-side) of the diode so the diode becomes forward bias and permits the flow of current and positive pulse is obtained across the output resistance R. During the negative half cycle of the input AC cycle the anode (p-side) of the diode is negative with respect to the cathode (n-side) of the diode so the diode becomes reverse bias and does not permit the flow of current and so that zero volt is appear across the load resistance R. Unidirectional output current is obtained for every positive half cycle of the input AC voltage.

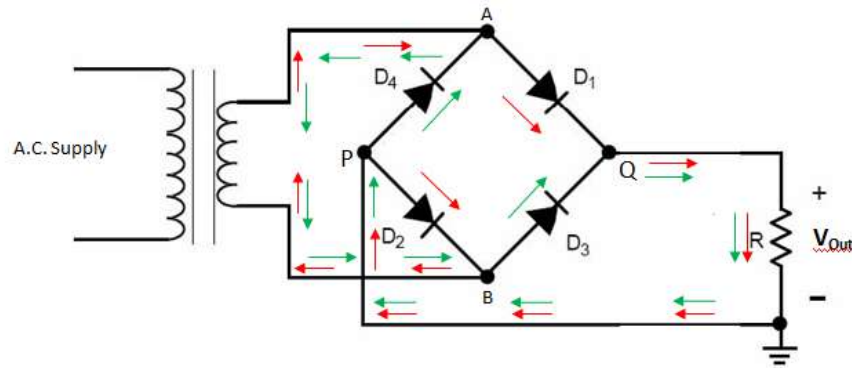
Full wave rectification:

Such type of rectification in which output current flows in the same direction for the positive and negative half cycle of the input AC voltage is called full wave rectification.

Construction:

It consists of 4 semiconductor diodes and load resistance R. Semiconductor diodes are connected in bridge type arrangement. The output is taken across the load resistance R.

In this circuit unidirectional DC current is obtained for both positive half and negative half cycle of the input AC voltage that's why it is called full wave rectifier.



Working

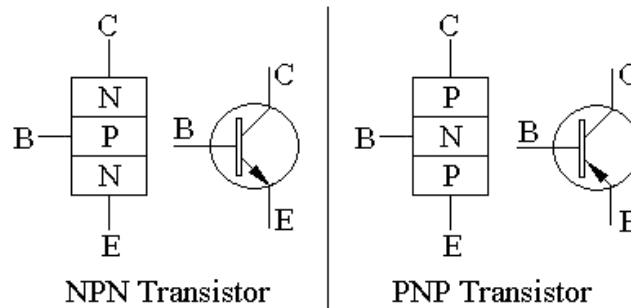
During the positive half cycle the terminal A of the bridge is positive with respect to the terminal B. The diodes D_1 and D_2 become forward biased and the current flows through the circuit and positive pulse appear on the load resistance. During the negative half cycle the terminal A of the bridge is negative with respect to the terminal B. The diodes D_3 and D_4 become forward biased and the current flows through the circuit and positive pulse appear on the output load resistance. Thus unidirectional pulsating DC output current is obtained for both positive and negative half of the input AC cycle.

Transistor

A transistor consists of two junctions. One junction acts as forward biased, so it transfer the charge while the second junction is reverse biased so it resists the flow of charge. OR

A transistor is a semiconductor which consists of a thin central layer of one type of semiconductor material sandwiched between two relatively thick pieces of the other types.

Transistor has three regions emitter, base and collector. Transistor is a semiconductor device which can amplify both current and voltage.

**Emitter:**

The emitter has greater concentration of impurity. It acts like a source of charges in the transistor. It is smaller than collector in size.

Base:

The central region is known as base. Generally the base is very thin of the order of 10^{-6} m as compared to emitter and collector.

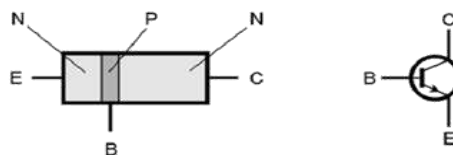
Collector:

The region to the right of the base is called collector. The collector has less concentration of impurity as compared to emitter. The collector is larger in size than emitter.

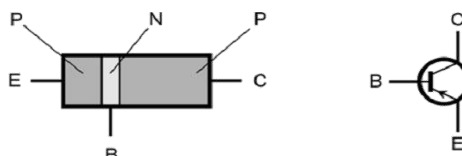
$$I_E = I_C + I_B$$

NPN Transistor:

In n-p-n transistor p-type material is sandwiched between two n-type materials. It is known as n-p-n transistor.

**PNP Transistor:**

In p-n-p transistor n-type material is sandwiched between two p-type materials. It is known as p-n-p transistor.



Advantages:

1. Transistors operate on low voltages so they do not require huge voltages.
2. They are exceedingly small and light.
3. They produce little heat.

Uses of Transistors:

1. Transistor can be used as an amplifier to amplify the input signal.
2. Transitory can be used as a switching device for ON/OFF.
3. Transistors are now widely using in radio, telecommunications, computers, television, telephones and cell phones for different purposes.

Limitations:

1. Power heating capability is limited to few watts.
2. Frequency response is limited.

AMPLIFIER:

Amplifier is a circuit or device that can raises the strength of a weak signal. The factor by which an amplifier increases the strength of a weak signal is called gain of an amplifier.

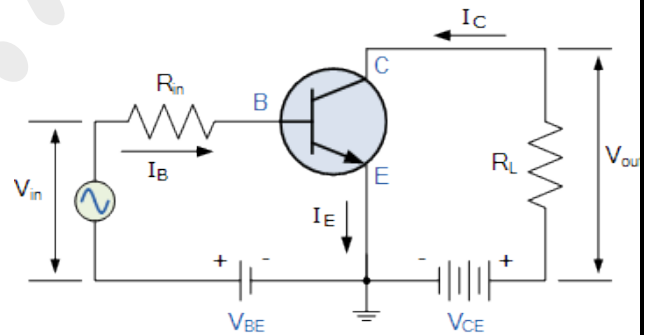
Transistor as an Amplifier:

A transistor can be used to amplify a weak signal. The transistor has following relation between base and collector current.

$$I_C = \alpha I_B$$

Where α is called the gain of the transistor.

Now if we apply a small voltage across the base emitter junction, this voltage allows a small base current to flow inside the base region resulting in large collector current to flow in collector. Now, if we connect a large resistance R_C at collector. Collector current will cause a large voltage to build across R_C . Hence, small voltage applied at BE junction is converted into large voltage across R_C .

**Effect of temperature on semiconductor:**

The semiconductor have negative temperature coefficient i.e. the resistance of semiconductor decreases with the increase in temperature for a specified range. As the temperature increases the impurities start to ionize and if sufficient thermal energy is acquired by them, electron hole pairs are created due to the bond breaking and conductivity increases.

Characteristics of EM waves:

Rectifier: (Text Book):

A device which converts alternating current voltage into pulsating direct current voltage is called a rectifier. The following two types of rectification can be obtained using a diode

1. Half wave rectification
2. Full wave rectification

A semiconductor diode, also known as crystal diodes, can be used for rectification purposes.

Half Wave Rectifier (Text Book):

In half wave rectification, the rectifier conducts current only during the positive half cycles of input ac voltage. The negative half cycles of a.c. supply are suppressed i.e. during negative half cycles, no current is conducted and hence no voltage appears across the load in the external circuit.

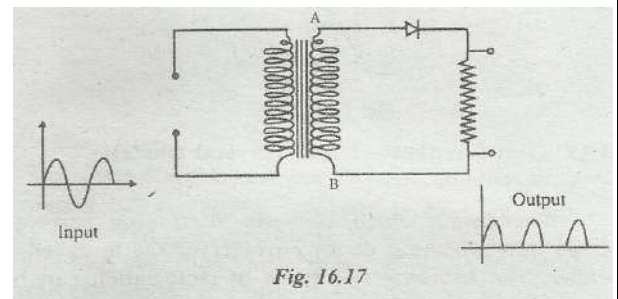
Construction:

Figure shows the circuit where a single diode acts as a half wave rectifier. The a.c. signal to be rectified is applied in series with the diode and load resistance R_L . The dc output is obtained across the load R_L . Generally a.c. supply is given through the transformer. The use of transformer has two advantages.

1. It allows us to step up or step down the ac input voltage.
2. Transformer isolate the rectifier circuit from power line and the reduces the risk of electric shock.

Working:

The AC voltage across the secondary winding AB changes polarities after every half cycle. During the positive half cycle of input a.c. voltage at A becomes positive with respect to end B. This makes the diode forward biased and hence it conducts current. During the negative half cycle A is negative with respect to the end B and this makes the diode reverse biased and it



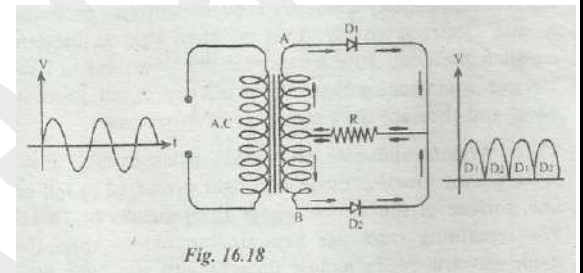
conducts no current, therefore the current flows through the diode during the positive half cycle of input a.c. voltage and it is blocked during the negative half cycles. In this way current flows through load always in the same direction. Hence dc output is obtained across the load resistor R_L .

Full Wave Rectifier (Text Book):

In full wave rectification, current flows through the load in the same direction for both half cycles of input a.c. voltage. This can be achieved with two diodes working alternately.

Construction:

The circuit employs two diodes D_1 and D_2 as shown in figure. A center tapped secondary winding AB is used with two diodes connected so that each uses one half cycle of input a.c. voltage. D_1 utilizes the a.c. voltage appearing across the upper half OA of secondary winding for rectification while diode D_2 uses the lower half winding OB.



Working:

During the positive half cycle of secondary voltage, the end A of secondary winding becomes positive and end B negative. This makes the diode D_1 forward biased and diode D_2 reverse biased. Therefore, diode D_1 conducts while diode D_2 does not. The conventional current flow is through diode D_1 , load resistor R_L and the upper half of the secondary winding, as shown in figure. During the negative half cycle, end A of the secondary winding becomes negative and end B positive. Therefore diode D_2 conducts while diode D_1 does not. The conventional current flows is through diode D_2 , load resistor R_L and lower half winding as shown in figure. Therefore current flows through load resistance for both half cycles of input a.c. voltage.

Full wave rectification:

Such type of rectification in which output current flows in the same direction for the positive and negative half cycle of the input AC voltage is called full wave rectification.

Construction:

It consists of 4 semiconductor diodes and load resistance R. Semiconductor diodes are connected in bridge type arrangement. The output is taken across the load resistance R.

In this circuit unidirectional DC current is obtained for both positive half and negative half cycle of the input AC voltage that's why it is called full wave rectifier.

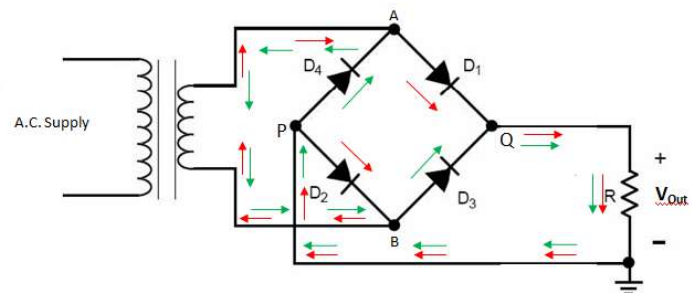
Working

During the positive half cycle the terminal A of the bridge is positive with respect to the terminal B.

The diodes D_1 and D_2 become forward biased and the current flows through the circuit and positive pulse appear on the load resistance. During the

negative half cycle the terminal A of the bridge is

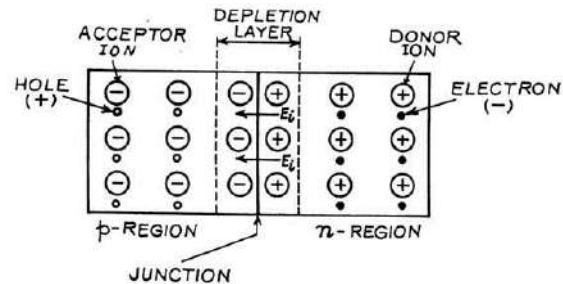
negative with respect to the terminal B. The diodes D_3 and D_4 become forward biased and the current flows



through the circuit and positive pulse appear on the output load resistance. Thus unidirectional pulsating DC output current is obtained for both positive and negative half of the input AC cycle.

Formation of the Potential barrier in PN-Junction:

In PN-Junction diode holes are majority charge carriers in p-side of the junction and electrons are majority charge carriers in the n-side of the junction. When diffusion of these electrons and holes take place across the junction the positive charge (due to the positive ions) developed at the n-side of the junction and negative charge (due to the negative ions) developed at the p-side of the junction. These newly formed positive and negative ions give an electric field which prevents the majority charge carriers flow across the junction is called "depletion region". Potential difference due to this electric field is setup at the junction which is called "potential barrier" or "junction barrier". This junction barrier is small in forward biasing and large in reverse biasing.



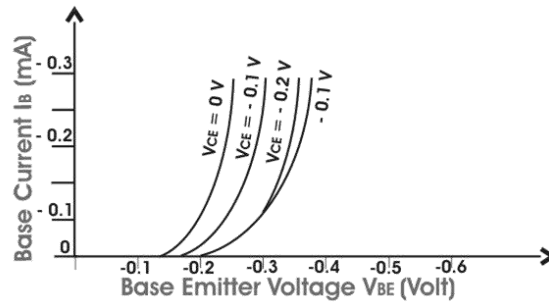
Characteristics Curves:

The characteristic contains two curves i.e. input characteristic and output characteristic.

Input Characteristics:

It gives the relationship between input voltage & input current

1. I_B (Base Current) is the input current, V_{BE} (Base - Emitter Voltage) is the input voltage for CE (Common Emitter) mode. So, the input characteristics for CE mode will be the relation between I_B and V_{BE} with V_{CE} as parameter.
2. The typical CE input characteristics are similar to that of a forward biased of p - n diode. But as V_{CB} increases the base width decreases
3. The characteristics curves is represented as,



Output Characteristics:

It gives the relationship between voltage & current in output circuit.

1. Output characteristics for CE mode is the curve or graph between collector current (I_C) and collector - emitter voltage (V_{CE}) when the base current I_B is the parameter.
2. CE transistor has also three regions named

1. Active region:

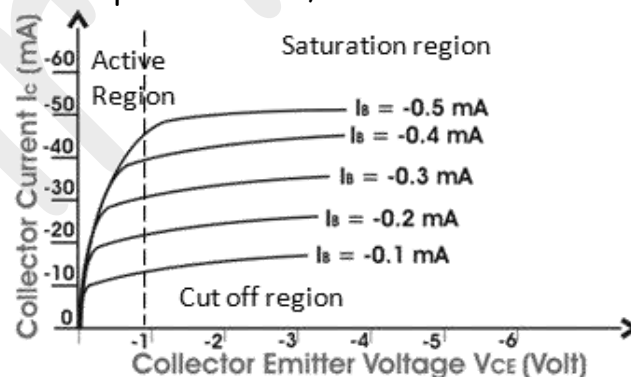
The active region has collector region reverse biased and the emitter junction forward biased.

2. Cut-off regions:

For cut-off region the emitter junction is slightly reverse biased and the collector current is not totally cut-off.

3. Saturation region:

For saturation region both the collector and the emitter junction are forward biased. The characteristics curves is represented as,



AMPLIFIER:

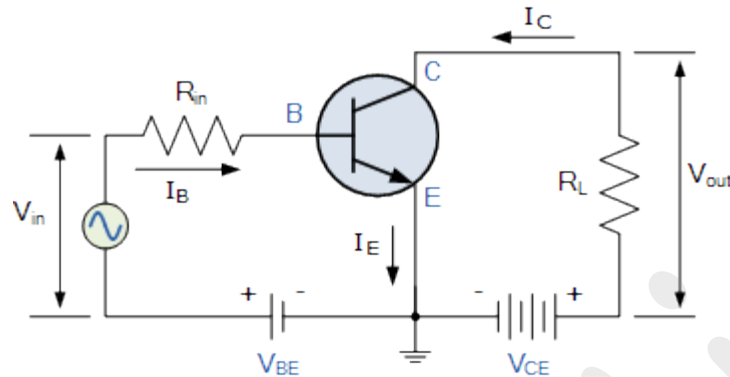
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Chapter 17 "The Advent of Modern Physics"

Relative Motion:

The rest position or the motion of an object is not same for different observers. Therefore we can't say whether an object is absolutely at rest or absolutely in motion. All motions are relative to a person or instrument observing it.

Examples:

1. The walls of the cabin of a moving train are stationary with respect to the passengers sitting inside it but are in motion to a person stationary on the ground.
2. If an observer sitting in a vehicle moving with a uniform speed throw an object vertically upward then the observer will see the object falls along the same vertical path but an observer at rest on the ground the path of the object will appear to be a curve.

Frame of Reference:

A frame of reference is any coordinate system relative to which measurement are taken.

Examples:

1. Position of a table in a room can be located relative to the walls of the room. The room is then the frame of reference.
2. The position of a spaceship can be described relative to the positions of the distant stars. A coordinate system based on these stars is then the frame of reference.

Inertial Frame of Reference:

An inertial frame of reference is a coordinate system in which the law of inertia is valid i.e. a body at rest remains at rest unless an unbalanced force produces acceleration in it. It is also called non-accelerated frame of reference.

Any frame of reference which is moving with uniform velocity relative to an inertial frame is also an inertial frame.

Example:

Earth can be considered as inertial frame of reference because of very small acceleration which arises due to spin and orbital rotation.

Special Theory of Relativity:

The special theory of relativity is concerned with the way in which observers who are in a state of motion describe physical phenomenon. The two postulates of special theory of relativity are given below:

1. The laws of physics are the same in all inertial frames.
2. The speed of light in vacuum is the same for all observers in uniform translational relative motion and is independent of the motion of the observer and the source.

Consequences of Special Theory of Relativity:

Time Dilation:

According to special theory of relativity, time is not absolute quantity. It depends upon the motion of the frame of reference.

Mathematically:

Suppose an observer is stationary in an inertial frame and measures the time interval " t_0 " between two events in his frame. This is known as proper time. If observer is moving with respect to the frame of events with speed " v " or if the frame of events is moving with respect to observer with a uniform speed " v ", the time interval measured by the observer would be " t " which is known as relativist time and is given by the following relation

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Conclusion:

Therefore time has dilated or stretched due to relative motion and we can say that clocks in moving frame run slowly.

Mass Variation:

According to special theory of relativity, mass of an object is a varying quantity and depends upon the speed of the object.

Mathematically:

An object whose mass when measured at rest is " m_0 " will have increased to be mass " m " when observed to be moving at speed v . They are related by

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Conclusion:

The increase in mass indicates the increase in inertia the object has at high speed. As " v " approaches " c ", it requires larger and larger force to change the speed of the object.

Length Contraction:

The length of an object in a frame of reference stationary with respect to the observer is not same when it is measured in the frame of reference moving with velocity with respect to the observer. If you are in motion relative to two points that are a fixed distance apart, the distance between the two points appear shorter than if you were at rest relative to them.

Mathematically:

If " l_0 " is the length of an object in a stationary frame of reference and " l " is the length of the same object when measured by the observer in the moving frame of reference with a relative velocity " v " then

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

Conclusion:

The distance become shorter along the direction of motion e.g. the distance from earth to a star measured by an observer in a moving spaceship would seem smaller than the distance measured by an observer on earth.

Energy - Mass Relation:

According to special theory of relativity mass and energy are different entities but are inter convertible.

Mathematically:

The total energy " E " and mass " m " of an object are related by expression

$$E = mc^2$$

Where m depend upon the speed of the object. At rest, the energy equivalent of an object's mass " m_0 " is called rest mass energy " E_0 ".

$$E_0 = m_0 c^2$$

As mc^2 is greater than $m_0 c^2$, the difference of energy is due to motion as such it represents the kinetic energy of the mass, hence

$$K.E = mc^2 - m_0 c^2$$

$$K.E = (m - m_0)c^2$$

Conclusion:

Any process that changed the mass by detectable amount would involve huge amount of energy.

Blackbody Radiations :

The radiations emitted by a body due to its temperature is called black body radiations. An object which can absorb all the radiations that fall on it is called black body.

Explanation:

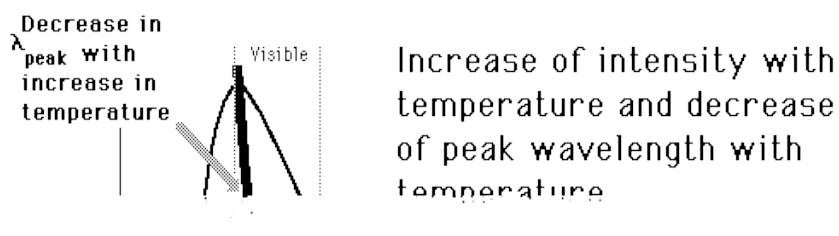
A hollow sphere of metal with a fine hole in it is called cavity as shown in figure. This cavity is approximately a black body because any radiation entering the hole is trapped by multiple reflections inside and very little of it is able to escape. Just as a black body is nearly a perfect absorber, so is the most effective emitter of radiation when heated.

Laws of Black Body Radiations:

Wien's Displacement law:

Statement:

"Wavelength at which maximum energy of radiation occurs is shifted towards the shorter wave length with the rise of temperature. "



Mathematically:

The wavelength for maximum radiation λ_{max} is related to the absolute temperature T of the black body by following expression

$$\lambda_{max} \times T = constant$$

The Wien's law is in agreement with the common observation that is a white hot furnace is hotter than one which is red hot.

Stefan's Boltzmann's law:

Statement:

"The total energy radiated per second per unit surface area is proportional to the fourth power of absolute temperature."

Mathematically:

$$E \propto T^4$$

$$E = \sigma T^4$$

$\sigma = \text{Stefan's boltzman' constant}$

Rayleigh and Jeans law:

Statement:

"Energy associated with a particular wavelength is inversely proportional to the fourth power of wavelength."

Mathematically:

$$E = \frac{\text{constant}}{\lambda^4}$$

Plank's law (Quantum Theory):

Statement:

"Radiant energy comes out in discrete amount or quanta of energy . The energy content of each quantum was directly proportional to the frequency ν ."

Mathematically:

$$E \propto \nu$$

$$E = h\nu$$

$$E = \frac{hc}{\lambda} \quad [\text{since } \nu = \frac{c}{\lambda}]$$

The constant, h is called plank's constant and the values of h was determined to be 6.634×10^{-34} J-s. On the basis of assumption that energy could only be emitted or absorbed by atomic oscillator in discrete quanta, the Plank's law would be

$$E = nh\nu$$

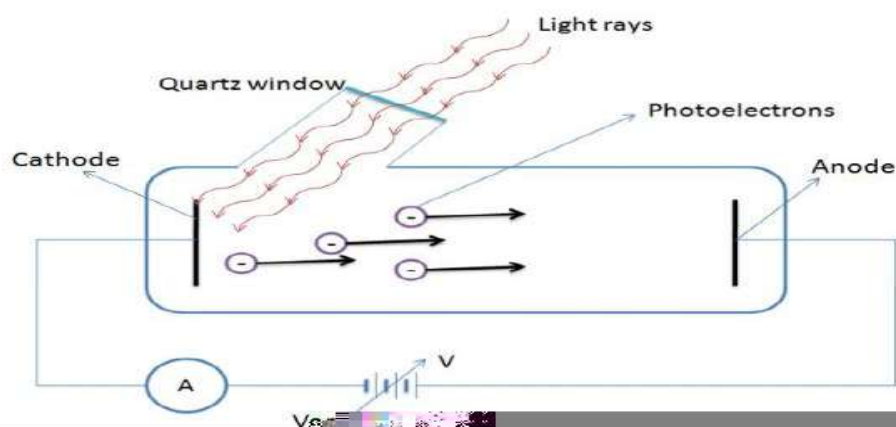
Where $n = 0, 1, 2, 3$ and so on.

Photoelectric Effect:

Hertz in 1887 discovered that when ultraviolet light falls on certain metals, electrons are emitted. The phenomenon in which certain metals emit electrons when exposed to high frequency light is called photoelectric current.

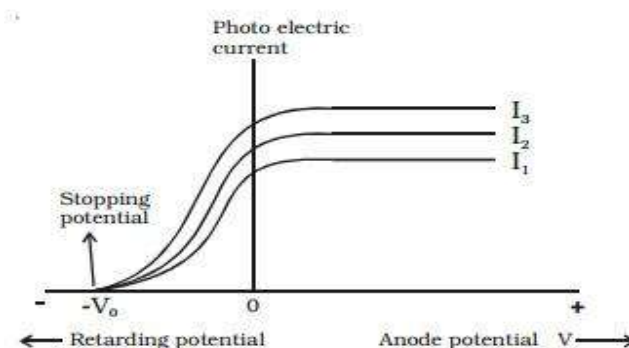
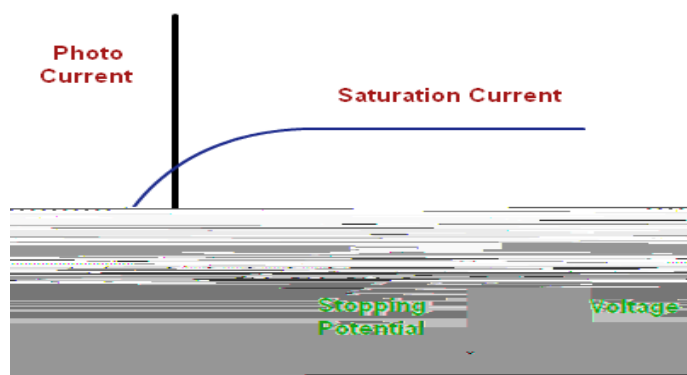
Explanation:

An evacuated glass tube contains two electrodes. The electrode connected to the positive terminal of the battery is known as anode. The electrode connected to the negative terminal is known as cathode. When monochromatic light is allowed to shine on the cathode, it begins to emit electrons. These photoelectrons are attracted by the positive terminal anode and the resulting current is measured by an ammeter. The current stops when light is cut off.

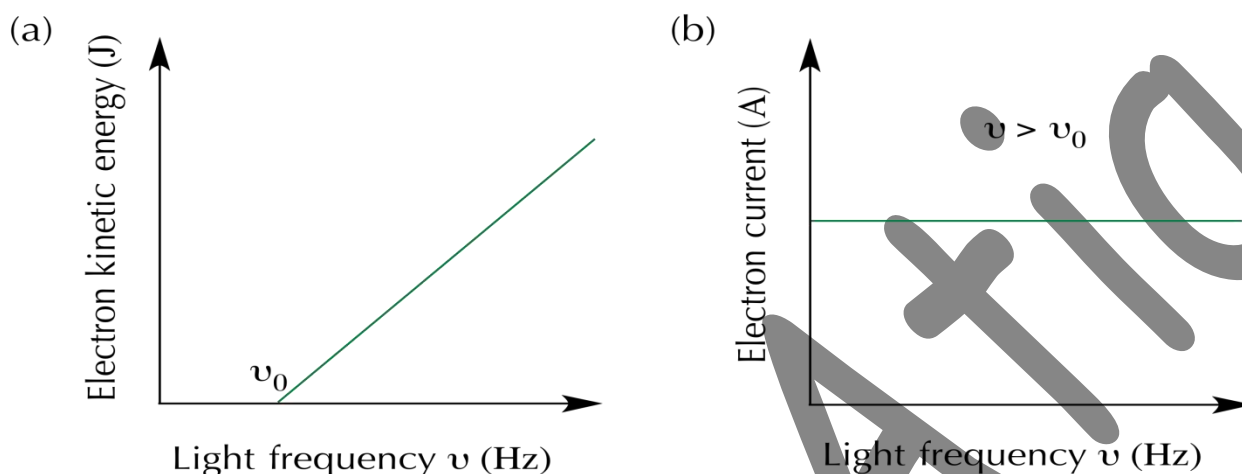


Experimental results of Photoelectric Effect:

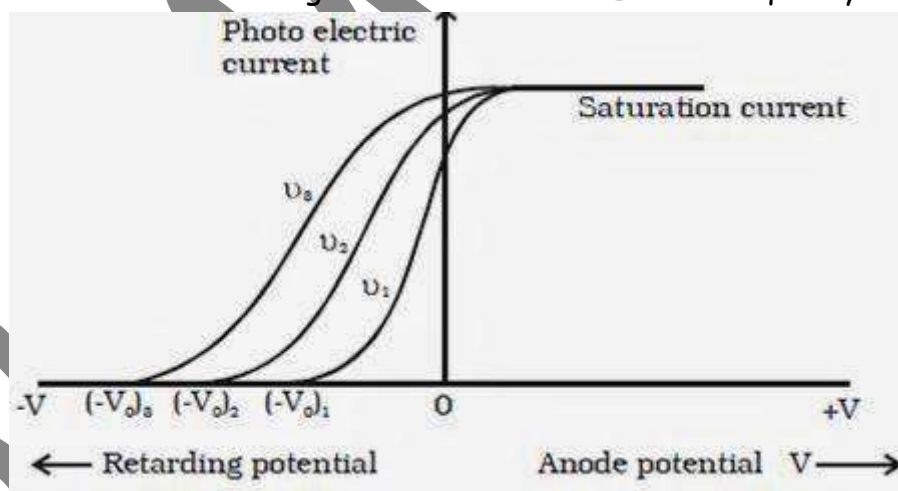
1. Increasing the intensity of the source of light, increases the number of photoelectrons i.e. electric current increases. It follows that there is a saturation current for different intensities.



2. For each substance there is a certain frequency called the threshold frequency " ν_0 " is a minimum frequency below which no electrons escape the metal surface. If the frequency increases beyond the threshold frequency the photo electric current remains constant.



3. The higher the frequency of the incident light, the greater the kinetic energy of photo electrons. The saturation current depends upon intensity and not on the frequency. However, the stopping potential becomes more negative with the increase in frequency.



Stopping Potential:

If the connection of the battery is reversed then the negative potential difference at which the photoelectric current becomes zero is called stopping potential. At this potential even the electrons of maximum energy are not able to reach the collector (anode plate).

Mathematically:

$$\text{Kinetic energy} = \text{Electric potential energy}$$

$$\frac{1}{2}mv_{max}^2 = V_0e$$

v_{max} = maximum energy of electrons

e = charge on electron

V_0 = stopping potential

Einstein Explanation of Photoelectric Effect on the basis of Quantum Theory:

An electron absorbs a photon of energy $h\nu$ and electron may lose some of this energy before leaving the metal surface and is ejected with a kinetic energy less than $h\nu$. The electron use some of its energy to overcome the force of attraction which holds the electrons within the metal. The energy required to overcome this binding force is called the work function of the particular metal and is denoted by ϕ which is a constant of the metal. Hence the Einstein's equation for photoelectric effect will be written as:

$$K.E_{max} = \frac{1}{2}mv_{max}^2 = h\nu - \phi$$

In this equation ϕ is a work function which directly depends on the threshold frequency ν_0

$$\phi = h\nu_0$$

Therefore,

$$\frac{1}{2}mv_{max}^2 = h\nu - h\nu_0$$

Since we know that the frequency and wavelength are related by

$$\nu = \frac{c}{\lambda}$$

Therefore,

$$\frac{1}{2}mv_{max}^2 = h\frac{c}{\lambda} - h\frac{c}{\lambda_0}$$

$$\frac{1}{2}mv_{max}^2 = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

Since

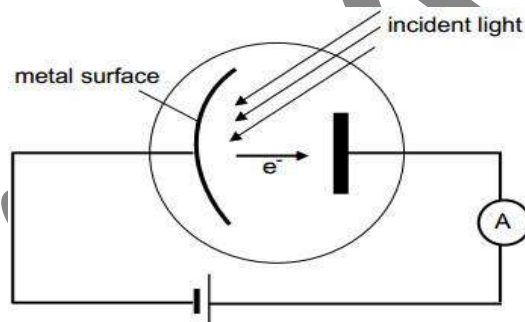
$$\frac{1}{2}mv_{max}^2 = V_0e$$

Therefore Einstein's photoelectric equation can also be written as

$$V_0e = hc \left(\frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$$

Photocell:

The photocell or phototube consists of a evacuated glass tube fitted with an anode plate and a concave metallic cathode of a appropriate surface. The material of the cathode can be chosen to respond to the frequency range over which the photocell operates. The response can be made proportional to the intensity of light source.



Applications of photocell:

A photocell can be used in any situation where beam of light falling on a cell is interrupted or broken.

1. To count vehicles passing a road or items on a conveyer belt.
2. To open doors automatically in a building.
3. To operate burglar alarms.

Photoconductors:

Photoconductors are materials in which an internal photoelectric effect may liberate free charge carriers in a material that is otherwise an insulator, and thereby increase its electrical conductivity by as much as 10000 times when it is illuminated by a light source. Such materials are called photoconductors.

Applications:

Photoconductive cells may be used in

1. Detection and measurements of infra-red radiation where the wavelength is of the order of 10^{-6}m .
2. As relays for switching on artificial lighting, such as street lights.

Photovoltaic cell:

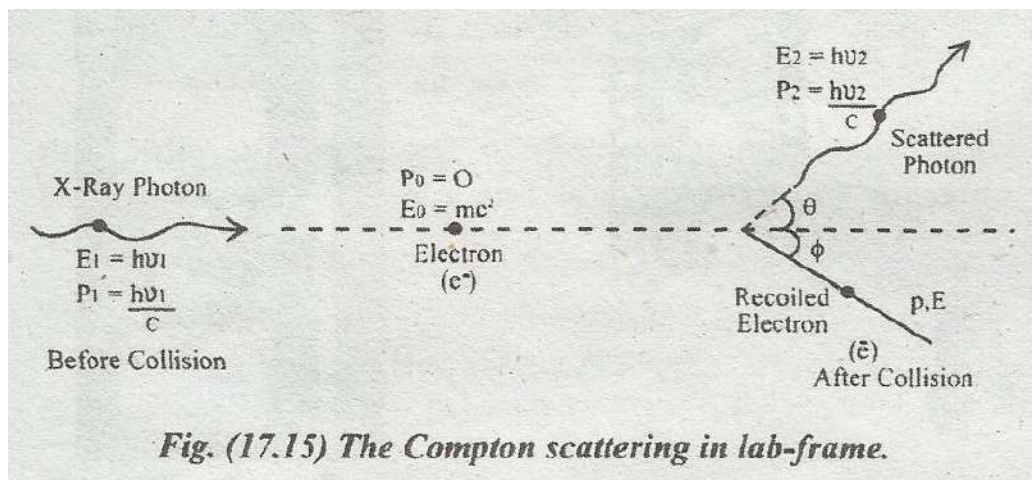
A photovoltaic cell which may consist of a sandwich of copper oxide and a thin film (layer) of translucent gold. An emf capable of giving a current of 1mA can be generated when the film is illuminated. Such cells need no source of e.m.f.

Compton's Effect:

When x-rays are scattered due to interaction with a light body such as an electron, the scattered rays exhibit lower frequencies (i.e. higher wavelengths) than the incident radiations. This phenomenon is called Compton's Effect.

Explanation:

The scattering is treated to be a two body collision between a photon and an electron. Compton's effect provided a solid support for photon theory of light. The figure below shows the event of scattering "before" and "after" the collision. The electron is treated at rest because for high energy photons, the initial motion of the electrons may be neglected.



Derivation for Compton's Shift:

The photon approaches towards the electron with a frequency ν_1 and is scattered at an angle θ with a lower frequency ν_2 . The photon energies before and after collision are $h\nu_1$ and $h\nu_2$. Whereas the corresponding momentum are $\frac{h\nu_1}{c}$ and $\frac{h\nu_2}{c}$. The energy and momentum of the recoiled electron are E and P respectively.

A/c to law of conservation of linear momentum

Total momentum before collision = Total momentum after collision

Along x-direction

$$0 + \frac{h\nu_1}{c} = p\cos\phi + \frac{h\nu_2}{c}\cos\theta$$

$$\frac{h\nu_1}{c} = p\cos\varphi + \frac{h\nu_2}{c}\cos\theta \text{ --- (1)}$$

Along y-direction

$$0 + 0 = \frac{h\nu_2}{c}\sin\theta - p\sin\varphi$$

$$0 = \frac{h\nu_2}{c}\sin\theta - p\sin\varphi \text{ --- (2)}$$

A/c to law of conservation of energy

Total energy before collision = Total energy after collision

$$m_0c^2 + h\nu_1 = E + h\nu_2 \text{ --- (3)}$$

To obtain an expression for the final frequency ν_2 as a function of initial frequency ν_1 and scattering angle θ we have to eliminate p and E from the above three equations. After solving these three equations, we get

$$\frac{1}{\nu_2} - \frac{1}{\nu_1} = \frac{h}{m_0c^2} (1 - \cos\theta)$$

since we know that, $\nu = \frac{c}{\lambda}$, therefore we get

$$\frac{\frac{1}{c}}{\lambda_2} - \frac{\frac{1}{c}}{\lambda_1} = \frac{h}{m_0c^2} (1 - \cos\theta)$$

$$\frac{\lambda_2}{c} - \frac{\lambda_1}{c} = \frac{h}{m_0c^2} (1 - \cos\theta)$$

$$\lambda_2 - \lambda_1 = \frac{h}{m_0c} (1 - \cos\theta)$$

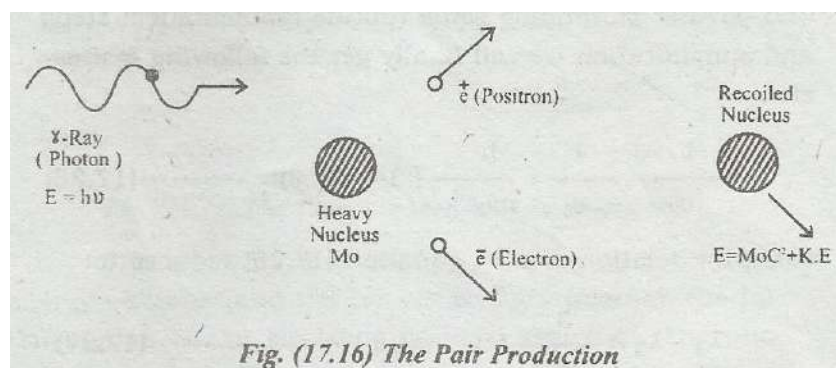
The above equation is known as the Compton's formula for the increase in wavelength for the scattered photon. In the above formula $\Delta\lambda = \lambda_2 - \lambda_1$ is called the Compton's shift and $\lambda_c = \frac{h}{m_0c}$ is called the Compton's wavelength.

Pair Production or Materialization of Energy:

The phenomenon in which photon in the vicinity of a nucleus disappear with the production of an electron-positron pair is called pair production or materialization of energy.

Explanation:

The pair production requires that the two particles created by the interaction of photon with matter must have equal and opposite charges and the photon from which the pair was produced must have an energy at least $2m_0c^2$.



The positron has been identified to be identical with an electron in mass and carries an equal positive charge and is called the anti-particle of electron. Due to large mass of the heavy nucleus the recoiled kinetic energy of the nucleus is negligible as compared with the kinetic energies of the pair. A/C to the law of conservation of energy

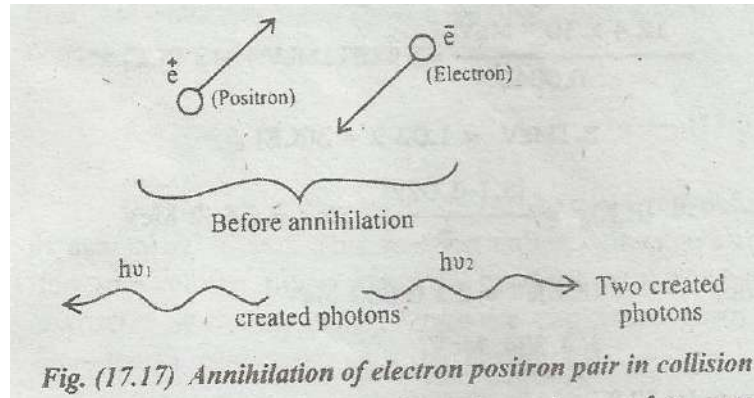
$$h\nu = 2m_0c^2 + K.E_{e^-} + K.E_{e^+}$$

Annihilation of Matter:

The destruction or annihilation of electron-positron pair with the creation of at least two or more photons is called annihilation of matter.

Explanation:

In annihilation of matter a particle and its anti particle come close enough to be converted completely into radiation energy of the two photons moving in opposite direction conserving the total momentum of the creation and annihilation process. Each photon will have energy equal to rest mass energy m_0c^2 of an electron.



Since the rest mass energy of electron-positron pair is 1.02 Mev, each photon created in the annihilation process will have an energy of 0.51 MeV. The energy conservation equation for the process will be

$$m_0c^2_{e^-} + K.E_{e^-} + m_0c^2_{e^+} + K.E_{e^+} = 2h\nu$$

De-Broglie Hypothesis:

"If light (electromagnetic radiation) can have particle behavior, then material particles, such as electron and proton etc. can also behave in a wave like manner."

Thus a particle like electron can possess a momentum given by

$$p = mv = \frac{h}{\lambda}$$

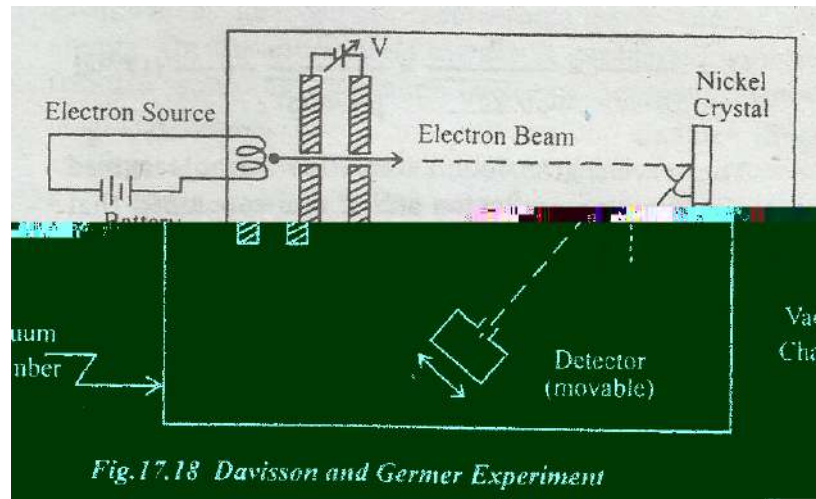
Where m is the mass of the particle. This relation called deBroglie's relation has related the electron (particle) and the wave character of a frequency. Therefore we can write the wavelength of a particle

$$\lambda = \frac{h}{mv}$$

Davisson and Germer Experiment:

Introduction:

The theoretical prediction of deBroglie's hypothesis was experimentally confirmed by Davisson and Germer. They were investigating the scattering of an electron beam by the metallic crystal of Nickel. The experiment setup is shown below



Construction and Working:

a beam of electron accelerated through the potential V were allowed to strike the nickel crystal. Measurements were made to count the number of electrons scattered by the crystal. It was observed that electrons reflected very strongly at certain angles only and not at other direction. They concluded that electron behaves as waves of wavelength λ . They calculated the wavelength of electron from the known accelerating potential V by applying the relation for kinetic energy of the electron i.e.

$$\frac{1}{2}mv^2 = eV$$

$$v = \sqrt{\frac{2eV}{m}}$$

Since we know that the deBroglie's relation

$$\lambda = \frac{h}{mv}$$

Substituting the value of v in deBroglie's relation, we get

$$\lambda = \frac{h}{m\sqrt{\frac{2eV}{m}}}$$

$$\lambda = \frac{h}{\sqrt{2emV}}$$

Heisenberg's Principle of Uncertainty:

"It is in principle impossible to measure with accuracy both position and momentum of a particle simultaneously."

Mathematically:

If we denote Δx and Δp_x the corresponding uncertainties in position: then according to Heisenberg's principle of uncertainty, the product of these uncertainties must always satisfy the inequality:

$$(\Delta x) (\Delta p_x) \geq \frac{h}{2\pi} = 1.05 \times 10^{-34} \text{ J-s}$$

Similarly if ΔE and Δt represent the uncertainties in the energy and time respectively then we have an expression

$$(\Delta E) (\Delta t) \geq \frac{h}{2\pi} = 1.05 \times 10^{-34} \text{ J-s}$$

Chapter# 18 "ATOMIC SPECTRA"

Atomic Spectroscopy:

The branch of physics which deals with the measurement of the wavelength and intensities of electromagnetic radiation emitted or absorbed by atoms is called Atomic spectroscopy.

Series of hydrogen spectrum:

Electron emits its energy in the form of electromagnetic radiation when it jumps from higher energy level to lower energy level. Frequency of emitted EM radiation depends upon the difference in energy between these energy levels.

These frequencies belong to the different regions of EM spectrum depending upon in which energy state electron is after losing its energy. This phenomenon gives rise to the different series of hydrogen spectrum. These series are given below

1. Lyman Series:

If an electron falls from higher energy level to the ground state ($n=1$) it emits radiation which are in the ultra violet region of EM spectrum. $|\bar{\nu}|$ for Lyman series is written as

$$|\bar{\nu}| = \frac{1}{\lambda} = R_H \left(\frac{1}{1^2} - \frac{1}{n_i^2} \right)$$

2. Balmer Series:

If an electron falls from higher energy level to the 1st excited state ($n=2$) it emits radiation which are in the visible region of EM spectrum. $|\bar{\nu}|$ for Balmer series is written as

$$|\bar{\nu}| = \frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n_i^2} \right)$$

3. Paschen Series:

If an electron falls from higher energy level to the 2nd excited state ($n=3$) it emits radiation which are in the infrared region of EM spectrum. $|\bar{\nu}|$ for Paschen series is written as

$$|\bar{\nu}| = \frac{1}{\lambda} = R_H \left(\frac{1}{3^2} - \frac{1}{n_i^2} \right)$$

4. Brackett Series:

If an electron falls from higher energy level to the 3rd excited state ($n=4$) it emits radiation which are in the far infrared region of EM spectrum. $|\bar{\nu}|$ for Brackett series is written as

$$|\bar{\nu}| = \frac{1}{\lambda} = R_H \left(\frac{1}{4^2} - \frac{1}{n_i^2} \right)$$

5. Pfund Series:

If an electron falls from higher energy level to the 4th excited state ($n=5$) it emits radiation which are in the far infrared region of EM spectrum. $|\bar{\nu}|$ for Pfund series is written as

$$|\bar{\nu}| = \frac{1}{\lambda} = R_H \left(\frac{1}{5^2} - \frac{1}{n_i^2} \right)$$

Bohr Atomic Model of hydrogen:

In order to develop a quantitative theory for the spectrum of the hydrogen atom, Bohr put forward the following postulates.

1. Total energy of the electron remains constant as long as it remains in the same orbit.
2. If an electron jumps from higher energy level (orbit) of energy E_i to the lower energy level (orbit) of energy E_f i.e. ($E_i < E_f$), a photon is emitted. whose frequency is given by

$$f = \frac{E_i - E_f}{h}$$

3. An electron moves only in those circular orbits for which its orbital angular momentum L is an integral multiple of $\hbar \left(\frac{h}{2\pi} \right)$.

$$L = n\hbar = \frac{nh}{2\pi}$$

Determination of Atomic Radii:

Consider a hydrogen atom as shown in figure as the electron revolving around the nucleus. Since the electron is negatively charged and the nucleus is consisting of a single proton there is a coulombic attractive force between

Squaring on both sides we get

$$v^2 = \frac{n^2 \hbar^2}{m_e^2 r^2}$$

Putting above in equation (a) we get

$$\frac{ke^2}{r^2} = \frac{m_e}{r} \times \frac{n^2 \hbar^2}{m_e^2 r^2}$$
$$ke^2 = \frac{n^2 \hbar^2}{m_e r}$$

Separating for r we get

$$r = n^2 \left(\frac{\hbar^2}{m_e ke^2} \right)$$
$$\boxed{r = n^2 a_o}$$

Where a_o is called Bohr radius and its value is given by

$$a_o = \frac{\hbar^2}{m_e ke^2}$$
$$a_o = \frac{(1.05459 \times 10^{-34})^2}{9.11 \times 10^{-31} \times 9 \times 10^9 \times (1.6 \times 10^{-19})^2}$$
$$\boxed{a_o = 0.53 \text{ \AA}}$$

n is called principle quantum number and it represents orbit number for the electron.

For ground state ($n = 1$) radius r_1 of orbit will be

$$r_1 = (1)^2 \times 0.53 \text{ \AA}$$
$$r_1 = 0.53 \text{ \AA} = a_o$$

There for Bohr radius can be defined as

"It is the radius of orbit of electron in of a hydrogen atom in ground state."

Determination of the Energy of an Electron in a "Stationary State":

According to Bohr atomic theory electron does not radiate energy as long as it remains in the same orbit or energy level.

The total energy E of the electron can be written as

$$\boxed{E = K.E + P.E}$$

Kinetic Energy:

$$K.E = \frac{1}{2} m_e v^2$$
$$\frac{ke^2}{r^2} = \frac{m_e v^2}{r}$$
$$\frac{ke^2}{r} = m_e v^2$$

Separating for v^2 we get

$$v^2 = \frac{ke^2}{m_e r}$$

Putting in expression for kinetic energy we get

$$K.E = \frac{1}{2} m_e \frac{ke^2}{m_e r}$$

$$K.E = \frac{ke^2}{2r}$$

Potential Energy:

Expression for electric potential is

$$V = \frac{W}{e}$$

$$P.E = -W$$

$$V = - \frac{P.E}{e}$$

$$P.E = -eV$$

Now Since

$$V = \frac{ke}{r}$$

Now potential energy can be written as

$$P.E = -e \frac{ke}{r}$$

$$P.E = - \frac{ke^2}{r}$$

Total Energy:

Total energy can be written as

$$E = K.E + P.E$$

$$E = \frac{ke^2}{2r} - \frac{ke^2}{r}$$

$$E = \frac{ke^2}{r} \left(\frac{1}{2} - 1 \right)$$

$$E = - \frac{ke^2}{2r}$$

Putting the value of r in above expression we get.

$$E_n = - \frac{ke^2}{2n^2 a_0}$$

$$E_n = - \frac{1}{n^2} \left(\frac{ke^2}{2a_0} \right)$$

Now putting values of constant and converting in to Electron-Volts we get

$$\frac{ke^2}{2a_0} = 13.6\text{eV}$$

Putting the values in expression for E_n we get

$$E_n = -\frac{13.6\text{eV}}{n^2}$$

Hydrogen Spectrum:

According to Bohr Atomic theory when an electron jumps from higher energy level (orbit) of energy E_i to the lower energy level (orbit) of energy E_f i.e. ($E_i < E_f$), a photon of is emitted Whose frequency is given by

$$f = \frac{E_i - E_f}{h}$$

Here

$$E_i = -\frac{1}{n_i^2} \frac{ke^2}{2a_0}$$

$$E_f = -\frac{1}{n_f^2} \frac{ke^2}{2a_0}$$

Putting values we get

$$f = \frac{1}{h} \left[-\frac{1}{n_i^2} \frac{ke^2}{2a_0} - \left(-\frac{1}{n_f^2} \frac{ke^2}{2a_0} \right) \right]$$

$$f = \frac{1}{h} \left(\frac{1}{n_f^2} \frac{ke^2}{2a_0} - \frac{1}{n_i^2} \frac{ke^2}{2a_0} \right)$$

$$f = \frac{ke^2}{2ha_0} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

Now since

$$c = \lambda f$$

$$f = \frac{c}{\lambda}$$

Putting in above we get

$$\frac{c}{\lambda} = \frac{ke^2}{2ha_0} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$\frac{1}{\lambda} = \frac{ke^2}{2cha_0} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \dots (b)$$

Now here

$$\frac{1}{\lambda} = |\bar{\nu}|$$

$|\bar{\nu}|$ is called wave number and

$$\frac{ke^2}{2hca_0} = R_H$$

Where R_H is called Rydberg constant and its numerical value is

$$R_H = 1.0970 \times 10^7 m^{-1}$$

Now equation (b) will become

$$|\bar{\nu}| = \frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

X-Rays:

X-rays are high frequency (short wavelength) electromagnetic radiation. Frequency range of X-ray is from 10^{16} Hz to 10^{20} Hz .

Frequency Range

The frequency range of these x-rays is from $30 \times 10^{15} \text{ Hz}$ to $30 \times 10^{18} \text{ Hz}$.

Wavelength range

The wavelength range of x-rays is from 0.001 nm to 1nm.

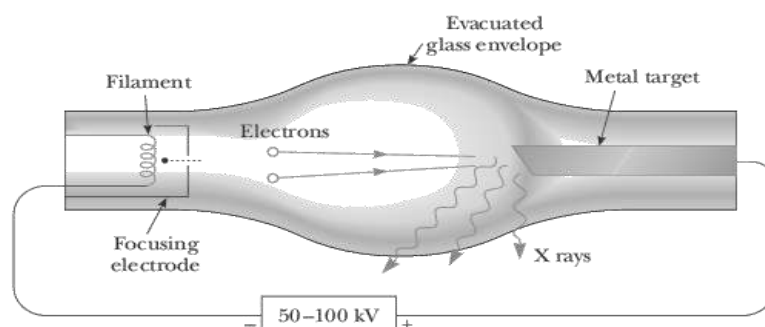
Properties of X-rays

1. X-rays have very short wavelength of the order of the same size as that diameter of an atom.
2. X-rays causes ionization in atoms or molecules.
3. They affect photographic film in the same way as visible light.
4. X-rays are stopped by metal and human bones when incident on them.
5. X-rays are not deflected by electric or magnetic field.
6. They posses high penetrating power.
7. X-rays can penetrate through optically opaque objects such as paper, wood, glass, etc.
8. X-rays travel in the straight line with a speed equal to the speed of light.
9. They can be diffracted by the crystal lattice.
10. X-rays can destroy the living cell and tissues.

Generation of X-rays:

X-rays are generated by accelerating electron to high velocity and then suddenly stopping them by collision with a solid body of atomic number greater than 10.

Typical apparatus used for X-ray generation is shown in figure.



High amount of current is passed through highly resistive filament at cathode. Cathode emits electron by thermionic emission. These electrons are accelerated to high speed by high potential difference between anode and cathode. These high speed electrons collide with Anode (of high atomic number). If potential difference is of several thousand of volt X-rays are emitted out of the anode.

Derivation:

X-rays are produced when electron strike a metal surface. These electrons are released from the heating filament and accelerated by a high voltage towards the metal target. When the electrons collide with the atoms and nuclei of the metal surface, these highly energetic particles are decelerated by the atoms and some of the kinetic energy is converted into electromagnetic energy and x-rays are produced. If the incident electrons lose the kinetic energy in more than one collision then less energetic x-rays are produced, called soft x-rays. If the incident electrons lose all the kinetic energy in one collision then high energetic x-rays are formed, called hard x-rays. If one electron completely loses its kinetic energy and a photon is emitted then

Loss in kinetic energy = energy of a single photon

$$(V)(e) = (h) (\nu_{\max})$$

$$[\nu_{\max} = (c/\lambda_{\min})]$$

$$(V)(e) = (h) (c/\lambda_{\min})$$

$$\lambda_{\min} = \frac{hc}{Ve}$$

X-Ray Spectra:

Spectral analysis of X-ray shows that

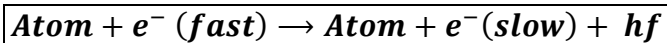
- i. X-rays have continuous spectrum also called X-ray Bremsstrahlung.
- ii. Under certain condition there is an additional line spectrum called Characteristic spectrum.

X-Ray Continuous Spectra (X-ray Bremsstrahlung):

In ordinary condition, a continuous spectrum is emitted with maximum frequency directly proportional to the acceleration voltage and is nearly independent of material of which electrodes are made of.

Explanation:

When a fast moving electron from cathode passes near to the nucleus of atoms in anode nucleus attracts it toward itself by coulombic force but due to the high speed of electron it passes away with some deflection due to this scattering process electron transfers some of its momentum to atom (or loses energy). Since electron can lose any amount of energy (from zero to its maximum energy eV) and therefore Continuous spectrum is obtained and whole process can be described by the equation



here; hf is the energy of emitted X-ray photon.

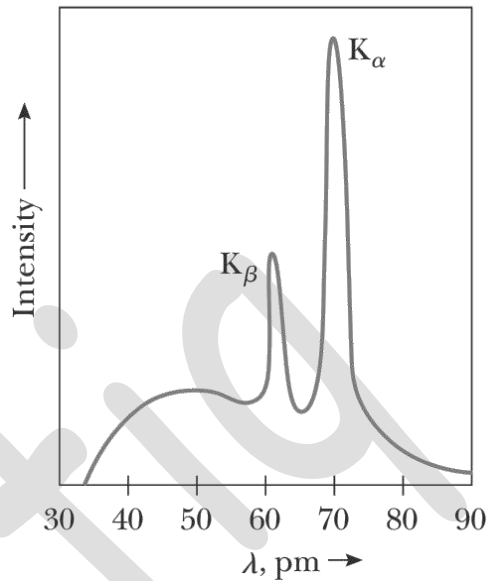
X-ray line spectra (Characteristic Spectra):

In X-Ray spectra certain peaks high intensity peaks appear that are called X-ray line spectra (Characteristic Spectra).

Explanation:

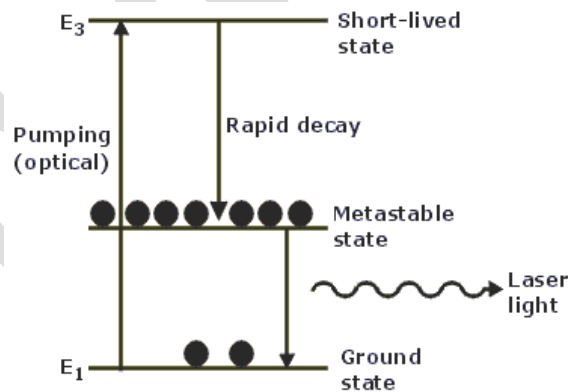
The process of emission of characteristic spectra takes place as follows. When a highly energetic incident electron knocks an electron from the k-shell, a vacancy occurs in that shell. This vacancy is filled by the arrival of an electron from outside the k-shell, emitting excess amount of energy in the form of photon.

If the electron jumps only one shell and returns with the emission of X-rays to Y shell, then X-rays are termed as ' Y_{α} ' X-rays. If the electron jumps two shells and returns with emission of X-rays to suppose 'Y' shell, then X-rays are termed as ' Y_{β} ' rays and so on, where Y may be K, L, M,.....



LASER

LASER stands for *Light Amplification by Stimulated Emission of Radiation*. Laser is a device used to produce very intense, highly directional, coherent and monochromatic beam of light.



Basic Conditions to Produce Laser

1. There must be a Meta stable state in the system.
2. The system must achieve population inversion.
3. The photons emitted must be confined in the system for a time to allow them further stimulated emission.

Principle of Laser

The principle of laser production is based on the fact that atoms of a material have a number of energy levels in which at least one is Meta stable state.

The laser operation

To understand the laser operation we have to study the following processes and definitions

1. Optical pumping of atoms
2. Spontaneous emission
3. Population inversion
4. Metastable state
5. Stimulated emission
6. Amplification

Explanation:

Consider a three level atomic system having energies E_1 , E_2 and E_3 respectively. Let the atoms are at ground state E_1 . If photons interact with an atom in ground state, the atom absorbs the photon and reaches the excited state E_3 . We know that the excited state is an unstable state, therefore, electron must return back to ground state E_1 but such transitions are not allowed and the electron first reaches the state E_2 . Atoms in the state E_3 which has a life time of about 10^{-8} sec decay spontaneously from state E_3 to state E_2 which is Meta stable and has life time of 10^{-3} sec. This means that the atoms reach state E_2 much faster than they leave state E_2 . This results in an increase in number of atoms in state E_2 , and hence population inversion is achieved. After achieving population inversion it is exposed to a beam of photons which causes induced emission of photons and a beam of laser is produced.

Characteristics of laser light

1. Laser produces an intense beam of light.
2. Laser beam is highly monochromatic i.e. single wavelength.
3. Laser light is unidirectional.
4. It is coherent beam of visible light i.e. all parts of beam are in phase.
5. It can be focused very sharply.

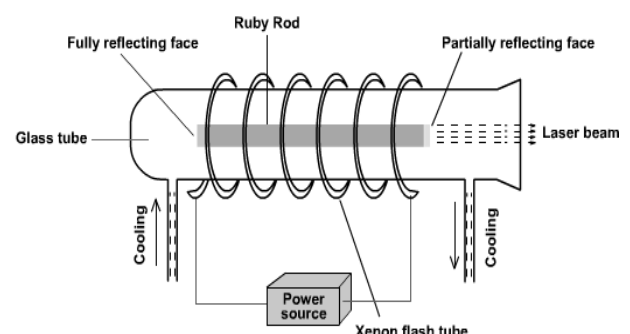
Applications of laser light

1. The laser light is used in holography in which three dimensional images of objects are formed.
2. It is used as a surgical tool for welding detached retina
3. Laser is used as a sensor to measure the distance.
4. It is used as a potential energy source including nuclear fusion reaction.
5. It is used in communication along optical fibers.
6. It is used to drill tiny holes in solid materials and for precise cutting of metals.
7. This focused beam of light is used to destroy cancerous and pre-cancerous cells.
8. The narrow intense beam of laser is used to destroy tissue in localized area.

RUBY LASER:

Construction:

Ruby is a crystal of Al_2O_3 , a small number of whose Al atoms are replaced by Cr^{+3} ions. A high intensity helical flash lamp surrounding the ruby rod is used as light source to raise Cr atoms from state E_1 to E_3 .



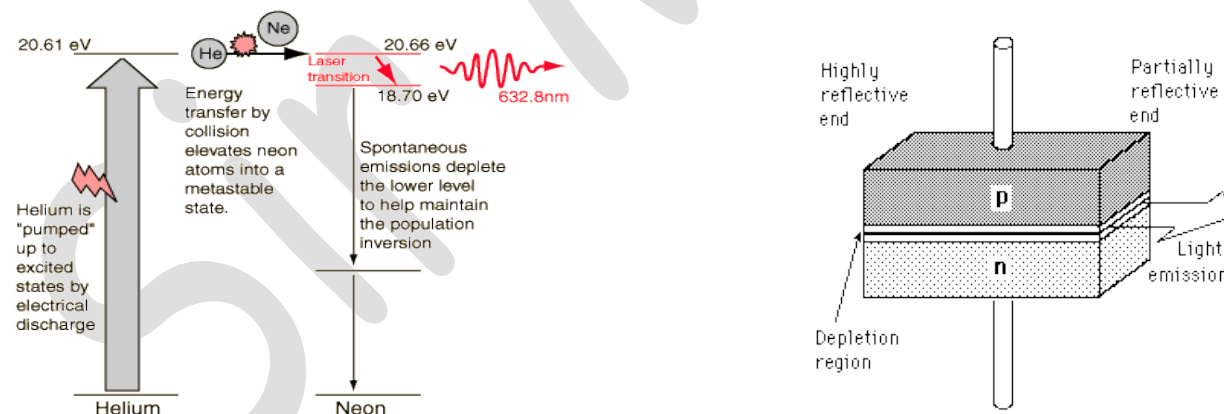
The ruby laser is a cylindrical rod with parallel, flat reflecting ends. One end is partially reflecting. The flash light is attached with the high voltage.

Working:

1. Electrons are raised from ground state E_1 to Excited state E_3 (optical pumping) which has a life time 10^{-8} sec.
2. The atoms from the state E_3 make transition to state E_2 . Since E_2 is meta-stable state having life time equal to 10^{-3} sec.
3. As atoms reach state E_2 much faster than they leave state E_2 . This results in an increase in the number of atoms in state E_2 and hence population inversion is achieved.
4. In this process few Cr atoms make spontaneous transition from E_2 to E_1 and emitted photons stimulate further transition. In this way we obtain an intense, coherent, monochromatic beam of red laser.

Helium Neon Laser

Helium Neon laser is a gas type laser and its discharge tube is filled with 85% helium gas which is used to be pumped up to excited state by electrical discharge and 15% neon gas which is used to produce lasing action in the range 632.8 nanometer in the visible spectrum. The Metastable state of helium gas is 20.61eV and that of neon gas is 20.66eV. The high voltage electric discharge excites the electrons in helium atoms to the 20.61eV. The excited atoms of helium collide with the neon atoms and transfer its energy 20.61eV to neon atoms along with the kinetic energy of 0.05eV of moving atoms. Now the electrons in the neon gas raised to the 20.66eV state. Stimulated in the neon gas de-excite the electrons from energy level 20.66eV to an energy level 18.70eV and red laser light of wavelength 632.8 nm corresponding to energy 1.96 is produced.



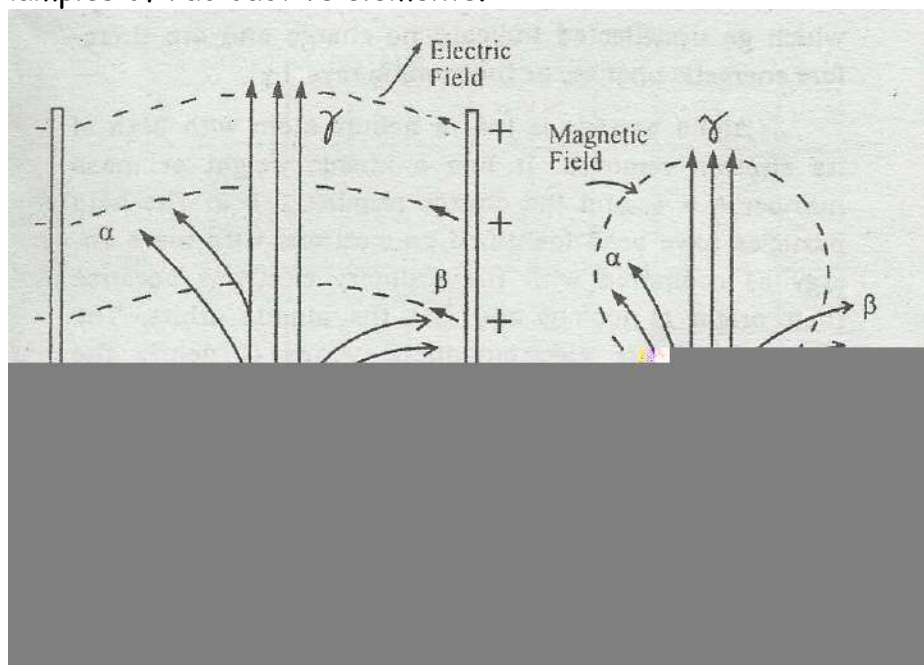
The Diode Laser:

Laser action can be achieved in a p-n junction formed by two doped gallium arsenide layers. The two ends of the structure need to be optically flat and parallel with one end mirrored and one partially reflective. The length of the junction must be precisely related to the wavelength of the light to be emitted. The junction is forward biased and the recombination process produces light as in the LED (incoherent). Above a certain current threshold the photons moving parallel to the junction can stimulate emission and initiate laser action.

Chapter 19 "The Atomic Nucleus" Compiled by: Sir M. Atiq

Radioactivity

The element whose atomic number is greater than 82 are unstable and emits radiation called α -rays, β -rays and γ -rays. The elements are called radioactive elements. Uranium, polonium and radium are the examples of radioactive elements.



Properties of α -rays

1. The ionization capability of α -rays is very large.
2. Penetration power of α -rays is very small.
3. These rays produce fluorescence in certain substances.
4. These rays can induce artificial radioactivity in certain nuclei.
5. These rays produce burns and sores on human body.
6. α -rays get absorbed after passing through a small distance in air because of high ionization power.
7. When α -rays are absorbed in matter, they produce heating effect.
8. These rays are deflected from electric and magnetic field showing that these rays are positively charge particles.
9. The velocity of these particles very small as comparable to the speed of light.
10. The atomic mass number of α particle is 4 and charge number is 2, e.g. He_2^4 .

Properties of β -rays

1. These rays affect the photographic plate.
2. β -rays has smaller in mass than a particles.
3. These rays produce fluorescence easily.

4. These rays are easily scattered by nucleus of atoms as comparable to a particles because of their small mass.
5. The velocity of β -rays much greater than the velocity of a particles but less than the speed of light.
6. The penetration power of β -rays much larger than the penetration power of α -rays.
7. The ionization power of β -rays is small.
8. The atomic mass number of β -rays is 0 and the charge number is $-1, e_{-1}^0$.
9. These rays are deflected by electric and magnetic field showing that these rays are of negative charge.
10. These rays also produce heating effect when absorbed by matter.

Properties of γ -rays

1. γ -rays produce feeble fluorescence.
2. The eject electron when incident on atom.
3. These rays also get absorbed in various elements.
4. Penetration power of γ -rays is very large.
5. These rays can penetrate even through a human body.
6. These rays are not deflected by electric and magnetic field showing their neutral nature.
7. These rays are dangerous can cause damage human cell.
8. They can also used to destroy can cell in human body.
9. They are diffracted by crystals like X-rays.
10. The ionization power is negligible.

Uses and applications of radioactive substance

1. In industry radio isotopes are used widely to check whether the thickness of a material is constant or not.
2. In industry ^{60}Co emits high energy gamma radiation and it can be used to detect cracks in welded joints.
3. In agriculture radio isotopes are used to kill bacteria and preserved food stuff.
4. In the field of agriculture radio isotopes determine the optimum amount of fertilizers.
5. In the field of medicine they have also helped in the diagnosis and cure of many complicated diseases.
6. Phosphorus-32 and iodine-131 are used as tracer to trace out the path of an element in the body or plant, animal or human being.
7. In geology as a radioactive dating the age of rocks, organism etc can be estimate.
8. Radioactive rays are widely used in science research laboratories.

Radioactive decay:

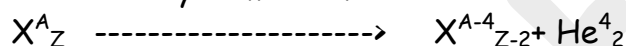
When radiation is emitted out of any radioactive element it always change into a new element and this phenomenon is called radioactive decay or radioactive transformation or radioactive transmutation.

Daughter Element and Parent Element

The new element found due to the process radioactive decay is called daughter element and the original was parent element.

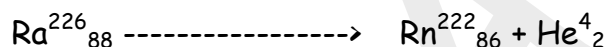
Alpha decay

When a particle is emit out of any nucleus the mass number of the nucleus decrease by number 4 and the charge number is decrease by number 2.



Example

When alpha particle is emit out from the nucleus of radium Ra_{88}^{226} the element changes into a radon gas Rn_{86}^{222}

**Beta decay**

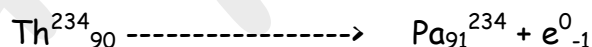
When a β particle is emit out of any nucleus the only charge number is increase by one



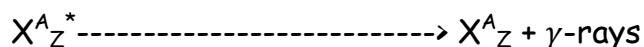
Example



Thorium changes into protactinium emitting beta particle

**Gamma decay**

When gamma radiation is emitting out of the nucleus then the charge number and mass number of the element remains same. When a nucleus is excited by absorbing energy and after when it de-excite and coming back to the ground state because it is unstable so it produce gamma rays.



Where

X_Z^{A*} represent excited state and X_Z^A represent ground state.

1. Protons - Induced Reactions

If lithium absorb a proton, two alpha particles are found to be produced in the reaction.

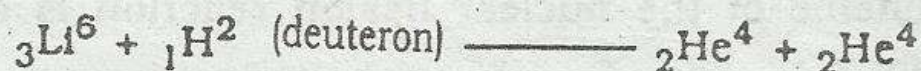


Two alpha particles.

This reaction is of great historical importance because it provided the earliest experimental verification of the Einstein's mass - energy relationship.

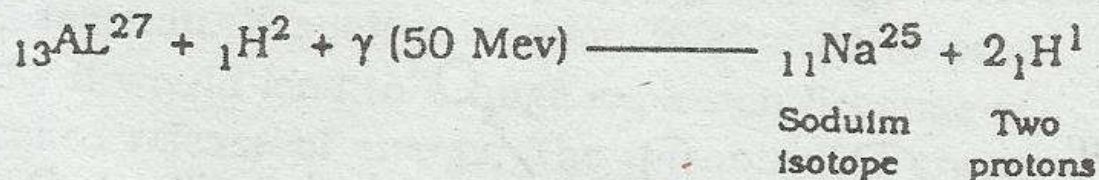
2. Deuteron - Induced Reactions

High energy deuterons may be absorbed by ${}_3\text{Li}^6$ to produce two alpha particles i.e



3. Gamma - Induced Reactions

High energy gamma rays also have been found to induce nuclear reactions by a process which is usually known as photo disintegration. Examples of such reactions are :



Mass Defect

The mass of the nucleus is always less than the total mass of all the protons neutrons making up the nucleus. The missing mass is called mass defect

$$\Delta m = zm_p + (A-Z)m_n - m_{\text{nucleus}}$$

Binding Energy

The missing mass is converted to the energy in the formation of the nucleus called binding energy. It is given by Einstein mass energy relation

$$B.E = (\Delta m)c^2$$

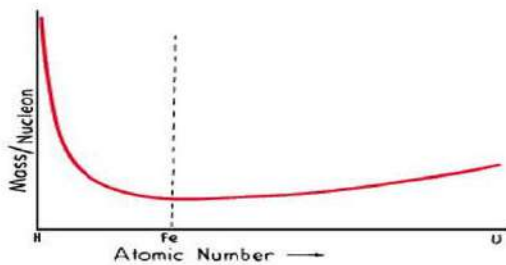
The binding energy is the amount of energy that must be supplied to a nucleus if the nucleus is to be broke up into protons and neutrons.

Packing Fraction

The mass defect per nucleons is called packing fraction.

$$F = \Delta m/A$$

Where A is the number of nucleons.



Binding Fraction

The binding energy per nucleons is called binding fraction.

$$f = B.E / A$$

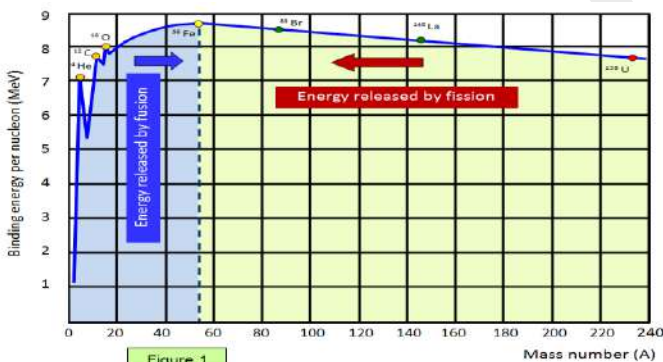


Figure 1



Half Life

The period of time in which half of the atoms of the radioactive substance decay is called half-life of radioactive substance.

Derivation

Let N is the number of atoms present in a radioactive substance and ΔN is the number of decaying atoms. It is proof that the decaying atoms are directly proportional to the time interval Δt and the number of atoms present initially N.

$$\Delta N \propto N \text{-----}(1)$$

$$\Delta N \propto \Delta t \text{-----}(2)$$

Comparing (1) and (2)

$$\Delta N = -(\text{constant}) N \Delta t$$

$$\Delta N = -\lambda N \Delta T \text{-----}(3)$$

Where λ is the constant of proportionality and is known as decay constant.

The negative sign indicates the decrease in number of atoms.

$$\lambda = -(\Delta N/N) / \Delta t$$

Here $\Delta N/N$ is called fraction of decaying atoms. The decaying constant is defined as the fraction of the decaying atoms per unit time. The S.I unit of decaying constant is s^{-1} .

By using integration the relation can also be written as

$$N = N_0 e^{-\lambda t} \text{-----}(4)$$

N_0 = number of atoms in parent elements at $t=0$

N = number of atoms in daughter elements

At half life $t = T_{1/2}$ and $N = N_0/2$

Substituting values in (4)

$$N_0/2 = N_0 e^{-\lambda T_{1/2}}$$

$$1/2 = e^{-\lambda T_{1/2}}$$

Take natural logarithm both sides of the equation

$$\ln(1/2) = \ln e^{-\lambda T_{1/2}}$$

$$T_{1/2} = 0.693/\lambda$$

By knowing the value of decay constant of radioactive element we can easily estimate the half life of radioactive element.

Activity

The number of decays per unit time is called decay activity.

$$A = \Delta N / \Delta t$$

We know that

$$\Delta N / \Delta t = \lambda N$$

So

$$A = \lambda N$$

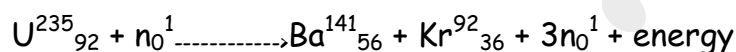
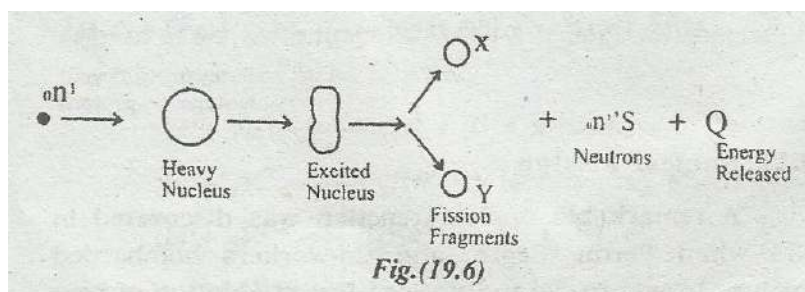
The S.I unit of activity is Becquerel (Bq), other units are curie and Rutherford.

Radioactive Isotopes

Isotopes of an element have the same atomic number and different atomic mass number i.e. they differ only in the number of neutrons.

Nuclear Fission

The splitting of a nucleus into fragments with the emission of energy when bombardment by a neutron is called a fission process.



Where ${}^{235}_{92}\text{U}$ isotope of uranium

${}^{141}_{56}\text{Ba}$ represent the barium

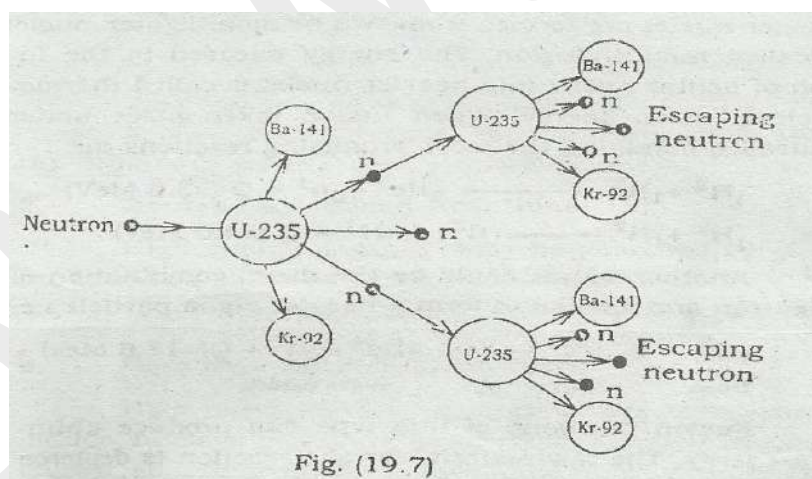
${}^{92}_{36}\text{Kr}$ represent krypton nuclei.

Examples

1. The atomic bomb is an example of uncontrolled nuclear fission reaction.
2. The reaction in nuclear reactors to generate the electricity is an example of controlled fission reaction.

Chain Reaction

In a fission reaction each nucleus emits about two to three neutrons. These neutrons may collide with the other uranium nuclei and cause fission reaction which will emit more neutrons. These neutrons will produce further fission in other nuclei. If this process continuous more and more neutrons are produced and a large amount of energy will released. This is called a nuclear chain reaction.

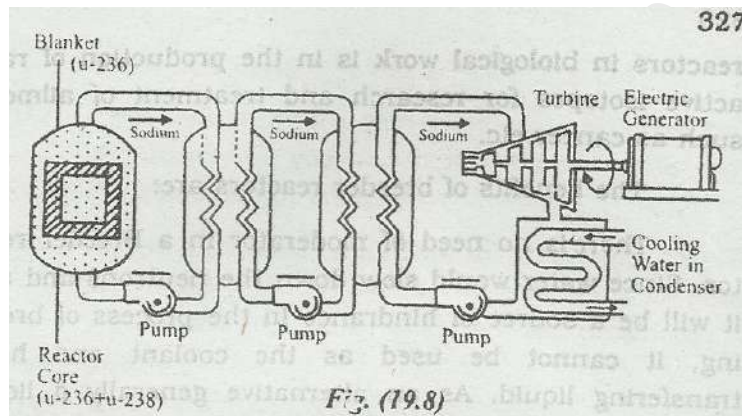


Controlled Chain Reaction

Basically nuclear chain reaction is uncontrolled fission reaction release large amount of energy. This energy is in the form of heat. If we controlled the chain reaction this heat can be used to run a turbine for the generation of electricity. A system used to control the fission chain reaction is called nuclear reactor.

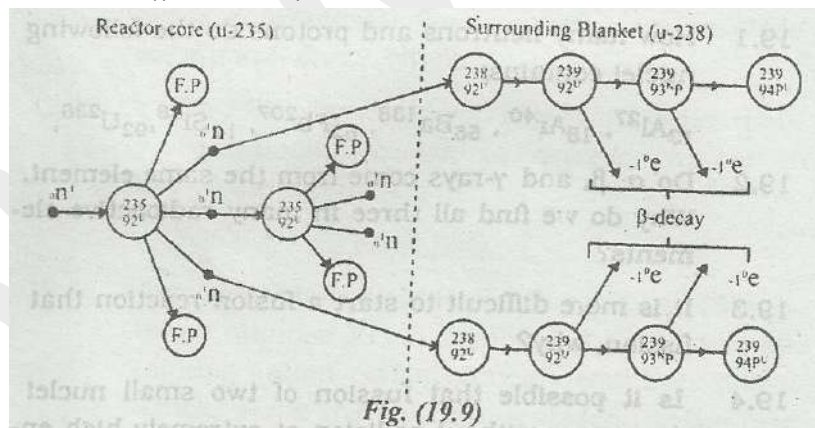
Nuclear Reactor

The fission reaction in a reactor is uranium isotope U^{235}_{92} and is called the fuel element. The neutrons released from fission move with high velocities. These fast moving neutrons are usually lost or absorbed by somewhere in the reactor assembly before producing further fission reaction. The fast moving electrons are slowed down before they are able to cause further fission reaction. The process of slowing down neutrons is called moderation. Heavy water is commonly used as a moderator.



Control Rods

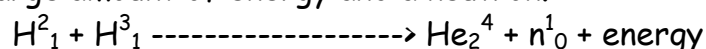
The rate of a nuclear chain reaction is controlled by inserting some substances that absorb neutrons. They are called control rods. Boron rods are used as control rods. If too many of the neutrons are absorbed by the control rods the nuclear reaction will stop. To start the chain reaction the control rods are moved out.



Nuclear Fusion

When lighter nuclei combined together to form a one heavy nuclei and the amount of energy is release in the process then it is called nuclear fusion reaction.

When deuterium and tritium isotopes of hydrogen are combined together, they form a helium nucleus with the release of large amount of energy and a neutron.



In nuclear fusion reaction a large amount of temperature is required to start the reaction greater than one million degree Celsius. In hydrogen bomb fission reaction is used to start the fusion reaction.

Examples

1. The chemical reaction in the stars and sun is nuclear fusion reaction.
2. Hydrogen bomb is also an example of nuclear fusion reaction.

Atomic mass unit (a.m.u)

To indicate the mass of atomic particles, instead of kilogram, unified mass scale (u) is generally used. By definition $1\text{u} = 1.67 \times 10^{-27}\text{kg}$

Mass of proton is 1.0078 u

Mass of neutron is 1.0086 u

Mass of electron is 0.00055 u

Mass of ${}_1\text{H}^1$ 1.007825 u

Mass of ${}_1\text{H}^2$ 2.014102 u

Mass of ${}_1\text{H}^3$ 3.016049 u

Mass of ${}_2\text{He}^4$ 4.002603 u

Mass of ${}_2\text{He}^3$ 3.016029 u

Mass of ${}_3\text{Li}^7$ 7.016003 u

Chapter 20 "Nuclear Radiations" Compiled by: Sir M.Atiq

Alpha particles:

1. Due to large mass α particles make very large number of collisions with the atoms and ionize them as it passes through atom.
2. When α particle passes through atom the strong electrostatic attraction force between it and an electron tears the electron off from the atom and ionize it.
3. The range of α particle is small and it is about 7×10^{-2} m in air and only 4×10^{-5} m in aluminum for the 7.7 Mev α particles.
4. The number of ions produced by an α particle or its range in air is a measure of its energy.
5. Alpha particles produce fluorescence on striking certain substances such as zinc sulphide and bariumplatinocynide.

Beta particles:

1. The mass and ionization of β is much less than the α particles.
2. Due to its small size the collisions are fewer and farther apart.
3. It can ionize an atom by strong electrostatics repulsion when it passes close to its electron.
4. The range of β particles in a medium is very large, nearly 100 times than that of α particles.
5. The ionization is less than the α particles and is one-hundred of that by α particle.
6. α Particles are stopped by an ordinary paper, but the β particles may pass through a thick book.
7. Small thickness of a heavy metal rich in electrons is enough to stop the β particles e.g. 5×10^{-3} m of aluminum.
8. Fluorescence is also produced when β particles strike calcium tungstate and barium platinocynide.

Gamma Rays:

1. Gamma rays are very high energy electromagnetic radiations of extremely short wavelength emitted from the nuclei of radioactive atoms.
2. They carry no charge and have no rest mass but possess very high energy of the order of several Mev.
3. They penetrate through far greater distance in material media as compared to α and β particles.
4. Very energetic gamma rays are capable of penetrating several centimeters of concrete.
5. Gamma rays can produce ionization in three ways e.g. photoelectric effect, Compton's effect and pair production.

6. Substance rich in electrons e.g. lead will stop most of the gamma ray photons and serve as a good shield against gamma rays.

Protons:

1. A proton is also a positively charged particle with properties similar to the α particles.
2. Its mass is one-fourth and charge is one half of that of an α particle.
3. It suffers fewer collisions with the atoms of the medium as compared with the α particle and penetrates the medium through a greater distance about (5 to 10 times) before stopping.
4. Its ionizing power is also much less, about one-fifth that of the α particle.

Neutrons:

1. A neutron is essentially emitted from the nucleus of an atom. It is so called because it is electrically neutral and carries no charge.
2. Its mass is very nearly equal to that of a proton.
3. It can neither experience nor exert any electrostatic force of attraction or repulsion.
4. When it hits an electron, it knocks it out from the atom (ionization) with practically no change in its own energy or direction of motion.
5. When it hits a nucleus, appreciable changes in its energy and direction of motion are likely.
6. A neutron is a highly penetrating but very slightly ionizing particle.

Geiger counter

A Geiger-Muller tube is one filled with a low pressure inert gas such as helium. The instrument works by ionizing radiation that passes through the tube making the gas molecules to be ionized creating charged ions and electrons. The energy created during the ionization process accelerates the ions to the cathode and the electron towards the anodes. This creates a current that is measured and counted by the device.

Structure of the Geiger counter

The main element of a Geiger counter is the Geiger-Muller tube, which is basically a chamber filled with inert gas or a mix of organic vapor and halogens. The tube contains two electrodes, the anode and the cathode, which are usually coated with graphite. The anode is represented by a wire in the center of the cylindrical chamber while the cathode forms the lateral area. One end of the cylinder, through which the radiation enters the chamber, is sealed by a mica window.

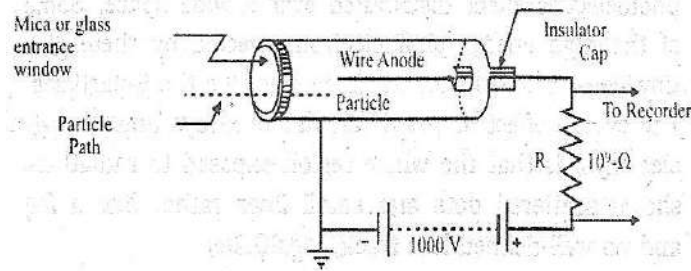


Fig.20.3 Geiger Counter

Working

As ionizing radiation coming from the surrounding medium passes through the mica window and enters the Geiger-Muller tube, it ionizes the gas inside, transforming it into positively charged ions and electrons. The electrons eventually migrate towards the anode of the tube detector, while the positively charged ions accelerate towards the cathode. As the positive ions move towards the cathode, they collide with the remaining inert gas thus producing more ions through an avalanche effect. When this happens an electrical current is established between the two electrodes.

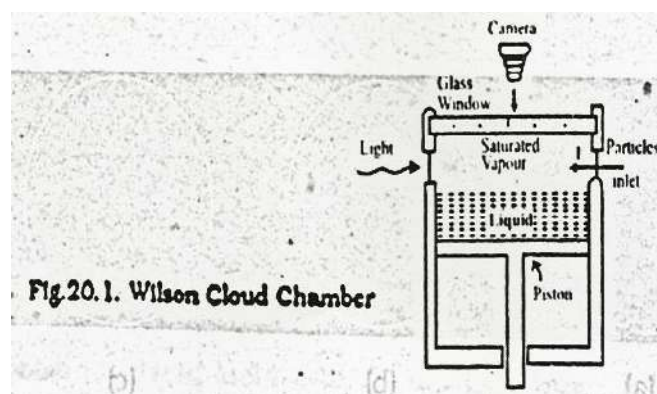
This current can then be easily collected, amplified and measured or counted and played in the form of an acoustic signal made out of clicks (available for most Geiger-Muller counters), each of which should correspond to the detection of a single ion (most of the times prevented by secondary avalanche processes). To improve the detection, multiple discharge stopping techniques can be used, either by removing the high voltage from the electrodes or by inserting additional organic or halogen gases in the inert mix.

Wilson Cloud Chamber

It is an instrument used for the detection and identification of the path of subatomic particles. In Wilson Cloud Chamber, paths of subatomic particles or ionized particles can be photographed.

Structure

It consists of a closed cylindrical chamber with the transparent glass top, a movable piston at the bottom. On the sides near the top, the cylinder is provided with a glass window. Inside the cylinder a liquid of low boiling point is placed. The piston can be moved up or down. The whole system is air tight. A strong light source is used to illuminate the chamber while the photograph is taken by the camera.



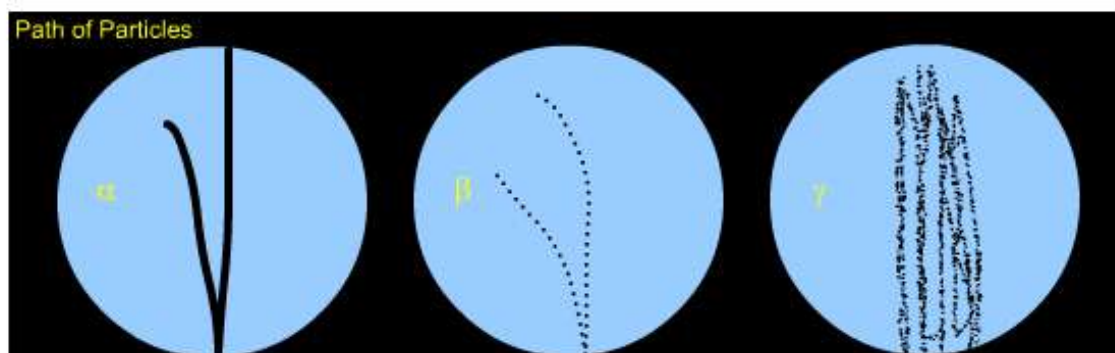
Working

Some volatile liquid having low boiling point (methanol CH_3OH or ethanol $\text{C}_2\text{H}_5\text{OH}$) poured on the inner surface of the chamber. The piston first is moved slowly up so the air inside the chamber is cleaned and then it is then moved down, so that the air pressure is dropped and the air gets vapors of the liquid and becomes supersaturated a fog is observed in the chamber.

At the right moment particles are allowed to enter into the chamber and a powerful intense beam of light is used to illuminate the track of the particles and photos are taken by the sensitive camera.

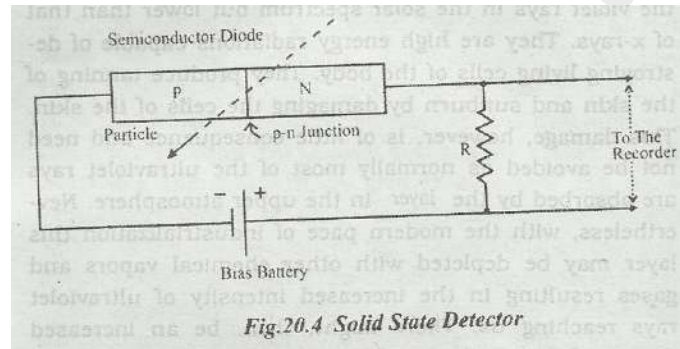
If a strong electric or magnetic field is applied to the particles (charged) then their path is altered. By the study of path's length, thickness, continuity or discontinuity the influence of magnetic field (curve) i.e. geometry the e/m ratio can be calculated and hence the particle is detected.

An alpha particle left a broad, straight path of definite length while an electron produced a light path with bends due to collisions. Gamma rays did not produce a visible track since they produce very few ions in air.



Solid State Detectors:

These devices basically make use of the solid state semiconductor diodes i.e. the p-n junction. It may be recalled that no current passes through a semiconductor diode when it is reverse biased. However if an energetic particle (or radiation) passes through the p-n junction region, a reverse current pulse passes through the diode due to the ionization of the atoms of the region. The current pulse so produced is then fed into an amplifier and the amplified pulse is applied to a loudspeaker or an electric counter which can register the number of clicks or counts respectively as the Geiger counter.



Advantages:

1. The advantage of this device over the Geiger counter is that it eliminates the use of Geiger tube, the special gaseous mixture in it and the low pressure it demands.
2. This device works at low potential difference up to 9 volts.
3. It can detect particles having energy only a few electron volts.

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Admin:

Abu Huraira Ahmad